The Geography of the Local Service Multiplier

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The draft will be fully updated with Census estimates within the next week.

Abstract. State and local governments spend approximately $20 billion a year on firm-specific subsidies, rationalized by the direct jobs these firms create and the indirect jobs their payroll supports. This paper studies, conditional on firm entry, the strength of the local service multiplier and how it depends on commuting and shopping networks in a city. Using employer-employee linked Census data, I find that about half of the entrant's new employees were not previously attached to the city labor market. Whether these workers migrate, commute from elsewhere, or were previously unemployed determines how much of the entrant's payroll becomes new local spending. I build a spatial equilibrium model across neighborhoods, calibrated to these rates of migration, commuting, and labor force participation, in which residents' choices of where to shop determine where their income is spent. In the model, the service multiplier is smaller than most existing estimates. For firm entries of the same relative size, the multiplier is about 3.7 times as large at the 90th percentile as at the 10th. Aggregate characteristics of the local economy explain only a small share of this variation, and much of the rest reflects the commuting and shopping flows around the entrant's specific location. Finally, I combine the model with a new dataset of subsidized firm entry events that records promised jobs, realized jobs, and the state and local portions of each subsidy award. In most events, the added residential property tax from the entry covers the local share of the award, and residents' welfare gains exceed the award's full cost, assuming the firm would not have entered otherwise. However, the same subsidy can deliver substantially different welfare gains at other locations in the same city.

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1 Introduction

State and local governments in the United States frequently provide subsidies to tradable firms as a development tool to boost local employment and increase economic activity in a locality. Recent estimates put this spending at roughly $20 billion per year over 2011–2021 (Hanson et al., 2025). In 2014, states spent between $5 and $216 per capita on business incentives, and in the typical state these outlays were close to 40 percent of corporate tax revenue (Slattery and Zidar, 2020).

Local officials argue that the benefits reach beyond the subsidized tradable firm because new workers spend at local service establishments such as restaurants, stores and salons. In this narrative, the entrant firm's own hiring is the first step in a broader local expansion. A city mayor expresses it this way:

> "A lot of them [new workers] live within the city of Reed City. So, they own homes here and pay property taxes. But even if they're not here inside the city, they're doing business here. They're supporting our grocery stores, they're supporting our gas stations. They're supporting our local restaurants."
>
> —Mayor of Reed City, Michigan

While the subsidized tradable firms this paper studies are among the largest half a percent of establishments in the United States, these firms are small relative to the labor market of the county and commuting zone (CZ) they enter. The median subsidized firm promises over $70 million in new payroll, yet it employs less than one percent of its county's workers and approximately 0.16 percent of its commuting zone's labor force. Because of this scale mismatch, the impact of firm entry is obscured by county- and CZ-level economic trends.

To overcome this challenge, I conduct my analysis at the neighborhood level. I combine employer-employee linked Census microdata on where each entrant's hires previously lived and worked, visits residents make to service establishments, a new subsidy dataset that distinguishes promised from actual jobs and local from state subsidies for a subsample of awards, and a quantitative general equilibrium model of Census tracts to ask three questions. Does the increase in service demand from the entrant's new hires lead to job gains and higher resident welfare? What mechanisms determine the scale of these gains? What proportion of the gains stays in the local jurisdiction paying for the subsidy, and how large is the financial leakage?

To help answer these questions, I trace the chain from the entrant's payroll to local service jobs. First, I use worker histories and earnings to approximate how much of the payroll becomes new local income. Second, I observe where residents of each census tract travel to buy services using cell phone ping data. I determine what percentage of total household shopping trips to service firms is done at local establishments. Third, I present quasi-experimental evidence on service employment from a difference-in-differences design using a tract-level exposure measure motivated by the two aforementioned pieces of motivating evidence.

The number of promised jobs does not deliver the correct metric for measuring an increase in local income and its subsequent service demand. Those are impacted by whether the hired workers

are existing residents, commuters, or migrants. I find that 58 percent of hires are continuing residents of the CZ, 28 percent live outside it after hire, and 14 percent move in. Additionally, 50 percent previously worked in the CZ. An approximate accounting from the Census microdata shows that, for every $100 of entrant payroll, $28 goes to workers living outside the CZ and commuting in. Subtracting a further $40 in pre-hire earnings of workers who previously lived in the CZ leaves $32 in additional CZ resident income. About $15 comes from earnings paid to workers who migrate into the CZ, while $17 represents the net income gain among workers who lived there before hire.

National accounts data show that approximately 48 percent of after-housing consumer spending goes to services that are consumed locally. Routing income through the full matrix of shopping flows conflates the entrant's payroll with every other income shock in the CZ. To better isolate the relationship between the income of an entrant's workers and their contribution to the nearby service sector, I use cell phone ping data. This shows that 85 percent of shopping trips to these services stay within the shopper's home CZ. A third of visits end in the home tract or a neighboring tract, and another third end at a shopping hub.[2] I use the third of spending that ends in the home tract or its neighbors to identify the impact of the entrant, because that is the piece that varies across space.

This paper presents quasi-experimental evidence on how service employment responds across neighborhoods after a firm enters, using a difference-in-differences design in which tracts differ in their exposure to the spending of the entrant's new hires. To define the exposure metric, I locate the prior residential tracts of entrants' new hires and allocate their potential service spending over their home and adjacent tracts, reflecting the pattern of spending from the cell phone ping data that varies within a CZ. Within each event, I rank tracts by this exposure and compare the top and bottom quartiles of tracts before and after entry in a stacked event study. I restrict the sample to commuting zones with no other large entrant. Seven years after entry, the top quartile has gained 60 service jobs and $5 million in service payroll relative to the bottom quartile of exposed tracts. The response is concentrated in the neighborhoods where the entrant's workers live, and gains of this size in a small number of tracts are difficult to detect in county totals.

These three pieces of evidence each trace one link – worker sourcing, shopping patterns, and service employment across neighborhoods. However, they cannot account for the full general equilibrium response of employment, wages and rents. To study these equilibrium effects, I build a model in which each CZ is a set of census tracts and an outside location. Individuals of two skill types decide where to live, work, and shop. Individuals can commute from the outside location to the census tracts within the CZ, and migrants enter the CZ from a national outside option. In the model, there are both tradable and nontradable firms of either high- or low-skill intensity. I model firm entry as a productivity increase in the entrant's tract and sector, with the increase scaled so that employment rises by the firm's promised number of jobs. Individuals choose whether to work

[2] I identify shopping hubs the way the urban literature identifies employment subcenters (Giuliano and Small, 1991; McMillen, 2001): a hub is a contiguous cluster of tracts that forms a local peak in service visits and receives at least 2 percent of its commuting zone's visits, with peaks within ten minutes' drive of one another merged into one hub (Appendix A.5).

and in which sector. Therefore, the entrant affects service firms through the increased demand for services resulting from higher incomes, and also through increased wage pressure and poaching of service workers. Additionally, individuals travel to consume services, incurring a cost that makes them sensitive to distance. These service trips can originate from both an individual's home and workplace. The migration, participation, sector choice, commuting, and shopping elasticities jointly determine how much of the entrant's payroll becomes local spending and where that spending lands. Incumbent tradable firms shrink as the entrant increases equilibrium wages and commercial rents, consistent with the modest employment decline in the tradable industries most exposed to the entrant's hiring I observe in the data. Service firms are monopolistically competitive and enter freely. Both residents and firms consume land, and residents earn rental income through ownership of residential land while commercial rents accrue to absentee landlords. The model yields welfare by skill, the fiscal cost of a job, and the share of the response that stays inside the jurisdiction that pays for it. The model isolates the effect of the entering firm, removing confounding trends that obscure the empirical comparisons at the county and CZ level.

I quantify the model separately for each of the firm entrant events in its CZ. The recruitment elasticities (migration, inbound commuting, and labor participation) are calibrated by GMM to the hiring shares I observe in the employer-employee linked data. The sector-choice elasticity is estimated with a shift-share design of industry growth instrumenting for cross-CZ variation in earnings, and the commuting and shopping elasticities are estimated from gravity equations on tract-pair flows. The remaining spending, cost, and supply parameters come from the national accounts, the shopping trip data, and the literature. Given these parameters, I invert the model to recover amenities, productivities, and fixed costs so that it reproduces each commuting zone's economy before entry. None of these steps directly uses the difference in service employment across tracts, so the event-study estimates are untargeted moments. To validate the model, I solve the equilibrium with and without each entrant, run the difference-in-differences design on the model's output, and compare the model-implied event-study coefficients with their empirical counterparts. The model is within 1.5 standard deviations of the empirical estimates.

With the model quantified, I return to the three initial questions, taking into account the full general equilibrium effects. First, does the increase in service demand from the entrant's new hires lead to job gains and higher resident welfare? It does, but by less than the promised jobs suggest. Across 55 events, an entry adds a median of 51 jobs to the commuting zone for every 100 it promises, because competition for workers and floor space removes 61 jobs at other tradable producers. The same 100 entrant jobs support 12 service jobs. The service response is small because little of the payroll becomes local spending. Of each \$100 of additional payroll at the entrant's site, residents' earnings rise by about \$30, close to the \$32 in the hired-cohort accounting, and only about \$16 reaches service firms in the CZ, because the rest goes to housing, tradable goods, and service firms elsewhere. Residents' welfare rises by about \$4,000 a year per entrant job, or 5 percent of the average wage in the entrant's tract and industry, because each hire gains only the new wage net of their outside option.

What determines the scale of these gains? Differences in entry size explain little. For entries of the same relative size, the multiplier is about 3.7 times as large at the 90th percentile event as at the 10th, ranging from about 7 to 24 service jobs per 100 entrant jobs. Aggregate characteristics of the local economy explain only about 20 percent of the variation within industries, and the clearest is reliance on workers who commute in from outside the CZ. The entrant's industry also matters. A high-skill-intensive entrant supports about 1.9 times as many service jobs per entrant job as a low-skill-intensive one and about 1.4 times as many per dollar of payroll, so higher pay explains about half of the gap. Much of the remaining variation comes from the particular site. Holding the promised jobs and the award fixed, I move each entry across at least ten sites in its host county. The observed site ranks at the 30th percentile of residents' gain at the median, and the best tenth of sites delivers about 2.1 times the gain of the worst tenth.

How much of the gain stays with the jurisdiction that pays for it? The award data I build separates the local share of each subsidy from the state share, and compares the number of jobs the firm promised with the number of jobs it actually creates. The host county holds 38 percent of the CZ's residents and receives 49 percent of the gain in resident income, but it keeps only 28 percent of the CZ's service job gain. The rest goes to the neighboring counties where the entrant's workers live and shop. The county can still afford its half of the award, because the added property tax revenue from new residents covers it in over 80 percent of events. The awards come to about $22,000 per promised job but about $44,000 per net job, because an entering firm adds only about half the jobs it promises. While this cost seems high, assuming the firm would not have entered otherwise, residents gain about $1.85 a year for each dollar of the annual subsidy.

Taken together, these results connect the entrant's employment commitment to the payroll needed to fill its jobs, the spending that reaches local businesses, and the gains accruing to residents. They distinguish regional benefits from the return to the jurisdiction offering an award. I find that the spillovers officials promise exist but are smaller than the promised jobs imply, and that their size depends on who fills the jobs, where those workers live and shop, and the entrant's industry. The multiplier is smaller where more of the workforce commutes in from outside the market. Evaluating subsidies requires accounting for both the employment and welfare created throughout the local economy and the share of those gains received by the residents bearing their cost.

Contribution to the literature. This paper makes three contributions. First, it measures the local multiplier of individual subsidized plants and traces it from payroll to service employment. Estimates from industry-wide growth (Moretti, 2010; Van Dijk, 2016), from shift-share designs (Bartik, 1991; Bartik and Sotherland, 2019; Autor et al., 2013, 2021; Hershbein and Stuart, 2024; Yagan, 2019; Hornbeck and Moretti, 2024; Acemoglu and Restrepo, 2020), and from regional variation in military procurement (Nakamura and Steinsson, 2014) treat the county, the CZ, or the state as a single point and do not observe who fills the new jobs (Goldsmith-Pinkham et al., 2020; Borusyak et al., 2022). Plant-opening and subsidized-firm designs (Greenstone et al., 2010; Patrick, 2016; Patrick and Partridge, 2022; Slattery and Zidar, 2020; Hanson and Rohlin, 2024) observe the

firm but not its workers, and county employment net of the entrant is too noisy to recover the effect of a single firm whose hiring is a small fraction of total employment. Evaluations of larger place-based programs do find local gains. Manufacturing employment in Tennessee Valley Authority counties was still growing faster than in comparison counties four decades after federal transfers lapsed (Kline and Moretti, 2014b), and federal Empowerment Zone designation raised employment in zone neighborhoods by 12 to 21 percent (Busso et al., 2013), although California's enterprise zones did not raise employment (Neumark and Kolko, 2010). These programs fund infrastructure, tax credits, or block grants across a region or neighborhood rather than a single firm. I follow the entrant's hires to where they live and shop and add up the links in equilibrium. The results reconcile the two literatures. An entrant's payroll supports local service jobs, but only about half of its own jobs are net additions, because it crowds out other tradable producers. The service response is spread over the neighborhoods where its workers live and shop, and 72 percent of it lands outside the host county, rendering a county-wide comparison incomplete. Because much of the payroll goes to workers who already had jobs in the CZ or who live outside it, the response to a single plant depends on the new income its payroll creates rather than on its headcount.

Second, I show that how open a local market is depends on where its boundary is drawn, and I measure the multiplier at the boundary of the county that pays as well as at the boundary of the CZ. The model has the neighborhood structure of Ahlfeldt et al. (2015) and Tsivanidis (2026) and adds two margins that the entry question requires. Individuals shop from home and from work, as in the consumption-trip model of Miyauchi et al. (2021) and in line with smartphone evidence that consumption trips are short and concentrated (Davis et al., 2019; Agarwal et al., 2020; Oh and Seo, 2023). I add a sector choice where individuals switch between tradable and service employment, so that the entrant's wage pressure can directly impact service firms. Both factor demand and output demand operate in a CZ whose boundary workers and shoppers cross in both directions, with leakage measured from data. Monte et al. (2018) show that local employment elasticities depend on commuting openness. I add shopping openness, which differs from commuting openness at every boundary, and show that the county that pays for an award is considerably more open than the CZ that contains it. The model is solved for a single large employer entering fine geography in 38 actual CZs rather than across counties.

Third, I separate the part of the multiplier that depends on the firm type and the part that depends on the local characteristics of the market, namely the commuting and shopping openness and the housing supply elasticity. I measure each channel by comparing the CZ total to the county that finances the employment. Theory treats mobility as a continuous relocation margin (Kline and Moretti, 2014a; Busso et al., 2013; Fajgelbaum and Gaubert, 2020), and the incidence of state business taxes is divided among workers, landowners, and firm owners with aggregate elasticities (Suárez Serrato and Zidar, 2016, 2023). Garin (2025) argues that local job counts are uninformative when the jobs go to migrants, commuters, or workers who would have been employed anyway, and the contrast between location-based and hiring-based incentives in enterprise zones (Neumark and Kolko, 2010; Freedman, 2013) shows that the concern is practical. I contribute to this analysis by

showing that while award programs measure performance by the jobs a firm promises, most of the variation in what those jobs contribute locally depends on the commuting and shopping openness of the place as well as housing flexibility.

The rest of this paper is organized as follows. Section 2 describes the policy setting and the data. Section 3 presents three facts that follow the entrant's payroll to whom it hires, where those workers shop, and where service employment responds. Section 4 develops the model and Section 5 quantifies it. Section 6 measures the net employment effect of entry, values the welfare effects and the fiscal cost of a job, and asks how much of the response the paying jurisdiction keeps. Section 7 decomposes the heterogeneity in the multiplier into the contributions of place and firm characteristics. Section 8 concludes.

2 Data and Context

Firm-specific incentives are one of the largest and fastest-growing forms of place-based economic development policy in the United States. State and local governments lost roughly $19.50 billion per year in potential revenue from 2011 to 2021 in incentive packages for business, double the level of the prior decade (Hanson et al., 2025). Good Jobs First, a nonprofit that tracks subsidies, counts 21 megadeals worth a combined $19.20 billion in 2022 alone (Tarczynska, 2022). A typical package combines income tax credits, job training credits, grants, infrastructure spending, and property tax abatements.

Firm entry. I assemble awards from Good Jobs First, state development agencies, and county records. I match each subsidized firm to its establishment record in Data Axle, a commercial dataset of U.S. businesses, and verify the location of the subsidized site with the Google Maps API. This matching allows me to ensure the firm actually enters. I then observe the exact address of its entry and can assign it an industry code. [3] For approximately 40% of the awards, I also separate the portion offered locally, as property taxes forgone by the county and other local governments, from the portion paid by the state through credits, rebates, and grants. For the remaining awards, I simply retain the amount spent by the state. I measure the jobs a firm actually creates from state compliance reports and other documented employment records. After location site verification, the subsidy dataset contains 226 firm-site events (Appendix A.1). Among the 64 firm entry events with reported employment at least four years after award, 39 percent of firms reach their promised employment, the median firm delivers 79 percent of the jobs promised, and 8 firms more than double their hiring promise (Figure A.1).

[3] Good Jobs First is a nonprofit research center in Washington, DC, that tracks economic development subsidies. Its Subsidy Tracker database compiles company-specific awards from federal, state, and local programs using government disclosures and public records requests. Data Axle, formerly Infogroup, maintains a commercial establishment-level database of U.S. businesses compiled from public sources and verified by telephone. Its annual snapshots, available from 1997, record each establishment's name, location, industry, employment, sales, year established, and parent company.

I focus on the entry and expansion of large, tradable firms. Consistent with the subsidy literature, I consider a firm "large" if it promises at least 500 new jobs, or 0.50% of the total employment in its commuting zone (CZ). I distinguish tradable industries, whose demand originates primarily in national or international markets, from nontradable industries, which mainly serve local demand (Moretti, 2010; Van Dijk, 2016). Throughout the paper, the service sector whose response I measure comprises retail, education, health, food services, and other services (NAICS 44–45, 61, 62, 722, and 81).[4]

For the motivating empirics, I examine all large tradable entries and expansions, not only subsidized ones. The labor demand and local spending shock depend on actual hires, whether or not the firm's entry decision was financed in part by subsidies. The subsidized firms are already a selected sample based on the firms' bargaining power and on the states that choose to disclose their awards (Slattery and Zidar, 2020). Using the Longitudinal Employer-Household Dynamics (LEHD) data, which link 95% of all private-sector workers to their employer, earnings, and residence, I identify large openings and expansions of tradable establishments in the 29 U.S. states for which I have data. I trace the hiring patterns of each firm, to the neighborhoods where its workers live, to the neighborhoods where they shop. I measure the response of nontradable employment, payroll, and establishments in those locations. Subsidized deals enter this sample when they meet the same size and industry screen I use to classify large, tradable firms in the public data.

The LEHD sample contains approximately 400 such entrants. I classify an entrant into high- and low-skill intensive tradables. [5] Table 1 compares the two firm types. I find that high-skill intensive tradable firms locate in substantially larger markets, with a median population of 1.21 million, compared with 291,000 for low-skill intensive entrants. High-skill intensive firms locate closer to the CZ center (defined as the group of tracts with the highest employment totals), with a median driving time of 18.2 minutes to the downtown area rather than 25.5 minutes. Lastly, median annual pay indicates that the entry of a high-skill intensive firm delivers a much larger payroll shock, but its entry is less common in less densely populated labor markets.

High-skill entrants draw more of their hires from other regions, but most of the other differences in where entrants find workers reflect the markets they enter rather than how they recruit (Table 2). The exception is the migrant share of hires, which is 2.2–2.4 percentage points higher at high-skill entrants in every specification. All other estimates (the share of hires who previously worked in the CZ, new commuters, and the share of workers within a 40-minute drive) attenuate when comparing within CZ or including controls for CZ size. Where an entrant locates therefore shapes the spatial distribution of whom it hires, while its skill-intensity shapes how much its hires gain.

Local markets and jurisdictional boundaries. I use the commuting zone (CZ) to define the local labor market, and set Census tracts as the unit for locations within it. I use CZs because they

[4] Because LODES and the Quarterly Workforce Indicators report only two-digit sectors, the public-data outcomes and the model's baseline use all of NAICS 72, which adds accommodation; the national spending shares in Section 3 also include it.

[5] A firm is classified as high skill when its four-digit industry had an above-median share of workers with a bachelor's degree in 2000 and as low skill otherwise.

Table 1: Characteristics of low- and high-skill entrants

	Low-skill entrants	High-skill entrants
Entrants (disclosure-rounded)	200	200
Hires		
Annual pay of high-skill hires (2023 $)	57,420	130,900
Annual pay of low-skill hires (2023 $)	44,690	87,240
Site and labor market		
Driving time to CZ center (minutes)	25.54	18.17
CZ population	290,600	1,211,000
Annual population growth, years −5 to −2 (percent)	0.68	0.61

Notes: Released estimates from the LEHD entrant sample. Characteristics are medians across entrants of each type; entrant counts are disclosure-rounded. Pay refers to the entrant's hires by skill and is annual, in 2023 dollars. Proximity to the CZ center is measured in driving minutes. Population growth is the annual percentage change in the CZ from five to two years before entry.

are defined as county groupings that maximize within residence–workplace connections (Tolbert and Sizer, 1996), but neither a county nor a CZ is a closed economy. Neither one of these boundaries captures the full effects of a firm entry shock. For high-skill entrants, the mean share of workers who live in the same county is 45%, compared with 48% for low-skill entrants. The ranking reverses at the CZ boundary (78% versus 75%). Thus the same high-skill sites appear relatively more open at the county boundary and less open at the CZ. In the model results, I use both boundaries to distinguish the effects of firm entry.

Shopping trips. To measure where residents shop, I use 2019 smartphone visit records from Veraset (Veraset, 2022). I weight each device equally within a month, so that the annual measure is a device-weighted distribution of visits from each home tract to each destination tract. Shopping is local: 84.80% of service visits stay within the shopper's home commuting zone, yet only 10.60 percent of within-CZ visits stay in the home tract (Table 4 in Section 3 and Appendix Table A.12) reflecting the need for a model with shopping trips to capture effects at granular scales. Because the visit data are observed in 2019, after all of my firm entry events, I use them to describe where entry-linked demand may be observable and to estimate shopping gravity. They are not used in the inversion of the model.

Prices, employment, and wages. Land data come from CoreLogic deed and assessor records over 2004–2020, which give residential and commercial repeat-sales and hedonic price indexes, building square footage, and property counts for each tract. Public employment outcomes come from LODES, which records jobs by workplace tract, resident workers by home tract, and the commuting flows between them. I use the Quarterly Workforce Indicators for county earnings by education that I allocate to tracts as workplace wages, and I use the County Business Patterns to provide establishment counts by county and industry. Appendix A documents each source and its construction.

Table 2: Entrant skill and hiring sources

	(1) No controls	(2) CZ fixed effects	(3) Log CZ employment
A. Share of hires (percentage points)			
Previously worked in the CZ			
High-skill entrant	6.35*** (2.34)	3.71 (3.08)	3.50 (2.68)
Log CZ employment			3.67*** (0.81)
New commuters			
High-skill entrant	−1.91 (2.04)	−1.16 (2.72)	0.54 (2.45)
Log CZ employment			−2.24*** (0.81)
Migrants			
High-skill entrant	2.20*** (0.84)	2.23** (0.91)	2.38** (0.97)
Log CZ employment			−0.52* (0.31)
Within a 40-minute drive at hire			
High-skill entrant	5.79*** (2.12)	5.26* (2.99)	3.31 (2.60)
Log CZ employment			1.35 (0.82)
B. Employer HHI ×100			
High-skill entrant	−2.47*** (0.42)	−0.38 (0.32)	
C. Annual earnings change, hire level (2023 dollars)			
High-skill entrant		6,419*** (2,294)	5,283*** (1,328)
BA+ hire		778 (1,298)	192 (1,027)
High-skill entrant × BA+ hire		3,186* (1,827)	5,935*** (1,994)
Log CZ employment			735 (517)
CZ fixed effects	No	Yes	No
Log CZ employment	No	No	Yes

Notes: Released estimates from the LEHD entrant sample. Each entry is a coefficient with its standard error in parentheses. Panels A and B are entrant-level regressions on an indicator for a high-skill entrant, one for each outcome. Panel C is a hire-level regression of annual earnings at the entrant minus annual earnings before hire, in 2023 dollars, on that indicator, an indicator that the hire holds a bachelor's degree, and their interaction. Column (1) has no controls, column (2) adds CZ fixed effects, and column (3) controls for log CZ employment. The regression classification counts current CZ residents with a prior job in the CZ. New commuters live outside the CZ and did not work in it before hire; within a 40-minute drive is the share of hires within a 40-minute drive of the site when hired. Employer HHI measures the concentration of employment across employers in the site's labor market. ***, **, and * denote significance at the 1, 5, and 10 percent levels, using the released p-values.

3 Motivating Facts

Fact 1. Only part of entrant payroll becomes additional local income. Of every \$100 of entrant payroll, only about \$32 is new income for residents of the CZ. About \$28 goes to workers who commute in and live outside of the CZ. An additional \$40 replaces what hires who already lived there earned before. Of the remaining \$32, about \$15 is paid to workers who move into the CZ and \$17 is the net gain of those who lived there before hire.[6]

[6]The earnings allocation is an approximation. It weights the joint residence–workplace counts by their quasi-median earnings, pooling prior industries and individual skills within entrant type and prior workplace, and then scales prior and after-hire allocations separately to the released earnings totals within entrant type and post-hire residence. This assumes the earnings weights transfer across residential histories and between the two tabulation samples. Hires with no observed prior job are assigned zero prior income. All amounts are annual 2023 dollars per \$100 of total entrant payroll.

Table 3 summarizes the new hires of large, tradable firms within the first five years of their entry into a local economy. The majority of hires are local residents. About 58 percent are continuing residents of the CZ, 14 percent move into it, and 28 percent live outside it after hire (Panel A). Those outside residents count as jobs in the CZ, but the income they earn leaves the local labor market (Panel C).

Table 3: Entrant hires, earnings, and payroll destinations

A. Residence one year after hire

	Percent of hires
In the CZ, lived there before hire	57.79
In the CZ, moved in	14.32
Outside the CZ	27.90
	Percent of continuing residents
No observed prior job	9.56

B. Earnings by prior workplace (2023 $ thousands)

Prior workplace	After hire	Increase
Inside the CZ	84.9	18.4
Outside the CZ	86.4	29.3
No observed prior job	72.2	—

C. Payroll by residence one year after hire

	Percent of entrant payroll
Paid to residents inside the CZ	71.69
Paid to residents outside the CZ	28.31

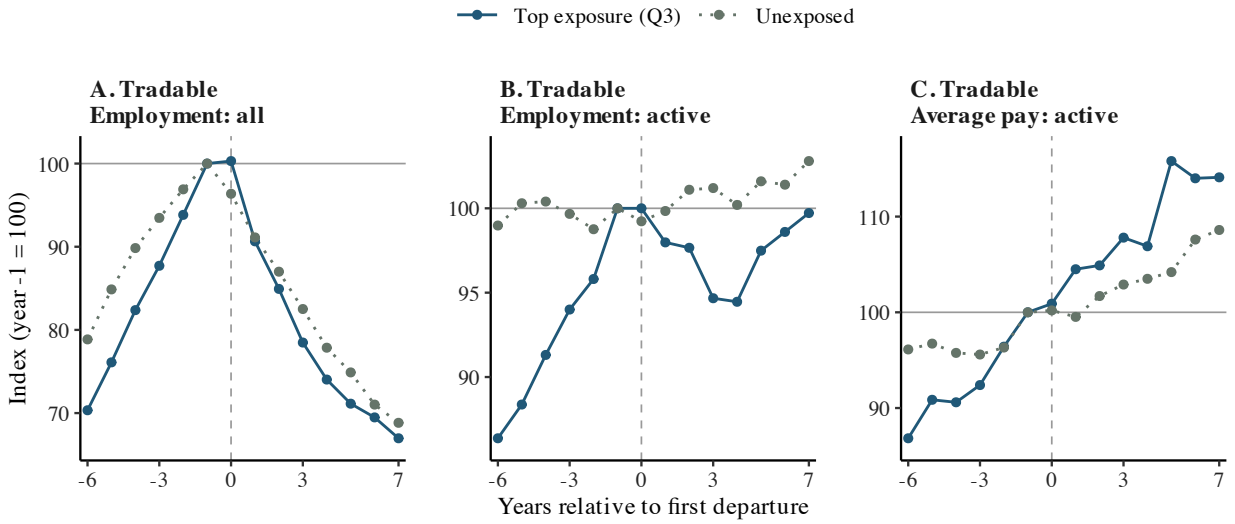
Notes: Released LEHD tabulations pooling low- and high-skill entrants. Panel A reports shares of hires by residence one year after hire. The no-observed-prior-job share uses continuing residents as its denominator. Panel B reports worker-weighted averages of the earnings medians, in thousands of 2023 dollars. After-hire earnings are annual earnings one year after hire; the increase is the difference between the after-hire and prior-year median earnings. Panel C uses group earnings totals by post-hire residence. Shares may not sum to 100 because of rounding.

The earnings summaries show why so much of the payroll replaces existing income. Hires whose previous job was in the CZ earn about $85,000 a year after hire, only $18,000 more than before, while hires from jobs outside the CZ earn about the same but gain $29,000 (Panel B). Movers bring their full pay into the CZ because their prior earnings accrued elsewhere, and continuing residents with no observed prior job add their full pay as well.

Not all of the $68 that does not become new resident income is lost to the CZ in the same way. The $28 paid to workers who live outside the CZ leaves with them. The remaining $40 is different. It is the pay that local hires earned at their previous jobs. When a worker leaves an incumbent employer for the entrant, her former job does not disappear. If the incumbent refills it with a worker who was not employed in the CZ, that worker's new earnings are also income the entry created. The $32 is therefore a lower bound. If every vacated job were eventually filled by a nonworker or a migrant, resident income would rise by about $72 per $100 of payroll instead.

The employment trajectories of other tradable firms in the same CZ show that incumbents who lose workers to the entrant generally refill these jobs, but their employment is still lower than that of unexposed establishments. Seven years after the first departure, total employment at the tradable establishments that lost the most workers to an entrant, counting those that later closed, was about 2 jobs per 100 baseline jobs below that of unexposed establishments (Figure 1 and Appendix A.4). The gap is widest three years after the first departure, at about 4 jobs per 100, and narrows as incumbents refill the vacated jobs. Among establishments still operating in year seven, the gap is about 3 jobs per 100, and average pay rose 14 percent at exposed establishments, compared with 9 percent at unexposed ones. Who fills the vacated jobs, and how much the higher wages cost incumbents, determine what total resident income is between these two bounds. The model in Section 4 answers both questions.

Figure 1: Tradable source-establishment trajectories by exposure



Notes: Released indices for the highest-exposure (Q3) and unexposed establishments. Exposure is the share of an establishment's baseline employment lost to the entrant (Appendix A.4). Year 0 is the first departure for exposed establishments; all indices equal 100 in year −1. Panels show employment across the full sample, employment among active establishments, and average pay among active establishments, each on its own scale. Missing active-employment values at years −1 and 0 are displayed at 100. No confidence intervals are released.

Fact 2. Local income is spent on services close to home. Fact 1 separates entrant payroll from new income for the CZ's residents. The next question is how much of that income reaches local service firms, and where. Nearly half of what households have left after housing goes to services that are mostly consumed locally.[7]

Workers will travel across a commuting zone for a job, but they are less willing to travel for groceries or a haircut. To directly compare commuting and shopping, I estimate, separately for

[7] In 2024, personal consumption of retail trade, education, health care, accommodation and food, and other services totaled $9.04 trillion, or 41.25% of disposable personal income and 48.14% of income left after housing. Author's calculations from the 2024 BEA PCE Bridge and NIPA Table 2.1; see Appendix A.5. These are average expenditure shares, not marginal propensities.

each commuting zone c and each travel mode $m \in \{S, C\}$, a gravity equation for shopping visits (S) and commuting flows (C) between tracts i and j,

$$\mathbb{E}\left[F_{ijc}^{m}\right] = \exp\left(\alpha_{ic}^{m} + \delta_{jc}^{m} + \beta_{c}^{m}\, t_{ijc}\right), \qquad m \in \{S, C\}, \tag{1}$$

where F_{ijc}^{S} is 2019 device-weighted service visits from residents of i to establishments in j, F_{ijc}^{C} is the 2019 LODES count of workers living in i and working in j, and t_{ijc} is driving time in minutes. I estimate equation (1) by Poisson pseudo-maximum likelihood (Silva and Tenreyro, 2006). The origin effects α_{ic}^{m} absorb the number of residents, workers, and devices in each tract, and the destination effects δ_{jc}^{m} absorb workplace size and destination attractiveness. β_{c}^{m} isolates how quickly each type of trip decays with travel time. Because travel time enters in levels, I report decay as the percentage decline in predicted flow from one additional minute, $100[1 - \exp(\widehat{\beta}_{c}^{m})]$. Appendix A.5 gives the estimation details.

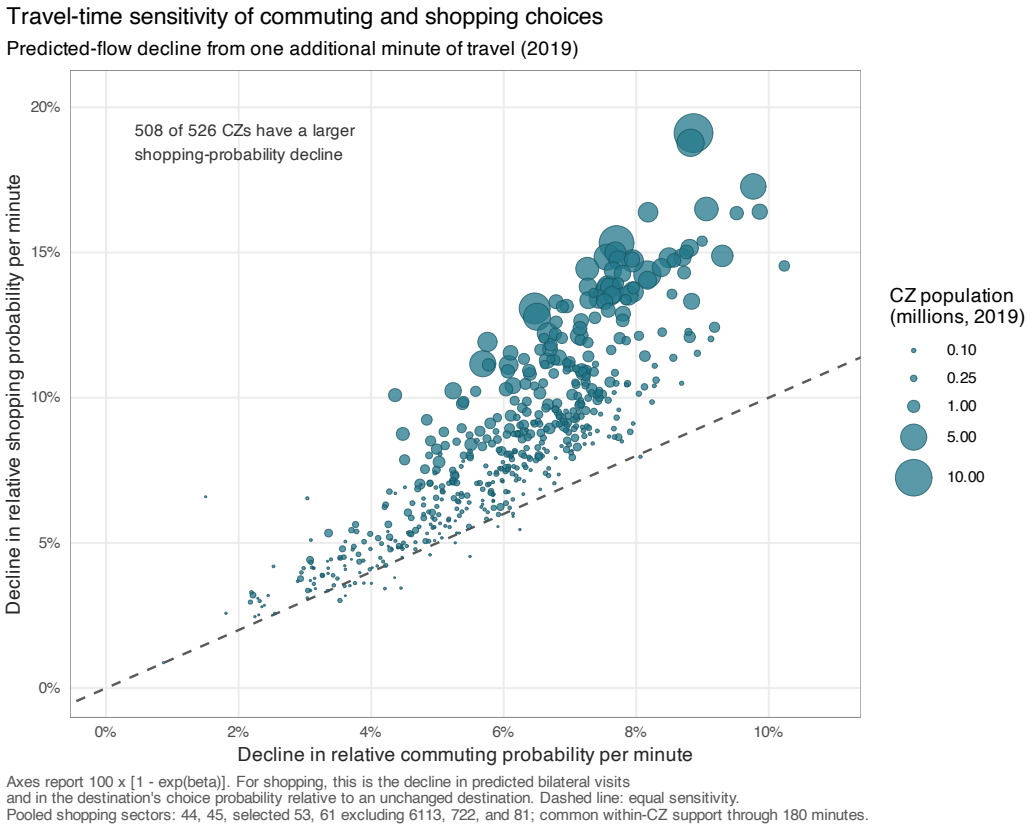
Figure 2 plots the two estimates for the 526 commuting zones. Shopping decays faster than commuting in 96.6% of CZs. At the median, an additional minute of driving time reduces predicted shopping flow by 8.1% and predicted commuting flow by 6.1%, a difference of a third. Over ten minutes, the median estimates imply that shopping flow falls by 57% and commuting flow by 46.7%. The two coefficients are strongly correlated across places (0.86), so markets in which commuting is sensitive to travel time also have concentrated shopping. These results show that the market area over which households shop is narrower than the area over which they supply labor. A new workplace can draw employees from a larger area of the commuting zone while the consumption their earnings generate remains concentrated near a smaller set of residential neighborhoods.

The same is true across jurisdictions. The median county holds 40 percent of its residents' jobs but 53 percent of their shopping visits (Appendix Figure A.5).

To measure where an entrant's workers spend in Fact 3, I have to assign their spending to destination tracts, and the geography of visits within a market determines how. Table 4 reports the shares of within-CZ visits that go to a resident's home tract, to adjacent tracts, and to shopping hubs outside the home neighborhood. Pooling the core service sectors, 11 percent of within-CZ visits end in the home tract and 24 percent in an adjacent tract, so a third of service trips end within one tract boundary of home. Nearly half of the rest goes to shopping hubs, clusters of destination tracts, merged within ten minutes' drive of one another, that each receive at least 2 percent of their market's service visits (Appendix A.5). Hubs receive 43.4% of within-CZ visits, and two-thirds of hub visits (67.5%) are to a hub away from the shopper's home neighborhood.[8] Because residents throughout a market shop at the same hubs, spending routed through the full shopping network lands in nearly the same places whichever entry generates it. Among the 85 entries in CZs with more than one entry, destination alone explains 76.6% of the variation in exposure routed through the network (Appendix A.6). The home tract and its neighbors instead carry the part of shopping that differs across tracts. In the data, 34.4% of a tract's visits end in its own or an adjacent tract,

[8] The number of hubs rises with market size, with an elasticity with respect to population of about 0.43 under the 2 percent definition (Appendix A.5).

Figure 2: Travel-time sensitivity of shopping and commuting across commuting zones



Notes: Each observation is a commuting zone. The horizontal axis is the percent decline in predicted commuting flow from one additional minute of driving time, and the vertical axis is the analogous decline in service visits, both computed as $100[1-\exp(\widehat{\beta}_c^m)]$ from the CZ-specific PPML estimates of equation (1). The dashed line denotes equal decay. The point size is proportional to 2019 population. Shopping pools NAICS 44–45, selected 53, 61 excluding colleges, 722, and 81. Both modes use 2019 flows, 2020 OSRM driving times, and the common within-CZ tract-pair support through 180 minutes; the sample is the 526 CZs in which data is available.

against 3.2% under the common pattern of its CZ (Appendix A.5). Fact 3 therefore assigns 30 percent of each worker's spending to the home tract and splits the remaining 70 percent equally among the adjacent tracts.

The visit data are from 2019, after every entry in the sample. If shopping itself responds to entry, for example because new stores open near the entrant or near its workers' homes, exposure routed through hubs and the full network would partly reflect that response. The home-and-adjacent rule does not have this problem, because which tracts it ranks as most exposed depends only on where hires lived before entry and on which tracts border theirs. This is the second reason for the rule.

Fact 3. Service employment grows where the income lands. The first two facts show that only part of entrant payroll is new resident income and that shopping concentrates near home. The third shows that service employment grows where entrant workers live. I place each hire at the tract where they lived before entry, assign their potential spending to that tract and its neighbors

Table 4: Allocation of within-CZ service visits between the home neighborhood and hubs, 2019

Sector	Same tract	Adjacent tract	Nonlocal 2% hub
NAICS 44	12.16	26.06	26.40
NAICS 45	8.98	23.58	33.47
NAICS 722	9.56	23.07	30.45
NAICS 81	13.48	22.88	22.74
Combined Core	*10.60*	*23.81*	*29.25*
NAICS 61	7.15	16.20	32.76

Notes: Entries are percentages of annual device-weighted visits, conditional on the destination remaining within the device's home CZ. The categories are mutually exclusive but not exhaustive; the remaining visits go to other destinations within the home CZ. "Adjacent tract" excludes the home tract; "Nonlocal 2% hub" is a shopping hub that excludes both the home tract and its adjacent tracts. Sector definitions are in Appendix A.5.

as in Fact 2, and compare the most and least exposed tracts within the same market. Seven years after entry, a tract in the top quartile of exposure has gained 59.4 more service jobs and $4.7 million more in annual service payroll, in 2023 dollars, than a tract in the bottom quartile, with 95 percent confidence intervals of 15.7 to 103.2 jobs and $1.6 to $7.9 million.

I restrict the sample to CZs with no other observed event, so that no comparison tract is exposed to a second entry. Within each event I compare the top and bottom quartiles whose mean exposure differs by 23.4 workers. Stacking the events, I estimate for service employment and payroll

$$Y^o_{ejt} = \alpha^o_{ej} + \delta^o_{et} + \sum_{k \in \{-6,\dots,7\} \setminus \{-2\}} \theta^o_k D_{ej} \mathbb{1}\{t - \tau_e = k\} + u^o_{ejt}, \tag{2}$$

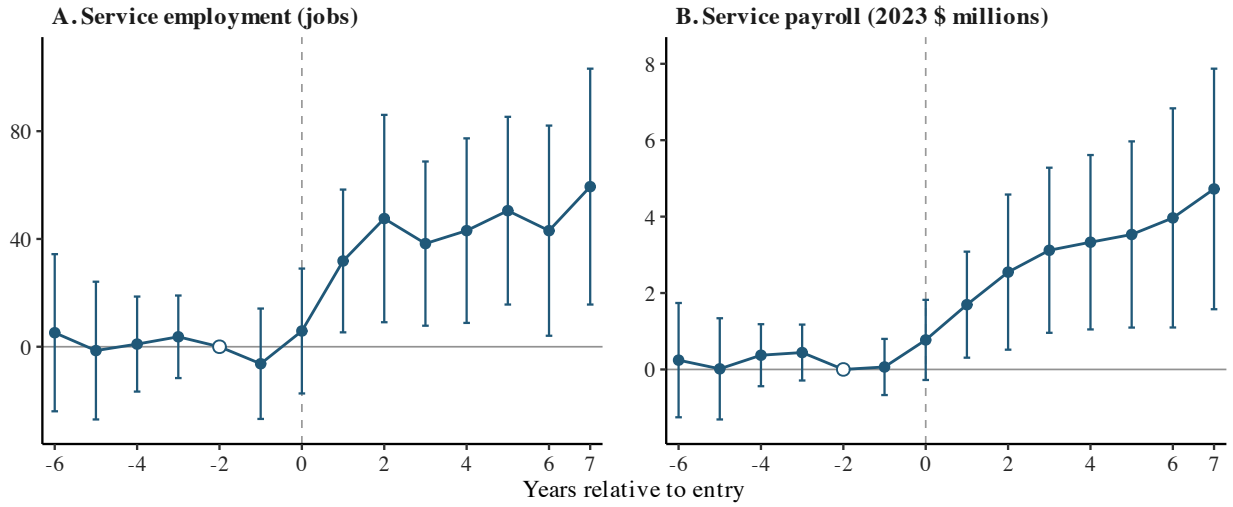
where D_{ej} indicates a top-quartile tract and τ_e is the entry year. Event-by-tract effects absorb permanent differences between tracts, and event-by-year effects absorb shocks common to the market, including the aggregate effect of the entry itself, so θ^o_k measures how the top-minus-bottom gap changes relative to two years before entry. The two groups receive equal weight within each event, the entrant's own tract is excluded, and standard errors are clustered by CZ.[9] Outcomes cover NAICS 44, 45, 722, 81, 61, and 62.

A causal reading requires that the two groups of tracts would have moved in parallel absent entry, and although fixing residence before entry rules out moves after hiring, it does not rule out hires coming from neighborhoods with different prospects. The comparison shows where service employment grows within a market, not the net gain to the CZ, which the model measures.

Figure 4 shows the continuous specification using actual-worker exposure across all events and exposure tracts. Each hire is located at their residence four quarters after hire; 30 percent of exposure is assigned to that home tract and 70 percent is shared equally among adjacent tracts.

[9]Entry or expansion is dated to the first year with positive employment. Employment is measured on March 12, so hiring that begins after March 12 of the preceding year can occur while recorded employment for that year is still zero.

Figure 3: Service employment and payroll around entrant workers' pre-entry residences



Notes: Stacked event-study coefficients from equation (2), comparing the top and bottom quartiles of pre-entry-resident worker exposure in commuting zones with no other observed events. The difference in mean exposure is 23.4 workers. Each worker's pre-entry home tract receives 30 percent of exposure, and adjacent tracts share 70 percent equally. Outcomes cover NAICS 44, 45, 722, 81, 61, and 62. Bars are 95 percent confidence intervals; year −2 is normalized to zero. Payroll is displayed in millions of 2023 dollars. The release reports 40 commuting zones.

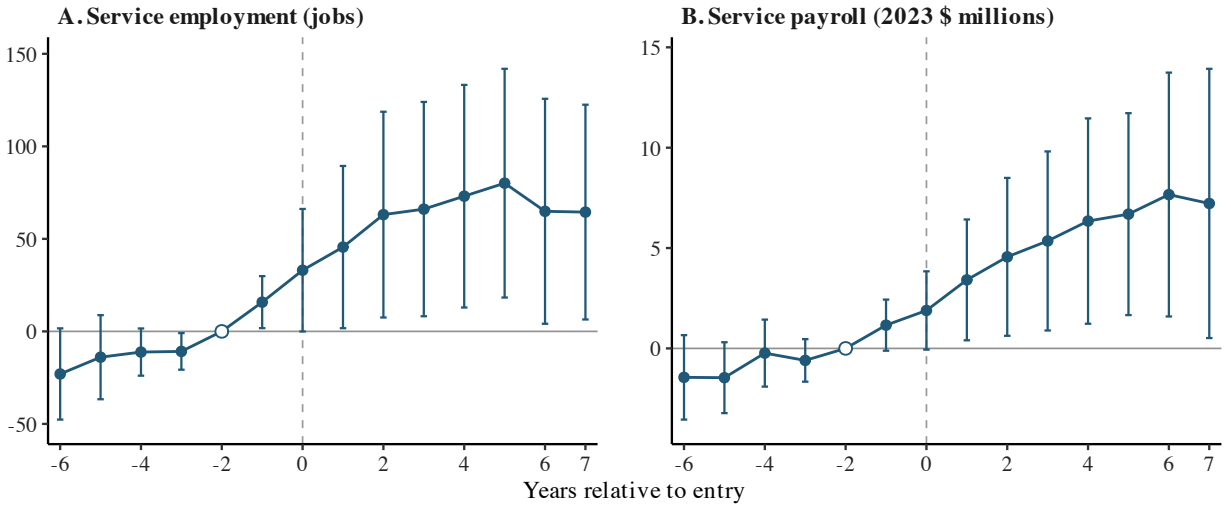
The coefficients average 69.8 service jobs and $7.2 million in annual service payroll per 100 exposed workers over years five through seven, and 65.3 jobs over years one through seven. These are slopes across the full exposure distribution, so their magnitude is not directly comparable to the stacked quartile contrast. Residence timing and the event sample also differ. Employment coefficients already rise before entry: the year-−3 interval is below zero and the year-−1 interval is above zero. Together with the use of post-hire residences, this limits a causal interpretation.

Appendix A.6 repeats the comparison with new-income exposure and with continuous exposure across all events and tracts, and Appendix A.6 uses residences predicted from public commuting data.

From the facts to the model. The three facts trace how entrant payroll reaches workers inside and outside the market, where residents shop, and whether service employment follows residential exposure. They do not aggregate to a net local employment effect. Hiring and residential moves can be endogenous to entry, and comparisons across neighborhoods combine growth with reallocation from elsewhere in the market.

Incumbent adjustment is another part of this accounting. Establishments that supply workers to an entrant may replace those workers, contract, or exit. The raw trajectories in Figure 1 distinguish employment across the full source sample from employment and average pay among establishments that remain active. Among the highest-exposure tradable sources, the year-seven employment index for active establishments is 99.72 and their average-pay index is 114.10, relative to a baseline of 100 in year −1. Employment across the full source sample falls to 66.97.

Figure 4: Continuous service responses to actual-worker exposure with 30/70 home-and-adjacent weights



Notes: Continuous event-study coefficients per 100 exposed workers from equation (A.13). Actual workers are located at their residences four quarters after hire. Each home tract receives 30 percent of exposure, and adjacent tracts share 70 percent equally. All events and exposure tracts enter, including CZs with multiple observed events. Outcomes cover NAICS 44, 45, 722, 81, 61, and 62. The regression includes tract and CZ-by-year fixed effects, is unweighted, and excludes the entrant's workplace tract. Bars are CZ-clustered 95 percent confidence intervals; year -2 is normalized to zero. Payroll is in millions of 2023 dollars. Each panel has its own vertical scale. The release reports 100 commuting zones.

The industry-level event studies in Appendix A.4 compare employment, payroll, and average pay by labor-flow and input–output exposure to the entrant. In the joint specification, tradable industries' labor-exposure employment coefficients are negative in years four through six, with confidence intervals below zero, while the year-seven interval includes zero.

4 A Model of Labor and Shopping

I develop a spatial general equilibrium model to measure the service multiplier across neighborhoods and compare it with resident welfare. The economy consists of a single commuting zone (CZ) partitioned into census tracts $i, j \in \mathcal{J}$ differentiated by exogenous amenities and productivity. Individuals choose where to live, work, and shop. Their shopping trips originate from both their home and workplace. The commuting zone is an open economy with workers living just outside its boundary and migration from elsewhere in the country. There are four sectors. Two tradable sectors, high and low skill intensive, produce with Cobb–Douglas technologies and sell at a national price, and two service sectors, also high and low skill intensive, are monopolistically competitive with free entry. A local government levies a property tax, and its revenue leaves the modeled economy. Residents own residential land, while commercial rents accrue to absentee landlords. Residential and commercial floor space are supplied with separate elasticities.

I model an entering tradable firm as a positive productivity shock in a sector and location. This

shock increases demand for labor and raises wages, encouraging individuals to join the labor force, migrate into, or work in the area. A portion of the increase in worker income is spent on local services. Service firms may see revenue gains that translate into increased demand for workers, but in an environment where labor and floor space may be more expensive. The spatial incidence of these net effects is determined by the interaction of workers' choices of where to live, work, and shop, so the gains from job creation accrue not at the firm's entry location but along the commuting and shopping networks that connect it to the rest of the commuting zone.

4.1 Individuals

Individuals differ by skill $k \in \{H, L\}$ and choose where to live, in what sector to work, whether to work, and where to work. The choices are nested. Individuals compare living in the CZ, in the accessible outside area, or elsewhere in the country. A migration elasticity governs their sensitivity to this utility difference across regions. Once they choose their region, they decide the sector in which to work given the opportunities in the CZ for high- and low-skill tradables and high- and low-skill nontradables. Within sector, the individual chooses whether to work or to be unemployed and collect unemployment income. Finally, the individual chooses their residence–workplace pair. Individuals' tastes are more similar among options within a nest than across nests, so individuals substitute more readily between two workplaces in the same sector than between sectors, and more readily between sectors than between regions.

This nested structure allows the model to distinguish new workers from new consumers. After a positive labor demand shock, substantial shares of new workers migrate, cross the CZ boundary for work, or come from nonemployment (Fact 1). Unemployed residents already consume in the CZ, and those who commute into the CZ spend much of their income outside of it. A semi-open city model with a single migration elasticity could approximate the employment openness of the local market, but it would treat every new worker as a new consumer, and in doing so, overstate the increase in service demand. In this model, these nested employment and location choices determine the labor available to local employers and the locations from which workers originate their shopping trips.

4.1.1 Preferences

Let ℓ index a joint residence and employment choice. For a working individual ω of skill k, this specifies residence i, workplace j, and sector g.[10] The individual consumes housing h, services n, and a tradable good x whose price is normalized to one. For an individual living and working within the CZ, preferences are Cobb–Douglas,

$$U_\ell(\omega) = z_\ell(\omega) \frac{B^R_{ik} B^W_{jgk}}{d_{ij}} h^{\alpha_H} \left(n^{s_N} x^{1-s_N}\right)^{1-\alpha_H}, \tag{3}$$

[10] For nonworkers, ℓ specifies residence i and a sector with amenities $B^R_{ik} B^U_{ik}$ and no commute. Appendix B.1 gives the corresponding amenities for choices involving the outside area.

where B^R_{ik} and B^W_{jgk} are residential and workplace amenities, with workplace attractiveness allowed to differ by sector. Additionally, $d_{ij} = \exp(\kappa t_{ij})$ is the utility cost of a commute of t_{ij} minutes, and $z_\ell(\omega)$ is an idiosyncratic taste for choice ℓ. The housing expenditure share is α_H, and s_N is the share of remaining income spent on services.[11]

Let y_ℓ denote disposable income, q^R_i the residential rent, and $\mathcal{P}_\ell$ the service price index. Maximizing utility over consumption gives indirect utility $z_\ell(\omega)\Phi_\ell$,[12] where

$$\Phi_\ell = \frac{B^R_{ik} B^W_{jgk}}{d_{ij}} \frac{y_\ell}{(q^R_i)^{\alpha_H} \mathcal{P}_\ell^{(1-\alpha_H)s_N}}. \tag{4}$$

Each individual chooses the option ℓ with the highest $z_\ell(\omega)\Phi_\ell$. The value Φ_ℓ is common to everyone of skill k who makes choice ℓ, while $z_\ell(\omega)$ is the individual's idiosyncratic taste for it. Tastes are drawn independently across individuals from a nested Fréchet distribution and are held fixed across equilibria. For an individual of skill k,

$$\Pr\{z_\ell(\omega) \le t_\ell \text{ for every } \ell\} = \exp\left\{-G_k\left((t_\ell^{-1})_\ell\right)\right\}, \qquad t_\ell > 0, \tag{5}$$

where G_k incorporates the nesting structure and the preference parameters (Appendix B.2). The shape parameters μ, ν, η, ψ, and χ govern migration, sector choice, labor-force participation, the choice between jobs inside and outside the CZ, and the residence–workplace choice, respectively.

4.1.2 Employment and location choice nests

Whether an individual moves to the CZ depends on the utility it offers, and that utility depends on the jobs in each sector they can reach from places they would like to live. A higher wage at one workplace raises not only the appeal of commuting there, but also the value of working in its sector, and the value of living in the CZ.

The value of working in sector g sums over every residence–workplace pair available in that sector, including workplaces outside the CZ, and the value of not working sums over residences,

$$\mathcal{W}_{gk} = \sum_{\ell \in \mathcal{L}^W_{gk}} \Phi^\chi_\ell, \qquad\qquad \mathcal{U}_{gk} = \sum_{\ell \in \mathcal{L}^U_{gk}} \Phi^\chi_\ell. \tag{6}$$

Working and not working then combine into the value of the sector,

$$\mathcal{S}_{gk} = \overline{B}^E_{gk} \mathcal{W}^{\eta/\chi}_{gk} + \overline{B}^U_{gk} \mathcal{U}^{\eta/\chi}_{gk}, \tag{7}$$

where $\overline{B}^E_{gk}$ and $\overline{B}^U_{gk}$ are preference parameters for employment and nonemployment, and the sector values in turn determine the value of living in the CZ in equation (8).

[11] For outside residents, the same preferences and demand expressions use the fixed housing share $\overline{\alpha}^o_H$ in place of α_H.

[12] For nonemployment, the amenity component is $B^R_{ik} B^U_{ik}$, income includes nonemployment income y^U_k, and the service price index is P_i, the price of services purchased from the home origin.

Individuals can also live outside the CZ, either in an accessible outside area or elsewhere in the country. The accessible outside area pools the neighboring CZs into a single location whose residents can work in the CZ, work outside it, or remain nonemployed. Moving between the outside area and the CZ is governed by the same parameter μ as migration from elsewhere.

For a resident of the outside area, the choice between a job inside the CZ and one outside it is a separate nest with shape parameter ψ, which governs how readily residents of neighboring CZs take CZ jobs when wages rise. CZ residents who work outside it instead treat the outside area as one more workplace in their residence–workplace choice, so ψ affects only outside residents. Commutes across the CZ boundary have no travel time in the model. The costs of travel into the CZ are part of the location-specific exogenous amenity term calibrated to the observed inbound and outbound flows (Appendix C.6.1). The term $\mathcal{O}_k$ in (8) collects the value of living in the outside area or elsewhere in the country. It can change with entry because outside residents benefit from access to CZ jobs, while utility elsewhere in the country is fixed.

Regional location choice. The outermost nest is the choice whether to live in the CZ. Let $\mathcal{N}_k$ be the fixed national population of skill k, $\overline{B}_k^R$ the preference parameter for CZ residence, and $\overline{B}_{gk}^S$ the preference parameter for sector g. The number of individuals residing in the CZ is

$$N_k = \mathcal{N}_k \frac{\overline{B}_k^R \left[\sum_{g\in\mathcal{G}} \overline{B}_{gk}^S \mathcal{S}_{gk}^{\nu/\eta}\right]^{\mu/\nu}}{\mathcal{O}_k + \overline{B}_k^R \left[\sum_{g\in\mathcal{G}} \overline{B}_{gk}^S \mathcal{S}_{gk}^{\nu/\eta}\right]^{\mu/\nu}}. \tag{8}$$

I assume utility elsewhere in the country is fixed. Because the CZ is small relative to the nation, the individuals it draws from do not change wages or prices in the nation. The migration elasticity μ is approximately the elasticity of the CZ's population with respect to the value of living there.[13] When μ is low, few individuals move, and a labor demand shock is absorbed by higher wages, prices and rents.

Sector choice. Within the regional choice, individuals compare employment opportunities in two tradable sector categories (high and low skill intensive) and two service sector categories (high and low skill intensive), so $\mathcal{G} = \mathcal{G}_T \cup \mathcal{G}_N$ with $\mathcal{G}_T = \{T_H, T_L\}$ and $\mathcal{G}_N = \{N_H, N_L\}$.[14] The number choosing sector g is

$$N_{gk} = N_k \frac{\overline{B}_{gk}^S \mathcal{S}_{gk}^{\nu/\eta}}{\sum_{g'\in\mathcal{G}} \overline{B}_{g'k}^S \mathcal{S}_{g'k}^{\nu/\eta}}. \tag{9}$$

[13] The exact elasticity is $\mu(1 - b_k^{\text{reg}})$, where b_k^{reg} is the share of the national population of skill k living in the CZ or the accessible outside area.

[14] Workers of both skill types can work in either the high or low intensity sectors. These sectors simply employ different shares of each worker type.

The labor demand shock increases wages in one of the tradable sectors, which draws individuals toward tradable employment, increasing labor competition across sectors for incumbent service firms. The choice of N_{gk} includes the ability to not work. Therefore, the sector choice is best described as an employment opportunity choice.

Labor-force participation. Within sector g, individuals compare working with nonemployment.[15]

The number employed in sector g among CZ residents is

$$E_{gk} = N_{gk} \frac{\overline{B}_{gk}^{E} \mathcal{W}_{gk}^{\eta/\chi}}{\mathcal{S}_{gk}}. \tag{10}$$

Higher wages can draw nonworkers into employment, expanding the workforce and raising the income available for local services.

Residence–workplace location choice. Conditional on living in the CZ and working in sector g, an individual of skill k chooses residence–workplace pair $\ell = (i, j, g)$ from the feasible set $\mathcal{L}_{gk}^{W}$ with probability

$$\pi_{\ell k} = \frac{\Phi_{\ell}^{\chi}}{\sum_{\ell' \in \mathcal{L}_{gk}^{W}} \Phi_{\ell'}^{\chi}}, \qquad F_{ijgk} = E_{gk} \pi_{\ell k}. \tag{11}$$

The numerator captures the attractiveness of the pair through residential and workplace amenities, disposable income, residential rent, service prices, and commuting costs. The denominator, $\mathcal{W}_{gk}$, sums these utility terms across all feasible residence–workplace choices in the sector, so the probabilities sum to one. Multiplying the probability by the number of employed CZ residents in sector g, E_{gk}, gives the corresponding commuting flow. Nonworkers choose among residences with the same parameter χ.

4.1.3 Income and spending

A working individual supplies one unit of labor and earns the wage w_{jk}^{g} paid to skill k in sector g at workplace tract j (Section 4.3), while a nonworker receives the externally financed payment y_{k}^{U}. Current CZ residents also receive a dividend D_k from ownership of residential land.[16] Disposable

[15] Nesting participation within sector treats each individual as attached to a sector even when not working, so that a nonworker compares not working with the within-sector wage. This follows the labor literature on sector-specific skills and attachment. I am assuming human capital is partly specific to an industry or occupation. This follows the literature in which displaced workers who switch industries suffer larger earnings losses (Neal, 1995), and Galle et al. (2023, Table 6), who estimate a larger shape parameter for the choice between nonparticipation and work than for the choice across sectors, consistent with $\eta \geq \nu$.

[16] In the quantification, housing expenditure includes the rental-equivalent value of owner-occupied housing. This essentially is a rent which residents pay to themselves. The assumption matters because rental income stays in the CZ, so part of it is spent on local services. Because the dividend is shared by all current residents, it spreads the

income is

$$y_\ell = w^g_{jk} + D_k \quad \text{for a worker,} \qquad\qquad y_\ell = y^U_k + D_k \quad \text{for a nonworker.} \tag{12}$$

Section 4.4 determines the dividend.[17]

At residential rent q^R_i, Cobb–Douglas preferences imply housing demand h_ℓ and service spending γy_ℓ, with

$$h_\ell = \frac{\alpha_H y_\ell}{q^R_i}, \qquad\qquad \gamma = s_N(1-\alpha_H). \tag{13}$$

Therefore, an additional dollar of disposable income raises service spending by γ.

Workers allocate a share ρ of service spending to purchases originating at home and the remainder to purchases originating at work, with $n = n_i^\rho n_j^{1-\rho}$. Let P_i denote the service price index at origin i. The service price index for workers is $\mathcal{P}_\ell = P_i^\rho P_j^{1-\rho}$. Meanwhile, nonworkers purchase all services from home, so their price index is $\mathcal{P}_\ell = P_i$.

Let y_{ijgk} denote the disposable income of a worker of skill k who lives in i and works in sector g at j, and let U_{ik} be the number of nonemployed individuals of skill k at residence i, summed across sectors. Total service spending originating in tract i is

$$\mathcal{E}_i = \gamma\Big[\underbrace{\rho \sum_{j,g,k} F_{ijgk} y_{ijgk}}_{\text{Workers' purchases from home}} + \underbrace{(1-\rho) \sum_{j,g,k} F_{jigk} y_{jigk}}_{\text{Workers' purchases from work}} + \underbrace{\sum_k U_{ik}\big(y^U_k + D_k\big)}_{\text{Nonworkers' purchases}} \Big], \tag{14}$$

where γ uses the outside housing share for inbound commuters.

4.1.4 Shopping destinations

From each shopping origin i, individuals purchase a CES bundle of services available across destinations j, with elasticity of substitution $\sigma_N > 1$. Tract j offers m_j differentiated service varieties, with prices summarized by p^N_j. Each variety is supplied by a firm operating under monopolistic competition with free entry.[18]

Travel from origin i to destination j takes t_{ij} minutes and raises the effective price of purchases by the iceberg factor $\exp(\kappa t_{ij})$, using the same travel-cost parameter κ as commuting. Under CES demand, these preferences and travel costs enter through $T_{ij} = \zeta_{ij} \exp\{-(\sigma_N - 1)\kappa t_{ij}\}$, which determines each destination's contribution to the service price index. The cost of one unit of the

gains from higher rents across the population rather than concentrating them on incumbent owners.

[17] Appendix B.3 gives the corresponding income accounting for outside residents.

[18] The model distinguishes two service sectors, N_H and N_L, with separate prices and entry responses. This section presents shopping and revenue aggregated by tract. Appendix B.6 gives the sector-specific equations and derives these aggregates.

service bundle purchased from origin i is[19]

$$P_i = \left[\sum_{j \in \mathcal{J}} T_{ij} m_j (p_j^N)^{1-\sigma_N} \right]^{1/(1-\sigma_N)}. \tag{15}$$

Firm entry expands the range of available services and, holding prices and travel costs fixed, lowers P_i. The expenditure share received by tract j is

$$\pi_{ij}^I = \frac{T_{ij} m_j (p_j^N)^{1-\sigma_N}}{P_i^{1-\sigma_N}}. \tag{16}$$

Individuals direct more spending toward destinations offering more varieties, lower prices, or better travel access. These purchases generate service revenue of $\sum_{i \in \mathcal{J}} \mathcal{E}_i \pi_{ij}^I$ in tract j.

4.2 Service Firms

A service firm in sector $g \in \mathcal{G}_N$ and tract j produces a differentiated variety under monopolistic competition and free entry.[20] With productivity A_{jg}, a firm combines labor $\mathcal{L}$, commercial floor space H, and intermediate inputs X purchased from outside the CZ to produce output Y, using the same Cobb–Douglas form as the tradable sectors (Section 4.3),

$$Y = A_{jg} \, \mathcal{L}^{\alpha_L^g} H^{\alpha_C^g} X^{\alpha_X^g}, \tag{17}$$

where the sector's expenditure shares on labor, commercial space, and intermediates satisfy $\alpha_L^g + \alpha_C^g + \alpha_X^g = 1$.

Under CES demand, firms charge a constant markup over their marginal cost c_{jg},

$$p_{jg} = \frac{\sigma_N}{\sigma_N - 1} c_{jg}. \tag{18}$$

Appendix B.7 derives unit costs and factor demands.

Each firm also incurs an annual fixed cost f_{jg} to maintain its establishment. The fixed cost uses the same inputs as production, so it is proportional to the unit cost c_{jg}, and the input requirement per firm changes with the number of firms m_{jg} in the tract,

$$f_{jg} = \varphi_{jg} c_{jg} m_{jg}^{1/\epsilon_m - 1}, \tag{19}$$

where φ_{jg} scales the requirement. At the CES markup, each firm keeps a share $1/\sigma_N$ of its revenue

[19] The term T_i^O summarizes shopping opportunities outside the CZ. Appendix B.6 describes these opportunities and derives the expenditure share spent outside the CZ.

[20] Appendix B.7 gives the full conditions for each service sector and their aggregation to tracts.

after variable costs, so free entry drives profits to zero when

$$\frac{R_{jg}}{\sigma_N} = m_{jg} f_{jg}, \qquad m_{jg} = \left(\frac{R_{jg}}{\sigma_N \varphi_{jg} c_{jg}} \right)^{\epsilon_m}. \tag{20}$$

The entry elasticity ϵ_m is the elasticity of the number of establishments with respect to revenue relative to unit cost. When $\epsilon_m = 1$, each firm's requirement is constant and establishments rise in proportion to revenue relative to cost. A value below one makes each firm's requirement rise as firms enter, so establishments respond less than proportionally.

Since revenue covers variable and fixed costs, total labor expenditure is $\alpha_L^g R_{jg}$. Allocating this expenditure across skills and dividing by the corresponding wage gives employment:

$$L_{jk}^g = \frac{\alpha_L^g \lambda_{jgk} R_{jg}}{w_{jk}^g}. \tag{21}$$

4.3 Tradable Firms and Entry

Tradable producers in sector $g \in \mathcal{G}_T$ (high and low skill intensive) sell a homogeneous good on the national market at price p^T, which the CZ takes as given.[21]

Tradable production in each tract and sector is competitive, with productivity A_{jg}. Production is Cobb–Douglas in labor, commercial floor space, and intermediate inputs. With the price of intermediate inputs normalized to one, cost minimization gives the unit cost

$$c_{jg} = \frac{(q_j^C)^{\alpha_C^g}}{A_{jg}} \prod_k \left(w_{jk}^g \right)^{\alpha_L^g \lambda_{jgk}}. \tag{22}$$

Production has constant returns and the output price is set nationally, so profits are zero in every location and sector,

$$c_{jg} = p^T, \tag{23}$$

and output Y_{jg} adjusts until the labor and floor-space markets clear. Labor demand is a constant share of revenue,

$$L_{jk}^g = \frac{\alpha_L^g \lambda_{jgk}\, p^T Y_{jg}}{w_{jk}^g}, \tag{24}$$

and demands for commercial space, H_{jg}^C, and intermediate inputs take the same form (Appendix B.8). Equation (23) means that a tradable producer cannot pass higher costs on to its consumers. Given its productivity, each producer can pay only the wages at which it breaks even, and its employment adjusts to the labor supply those wages attract. When the entrant's

[21] The high-skill industry comprises NAICS 51, 52, 54, and 55, and the low-skill industry comprises manufacturing (NAICS 31–33). The labels refer to industries' skill intensity, and individuals of either skill can work in either industry.

productivity gain raises the wages workers can earn elsewhere, incumbent tradable producers cannot match them, so they lose workers and shrink.

Entry. I model the entry of a tradable firm as an increase in the productivity $A_{j_E g_E}$ of its sector g_E in its tract j_E, sized so that employment in this tract and sector rises by J^E, the entrant's promised jobs. Higher productivity raises the wages the location-sector can pay, and its employment rises along the upward-sloping labor supply described in Section 4.1. The model does not distinguish the entrant from incumbent producers in the same tract and sector, so J^E measures the location-sector's net growth. Because the shock raises productivity for the whole location-sector, the higher wage it supports is paid to every worker here, not only to new hires. The model has no separate premium for the entrant's workers.

4.4 Land Market Clearing

Each tract has separate markets for residential and commercial floor space. Space cannot be converted between uses, so each tract has two rents. Owners receive r_i^R and r_j^C, and tenants pay these rents plus the property tax rate t,

$$q_i^R = (1+t) r_i^R, \qquad q_j^C = (1+t) r_j^C. \tag{25}$$

Floor space is built by combining land, which is fixed in each tract, with structures, so its supply slopes upward in the rent owners receive. I take this supply to have a constant elasticity,

$$H_i^R = \bar{H}_i^R (r_i^R)^{\epsilon_{R,i}}, \qquad H_j^C = \bar{H}_j^C (r_j^C)^{\epsilon_C}, \tag{26}$$

as in quantitative urban models with a construction sector (Ahlfeldt et al., 2015). The elasticities measure how much floor space expands when rents rise, and $\epsilon = 0$ gives the fixed supply of Monte et al. (2018). The shifters $\bar{H}_i^R$ and $\bar{H}_j^C$ capture each tract's land area and the ease of building on it.

Residents spend a share α_H of disposable income on housing (13). Let $Y_i^R = \sum_k \sum_{\ell:\, i_\ell = i} n_{\ell k} y_\ell$ denote the total disposable income of tract i's residents, workers and nonworkers alike. Commercial space is demanded by tradable producers, including producers in the entrant's tract and sector, and by service firms. Equating supply and demand in each market gives

$$r_i^R = \left[\frac{\alpha_H Y_i^R}{(1+t)\bar{H}_i^R} \right]^{\frac{1}{1+\epsilon_{R,i}}}, \qquad H_j^C = \sum_{g \in \mathcal{G}} H_{jg}^C. \tag{27}$$

Because the entrant's location-sector demands floor space in proportion to its output, entry competes with other producers for commercial space.

Residential land is owned by the CZ's residents. Rental income $\mathcal{R} = \sum_{i \in \mathcal{J}} r_i^R H_i^R$ is paid out through a local portfolio according to fixed housing-value shares ω_k^0, so each current resident of skill

k receives

$$D_k = \frac{\omega_k^0 \mathcal{R}}{N_k}, \tag{28}$$

where N_k counts all current residents of skill k, including renters, nonworkers, and new arrivals. Commercial rents accrue to absentee landlords.

4.5 Government

The government levies the property tax in (25) on annual pretax rent from all residential and commercial space. It collects

$$\mathcal{T} = t\left(\mathcal{R} + \mathcal{R}^C\right), \tag{29}$$

where $\mathcal{R}$ is residential rental income and $\mathcal{R}^C = \sum_{j \in \mathcal{J}} r_j^C H_j^C$ is commercial rental income. The revenue leaves the modeled economy and is not rebated or spent locally, so a CZ resident's resources are earnings or nonemployment income, which comes from outside the CZ, plus the residential dividend. The government also pays the recorded award to the entrant. The award does not change the entrant's hiring, so Section 7 treats it as a public cost. Appendix B.9 gives the full property and government accounts.

4.6 Equilibrium and the Effects of Entry

Given the parameters in Table 5, the national populations $\mathcal{N}_k$ and utility elsewhere in the country, and the location fundamentals (productivities A_{jg}, amenities and preference parameters, floor-space shifters $\bar{H}_i^R$ and $\bar{H}_j^C$, fixed-cost scales φ_{jg}, and travel times t_{ij}), an equilibrium is a set of wages w_{jk}^g, rents r_i^R and r_j^C, service prices p_{jg}, establishment masses m_{jg}, and allocations of individuals N_k, N_{gk}, E_{gk}, F_{ijgk}, and U_{ik} such that

(i) individuals choose residence, sector, participation, and commute as in (8), (9), (10), and (11);

(ii) tradable producers break even, (23), and employ workers according to (24);

(iii) service firms price at a markup and enter until profits are zero, (18)–(20), and employ workers according to (21);

(iv) labor markets clear at each workplace, sector, and skill, so the commuting flows from all feasible residences, including inbound commuters, equal employment,

$$\sum_i F_{ijgk} = L_{jk}^g, \qquad g \in \mathcal{G}, \quad j \in \mathcal{J}; \tag{30}$$

(v) residential and commercial floor-space markets clear, (27);

(vi) service revenue in each tract and sector equals the spending directed to it, $R_{jg} = \sum_i \mathcal{E}_i \pi^I_{ijg}$; and

(vii) residential rental income is paid out through the portfolio, (28).

The tradable price and the price of intermediates are set nationally and utility elsewhere in the country is fixed, so the CZ is a small open economy. Appendix B.10 writes out the full system and the feasible choices.

Each workplace tract has its own labor market for each sector and skill, and these markets are linked through individuals' choices, as in quantitative urban models with workplace-specific wages (Ahlfeldt et al., 2015; Monte et al., 2018). A higher wage at one workplace draws commuters from other workplaces, workers from other sectors, nonworkers, and migrants, so the employers they leave must respond.

Entry raises the productivity $A_{j_E g_E}$ of the entrant's tract and sector by the factor that raises employment there by J^E. I compare the equilibrium with this higher productivity to the equilibrium without entry, holding fixed all other productivities, amenities, preference parameters, housing expenditure shares, ownership shares, national populations, and the taste distribution. Wages, rents, floor space, prices, individual choices, service entry, and dividends differ between the two equilibria. I solve both in changes relative to the observed pre-entry economy (Section 5).

5 Quantification

I quantify the model separately for each event in its commuting zone (CZ), indexed by c, and its event year. The main parameters are the recruitment elasticities (migration, inbound commuting, and participation), which jointly determine how readily the CZ attracts workers not previously tied to its labor market. Also central are the sector-choice elasticity, which determines the wage pressure service firms feel from the labor demand shock in the tradable sector; the commuting elasticity, which relieves pressure from an inelastic residential land supply; and the shopping elasticity, which relieves pressure from an inelastic commercial land supply and from wage increases. I estimate the response of sector choice from public employment and earnings data that vary across commuting zones, and the responses of commuting and shopping from flows that vary across tract pairs. The recruitment elasticities are calibrated to the hiring sources that Fact 1 measures. The remaining cost and supply parameters are set from BEA national accounts, shopping trip data, and the literature. Given these parameters (Table 5), I recover the remaining primitives (amenities, productivities, fixed costs, and the service spending share) by inversion so that the model reproduces each commuting zone's economy before entry. None of these steps uses the service responses of Fact 3, which leaves those estimates free to validate the model in Section 5.4.

Table 5: Parameters used in the equilibrium calculations

Object	Value	Basis
Behavioral parameters		
Migration, μ	1.38	Calibrated jointly to five recruitment moments
Sector choice, ν	1.64	Estimated; 2012–2019 QWI shift-share 2SLS (Appendix C.2.1)
Participation, η	3.02	Same five-moment calibration
Inbound-to-pair ratios, (q_S, q_M, q_L)	$(0.665, 0.811, 1.000)$	Small/medium/large CZs; $\psi_e = q_{\text{size}(e)}\chi_{c(e)}$
Residence–workplace choice, χ	5.01–10.59	By CZ; 2019 LODES PPML
Service substitution, $\sigma_{N,c}$	5.60–16.14	Pooled CZ shopping PPML; $\kappa = 0.01$
Establishment entry, ϵ_m	0.40	Maintained benchmark; independent estimation pending
Accounting and individual calibration		
Service cost shares (L, C, X)	(0.53, 0.13, 0.34)	2012 BEA intermediates; 80/20 labor–space split
High-skill tradable cost shares	(0.49, 0.12, 0.39)	Same; NAICS 51, 52, 54, 55
Low-skill tradable cost shares	(0.27, 0.07, 0.67)	Same; NAICS 31–33
Housing-value shares, ω^0_{ck}	By event	ACS owner values; split equally between partners
Nonemployment income, y^U_k	By event and skill	ACS transfers among nonworkers
Housing share, $\alpha_{H,c}$	0.18–0.25	CZ household-median expenditure share
NT share of nonhousing spending, $s_{N,c}$	0.63–0.86	By CZ, 10th–90th percentile; service revenue inversion (BEA benchmark 0.48)
Maintained spending, travel, technology, and policy parameters		
Home-origin shopping share, ρ	0.70	Maintained 70/30 home/work spending split
Travel cost, κ	0.01 per minute	Common to commuting and shopping
Residential supply, $\epsilon_{R,i}$	Tract-specific	Mapped Baum-Snow–Han values; eight-neighbor extension
Commercial supply, ϵ_C	0.60	Maintained commercial-space benchmark
Property tax, t_c	0.28	Tax on pretax annual rent
Entrant jobs, skills, site, award, floor space	By event	Ten-year award; 50% local; floor-space proxy

Notes: Migration, sector choice, and participation are common across events and skills. Border scales vary by event through three CZ-size ratios; commuting and shopping scales also vary by CZ. The current five-moment recruitment RMSE is 1.73 percentage points, with the originally fitted coefficients retained after the fiscal-rule revision. Cost shares list labor, commercial space, and purchased intermediate inputs (Appendix C.5.2). Appendix C gives the estimands and diagnostics.

5.1 Individuals

Sector choice. Individuals' sector choice determines whether the increase in tradable labor demand is met by workers in the same sector or workers who leave other sectors, governed by the elasticity ν. Using the Quarterly Workforce Indicators for 2012–2019, I aggregate private employment and earnings into commuting-zone–sector–skill groups, with sectors defined as tradable and nontradable industries. I estimate the pooled specification

$$\Delta \log E_{csk} = \beta^{\text{QWI}} \Delta \log \bar{w}_{csk} + \alpha_{ck} + \delta_{d(c)s} + \gamma_{sk} + \varepsilon_{csk}, \tag{31}$$

where c indexes commuting zones, s sectors, and k worker skill, distinguishing workers with and without a bachelor's degree. Employment is denoted by E_{csk}, average earnings by $\bar{w}_{csk}$, and Δ denotes the change between 2012 and 2019. The pooled specification imposes a common coefficient across skills. The fixed effects α_{ck}, $\delta_{d(c)s}$, and γ_{sk} absorb CZ-by-skill, Census-division-by-sector, and sector-by-skill differences in growth, respectively, where $d(c)$ is the division containing CZ c.

Because earnings and employment are jointly determined, I estimate equation (31) by weighted two-stage least squares, instrumenting earnings growth with 2007–2011 within-sector industry employment shares interacted with subsequent industry employment growth outside the focal CZ. The pooled estimate is $\widehat{\beta}^{\mathrm{QWI}} = 1.64$, with a CZ-clustered standard error of 0.51 and a first-stage F of 15.03. Under the identifying assumptions, a 1 percent rise in one sector's earnings relative to another's raises the employment ratio by about 1.64 percent over seven years. I use seven years to match the length of time in my event-study designs. Appendix C.2.1 provides the full specification, the instrument, and the robustness checks.

Migration, participation, and inward commuting. Each recruitment elasticity governs one source of hiring, with migration drawing new residents, participation drawing nonemployed residents, and inbound commuting drawing workers who live outside the CZ. I fix $\nu = 1.64$ and calibrate common migration and participation elasticities and three CZ border elasticities for small, medium, and large CZs based on their initial total employment. The CZ border elasticity is expressed as a share of the residence–workplace choice elasticity: for event e it is $\psi_e = q_{\mathrm{size}(e)}\chi_{c(e)}$, where $c(e)$ is event e's CZ, subject to $0 < \mu \leq \nu \leq \eta \leq \psi_e \leq \chi_{c(e)}$. The calibration procedure targets the sources of new workers entering the entrant's location-sector, rather than the composition of all workers employed in the CZ after entry. I estimate common parameters across worker skill, and the nonemployment moment uses continuing residents as its denominator. For each potential parameter vector, I recalibrate each event's no-entry economy to the observed pre-entry allocation and solve the corresponding equilibrium with entry. I calculate the composition of gross worker inflows into the location-sector of the productivity shock. I aggregate the recruitment flows across 55 events in 38 CZs using the observed hiring weights. I adjust the parameters to minimize the equally weighted sum of squared differences between five model-implied shares and their empirical targets. For each parameter update, I recompute the event equilibria and the recruitment shares in the model. The calibrated values are $\mu = 1.38$, $\eta = 3.02$, and $(q_S, q_M, q_L) = (0.665, 0.811, 1.000)$. The original five-moment RMSE is 1.72. Appendix C.2.2 defines the source accounting, weights, numerical procedure, and remaining share errors.

Commuting. Within the commuting zone, residential and workplace choices determine how far a labor demand shock spreads and where additional income accrues. I estimate commuting gravity separately by CZ and year using annual LODES flows from 2007–2022 and a fixed matrix of driving times between tract centroids using driving routes from OpenStreetMap. Fact 2 in Section 3 presents the 2019 comparison with shopping flows. The commuting coefficient satisfies $\beta^C_{ct} = -\chi_{ct}\kappa$, so gravity identifies only the product of the choice scale and the per-minute cost. Following Ahlfeldt et al. (2015) and Tsivanidis (2026), I normalize the per-minute commuting cost to $\kappa = 0.01$, which gives $\chi_{ct} = -\widehat{\beta}^C_{ct}/0.01$. In the 526 CZs included in the 2019 comparison, χ has a median of 6.29 and ranges from 0.87 to 10.80.

Shopping. Where workers shop determines where the entrant's payroll becomes service revenue. With CES demand across service destinations and the same per-minute travel-cost parameter κ as commuting, the coefficient on travel time in a gravity equation for shopping visits is $\beta_c^S = -(\sigma_{N,c}-1)\kappa$, so the rate at which visits decay with travel time reveals how substitutable destinations are. I estimate β_c^S for each commuting zone from 2019 Veraset data restricted to the modeled service sectors (Appendix C.3.2 gives the estimating equation). With $\kappa = 0.01$ per minute, the estimates imply $\sigma_{N,c}$ between 5.60 and 16.14, with a median of 12.13. The same elasticity sets service markups and the share of revenue available to cover fixed costs under free entry.

Individual budgets. I calibrate the housing expenditure share $\alpha_{H,c}$ using household and person records from the American Community Survey (ACS) Public Use Microdata Sample (PUMS).[22] Housing expenditure is annual rent for renters and predicted rent for homeowners, estimated from renter households using dwelling characteristics and location effects. I set $\alpha_{H,c}$ to the weighted median of housing expenditure divided by disposable income within each CZ, where weights combine ACS household survey weights with geographic allocation to the CZ. Across CZs, the median housing expenditure share ranges from 13 to 35 percent, with a cross-CZ median of 20.2 percent. Figure A.10 compares the housing budget share of a typical household in each CZ with the share of the CZ's total household income devoted to housing. Appendix A.7 provides the measurement details.

After housing, individuals in CZ c spend the fraction $s_{N,c}$ of their remaining resources on services, common to both skills. I set $s_{N,c}$ in the inversion of Section 5.3, so that household spending covers the service revenue implied by each CZ's payroll. Between the 10th and 90th percentile CZs, the calibrated shares range from 63 to 86 percent. Thus, at fixed prices, an additional dollar of disposable income generates $s_{N,c}(1-\alpha_{H,c})$ dollars of service expenditure. I maintain that workers allocate the fraction $\rho = 0.70$ of this spending to shopping trips originating from home and the remainder to trips originating from work. This 70/30 split is a model assumption.

I calibrate nonemployment income y_k^U using ACS PUMS person records, taking average public assistance, Supplemental Security Income, and Social Security income among nonworkers of skill k, including those receiving no payments.

I construct the housing ownership shares ω_{ck}^0 from ACS PUMS housing records linked to the individual's education. The model allocates residential rental income across skills in proportion to these ownership-value shares and equally among individuals within each skill group. Appendix C.4.3 gives the rental-income distribution rule.

5.2 Firms and Space

Production cost shares. I measure purchased intermediate inputs, net of purchased real-estate services, in the 2012 BEA supply–use accounts and split the remaining share of gross output 80/20

[22] I use one-year ACS samples collected before the firm opens. Where these files are unavailable, I use the 2008–2012 five-year sample for firms opening in 2013 or later.

between labor and commercial space, following the labor–floor-space calibration in Ahlfeldt et al. (2015), Hoelzlein (2023), and Miyauchi et al. (2025) (Appendix C.5.2). Intermediate inputs account for 0.34 of gross output in the nontradable composite, 0.39 in high-skill tradables (NAICS 51, 52, 54, and 55), and 0.67 in manufacturing, which gives labor shares of 0.53, 0.49, and 0.27. The model treats these intermediates as supplied from outside the CZ at a fixed price. Service firms and tradable producers have the same Cobb–Douglas form, and each sector uses its group's shares. With constant returns and Cobb–Douglas production, the average wage a tradable producer can pay at given rents has an elasticity of $1/\alpha_L^g$ with respect to its productivity, so a 10 percent productivity increase raises it by about 21 percent in high-skill tradables and 43 percent in manufacturing.

Establishment entry. Free entry sets the number of service establishments in a tract with elasticity ϵ_m with respect to net revenue, and the fixed-cost shifter φ_j is recovered from the observed establishment count (equation (C.43)). I set $\epsilon_m = 0.40$, in line with a county regression of five-year growth in service establishments on growth in service revenue relative to wages. The regression pools the five service sectors (NAICS 44–45, 61, 62, 72, and 81) over 9,798 county five-year windows between 2002 and 2022, with end-year effects. Establishments and payroll come from County Business Patterns, revenue is payroll divided by the national service labor share of value added in the BEA accounts, and wages come from the Quarterly Workforce Indicators. The OLS coefficient is 0.39 (standard error 0.01), so a 10 percent rise in net revenue adds about 4 percent to the number of establishments (Appendix C.5.3). Because entry also raises revenue by attracting spending, and the revenue measure moves closely with employment, I treat the value as a descriptive calibration rather than a causal estimate. The value is also close to the response implied by Almagro and Domínguez-Iino (2025) for Amsterdam, where congestion in commercial real estate makes the number of establishments rise with amenity spending with an elasticity of about 0.40. They calibrate the congestion parameter from the range of housing supply elasticities in Saiz (2010).

Housing and commercial space. Residential supply uses tract-specific elasticities $\epsilon_{R,i}$ mapped from Baum-Snow and Han (2024). I retain each valid nonnegative prediction and replace a missing or negative value with the unweighted mean of the eight nearest tracts with valid predictions. Distance is measured between 2020 tract centroids. This extends the existing estimates without using employment or payroll validation outcomes. In the almost six thousand distinct tracts covered by the event baselines, the unweighted mean is 0.56 and the median is 0.60. I set the commercial space elasticity as $\epsilon_C = 0.60$, at the upper end of the metropolitan office estimates of Marcato and Tong (2025, p. 412), which range from 0 to 0.61, and above the retail estimate of 0.49 in Benjamin et al. (1998, p. 9). This value equals the median residential elasticity across tracts. Greaney et al. (2024, pp. 29, 52) take a similar approach. Noting the absence of local estimates for commercial floor space, they set commercial supply elasticities in their model of the San Francisco Bay Area from residential ones.

Property taxes. Users of housing and commercial space pay the property tax $t_c = 0.28$ on top of the rent. Because the model has annual rents but no property values, t_c is the tax bill per dollar of annual rent rather than per dollar of value. A tax of 1 percent of value on a property worth 20 years of rent is 20 percent of a year's rent. Couture et al. (2024, pp. 878–879) measure taxes the same way and find that they are 20 percent of the annual user cost of housing downtown and 30 percent in the suburbs. The 2010–2014 American Community Survey implies about 32 percent of annual rent.[23] I use an average of these estimates and apply the same rate to commercial space.

5.3 Constructing the Pre-Entry Economy

I recover the remaining fundamentals so that each event's baseline equilibrium reproduces its pre-entry economy. Given observed wages, rents, and commuting costs, residential amenities B^R_{ik} and workplace–sector amenities B^W_{jgk} match residence and workplace employment by skill. Tracts that have more residents than their rents and access to jobs predict must offer high amenities. The same logic applies for workplaces that employ more workers than their wages predict. The taste shifters $\overline{B}^R_k$, $\overline{B}^S_{gk}$, $\overline{B}^E_{gk}$, and $\overline{B}^U_{gk}$ match the CZ's population, sector shares, and participation rates by skill, and tradable productivity sets baseline unit costs equal to the national price.

Service productivities are recovered so that the model reproduces where service purchases occur, adapting the procedure of Oh and Seo (2023). I infer each tract's service revenue from its worker payroll, which equals the labor share of costs under free entry. I then set the service share of individuals' nonhousing spending, $s_{N,c}$, so that the spending of the CZ's residents, net of the share of shopping that leaves the CZ (observed in the Veraset data), covers total CZ service revenue. Given this total budget, I invert service productivities so that consumers' demand, which follows the gravity structure of shopping, matches the service revenue observed in each tract. The fixed-cost parameter then makes revenue per establishment consistent with free entry at the observed establishment count.

The calibrated shares, from 0.63 to 0.86 at the 10th and 90th percentiles of CZs, are well above the 0.48 share of nonhousing consumer spending that goes to local services as measured in the BEA accounts. The residents of the CZ are all workers observed in the LODES data plus the number of working-age nonemployed residents from the ACS data. This is not an accurate reflection of all of the people living there nor does it capture the full demand for local services. Local service establishments also sell to retirees, to visitors from outside the CZ, to businesses, and to the public and private insurers that pay for much of health care and education. The spending share of the model's individuals must support the demand of these other groups, so I scale $s_{N,c}$ to observed service activity rather than households' budgets. This choice materially matters for the multiplier, because each new dollar of resident income raises local service spending by $s_{N,c}(1 - \alpha_H)$ in the model. With this calibration, I am assuming the spending that $s_{N,c}$ absorbs moves in the same proportion as the individuals in the model. Employer-provided insurance pays for workers' health

[23] Because t_c is levied on top of the rent, taxes are $t_c/(1 + t_c) \approx 22$ percent of what users pay, which lies within this range on either reading.

care, public insurance and school budgets grow with the resident population, and local businesses buy more services as local employment grows, so I view this as a reasonable assumption.

I recover the amenity, productivity, fixed-cost, and taste fundamentals to rationalize the observed baseline economy as an equilibrium. These fundamentals are held fixed across counterfactuals. I solve for the entry equilibrium in changes relative to the baseline using exact hat algebra. Writing $\widehat{x} = x^1/x^0$, the fundamentals cancel, so the equilibrium conditions depend only on baseline shares and quantities, the parameters described above, and the entry shock. I solve jointly for the changes in wages, rents, service establishment counts, and dividends that clear all markets.

5.4 Validation Against the Event Study

The results and counterfactuals rely on where within the CZ the service gains occur, so the model must capture how service employment responds around the places where entrant workers live. Fact 3 measures that response, and no part of the quantification uses it. The elasticities come from public data that vary across commuting zones, counties, and tract pairs, while the recruitment scales from hiring sources. The event-study estimates are therefore untargeted moments. Comparing them with their model-implied counterparts assesses the magnitude of the spatial response on which the results rely. The benchmark here is the released continuous specification that assigns entrant hires to their pre-entry residence tract, with no exposure assigned to adjacent tracts. Outcomes cover expanded services (NAICS 44, 45, 722, 81, 61, and 62). The released continuous estimates cover 100 CZs; the model sample below covers 38, so this is not a comparison on identical event samples.

To construct the model equivalent, I solve the equilibrium with and without each entrant, record the change in service employment, payroll, and the number of service establishments in every tract, and calculate the same-tract exposure slope. Let x_{ei} be the saved exposure of tract i to event e, in hundreds of resident entrant workers, and Δy_{ei} its equilibrium change. The pooled slope is

$$b_y^{\text{model}} = \frac{\sum_e \sum_{i \in \mathcal{I}_e} (x_{ei} - \bar{x}_e)(\Delta y_{ei} - \overline{\Delta y}_e)}{\sum_e \sum_{i \in \mathcal{I}_e} (x_{ei} - \bar{x}_e)^2}, \tag{32}$$

where $\mathcal{I}_e$ leaves out the entrant's workplace tracts. The pooled sample covers 55 events and has 30,858 event–tract observations from the solved events in 38 commuting zones.[24]

Table 6 reports the comparison. The full-model same-tract employment slope is 13.36 jobs per 100 exposed resident workers, against 29.40 in years 5–7 and 28.01 in years 1–7. The corresponding payroll slope is \$798.06 thousand, against \$2.02 million and \$1.82 million. The model reaches 45.44 percent of the years 5–7 employment benchmark and 39.43 percent of its payroll benchmark. Using years 1–7 changes those comparisons to 47.69 and 43.92 percent. Both windows leave gaps in point estimates.

[24] All shopping comparisons hold exposure fixed at the residence distribution of entrant workers in the full model with 70 percent home-origin shopping. Workers living outside the CZ do not contribute to within-CZ same-tract exposure. The workplace tract is excluded after assignment, without reallocating its exposure.

Table 6: Pre-entry same-tract benchmarks and model diagnostics

	Pre-entry residence benchmark		Model diagnostic	
Outcome	Years 5–7	Years 1–7	Full model	All-home
Employment (jobs)	29.40	28.01	13.36	7.37
Annual payroll ($m)	2.02	1.82	0.80	0.48

Notes: Empirical benchmarks are equal-weight averages of the released continuous pre-entry same-tract worker-exposure coefficients for expanded services over years 5–7 and 1–7, relative to year −2. Employment is jobs and payroll is millions of annual 2023 dollars, per 100 exposed workers. Cross-year covariances are not released, so standard errors and significance stars for these averages are unavailable. Model slopes use the saved post-entry same-tract residence distribution for 55 events, with within-event demeaning and workplace exclusion. All-home refers to the shopping-origin assumption, not a different exposure geography. It uses its own no-entry equilibrium and the same residence exposure as the full model.

6 Results

I use the quantified model to compare the baseline economy with the economy after a productivity increase in the entrant's tract and tradable sector for 55 events in 38 commuting zones. This sample of firm entry events comes from the subsidy awards data described in Section 2 and Appendix A.1. Each firm entry event identifies a subsidy amount, a tract, an industry, and the number of jobs the firm promises to create. I set the productivity increase so that employment in that tract and industry rises by the promised number of jobs, and I report employment changes per 100 of these entrant jobs. Amenities, productivity elsewhere, preference parameters, and individuals' tastes are held fixed. Comparing the two equilibria captures CZ-wide responses, including competition for workers and space, that the event study in Fact 3 absorbs in its CZ-by-year fixed effects.

Section 6.1 reports employment and welfare across resident types. Section 6.2 traces the spatial incidence across neighborhoods, and Section 6.3 compares locations and entrant industries under common shocks. Section 7 then adds the award, asking what residents gain per public dollar, what a job costs, and how much the same award would buy at other sites.

6.1 How many jobs and how much welfare does an entry create?

An entry adds about half of the jobs it promises. For every 100 entrant jobs, the median event adds 51 jobs to the commuting zone. Other tradable firms lose approximately 61 jobs. In the model, all tradable sectors produce the same good. Tradable producers compete with the entrant for workers and floor space but sell at a national price, so they cannot pass higher costs on to consumers and shrink instead. As a result, increased demand for services translates into increased employment of 12 jobs in the service sector. Measured per net tradable job, the median service-job gain is 32 per 100. That is at the low end of the 30 to 80 in Bartik and Sotherland (2019) and about a fifth of the 160 in Moretti (2010). Overall, crowd-out of other tradable firms is about five times as large as the service gain, and it is the reason the entrant's promised job count is not added in net to CZ total employment.

A decomposition of the payroll to service spending in the CZ illustrates where the losses occur.

Following $100 of additional payroll at the entrant's site, total CZ workplace payroll rises by $62, because of employment losses elsewhere. Residents' income rises by about $30. The gap reflects two flows across the CZ border. Part of the new payroll goes to workers who live outside the CZ and commute in. The remainder of the gap goes to residents who used to work outside the CZ, and for them, the new paycheck mostly replaces earnings they already had. Rental dividends from ownership of residential land add about $6, so resident disposable income rises by about $37. Housing absorbs about $7 and tradable goods part of the rest, leaving residents' budget for services about $21 higher. Purchases that inbound commuters make near work add about $6, while approximately $10 goes to service firms outside the CZ, so total spending at service firms inside the CZ rises by about $16. This decomposition shows that the largest losses are from crowd-out and workers from outside the CZ, as about $70 of every $100 of new payroll never becomes income for residents in the CZ.

Residents' equivalent variation, the annual cash transfer without entry that leaves them as well off as entry, is about $4,000 per entrant job at the median, or 5 percent of the average annual wage in the entrant's tract and industry after entry. The welfare gain is small relative to the entrant's payroll because each hire gains the new wage net of the worker's outside option. The entrant's payroll that replaces earnings from a job the worker would otherwise have held adds little to welfare, and payroll earned by inbound commuters accrues to nonresidents. Additionally, rents and service prices offset part of the remaining income gain. Welfare per baseline resident is $3.9, with a 10th–90th percentile range of $1 to $14 across firm entry events. Higher wages contribute about 131 percent of the welfare increase and rental dividends add 116 percent on average. Higher housing costs and service prices reduce it by about 77 and 70 percent. Residential ownership is important because much of what individuals pay in higher rent returns to them as dividends. However, the return is unequal for high- and low-skill residents.

I quantify the model so that it assigns total residential rental income to each skill group in proportion to its share of owner-occupied housing value in the ACS (Section 5.1 and Appendix C.4.3). High-skill residents are about 23 percent of adults but hold about 47 percent of housing value at the median, so each receives about 3 times the rental dividend of a low-skill resident. High-skill residents account for about 42 percent of housing expenditure but receive about 47 percent of rental income. Wages differ as well. Employed high-skill residents earn about $97,000 a year on average in the median event, against $50,000 for low-skill residents, about 1.9 times as much. As a result, the gain for high-skill residents is $8.9 per person at the median, more than four times the $2 for low-skill residents.

Allowing workers to live across the CZ boundary and allowing residents to shop near work account for more than half of the service response. Removing both lowers the median service gain from 12 to 5 jobs per 100 entrant jobs. To determine this, I compare the full model with three restricted versions, maintaining the same productivity shock even as it translates into different employment increases in the entrant's tract and sector. In the first version, I impose the restriction that all shopping originates from the worker's home tract compared to just 70 percent at baseline.

This removes any local shopping by workers who live outside the CZ. Second, I fix the number of workers of each skill who commute in from outside the CZ at its no-entry level, so the entering firm must draw new hires exclusively from local residents and migrants. Then I impose both restrictions on a third version of the model.

Table 7: What inbound commuting and workplace shopping add at a common productivity shock

	Full model	Home-only shopping	Fixed inbound headcounts	Both restrictions	Interaction
Service jobs	12.35	9.73	7.91	5.43	1.28
Net jobs	50.54	51.13	34.24	34.50	0.69
Resident income ($ / $100 payroll)	37.31	36.53	40.79	39.78	0.08
Welfare ($ / resident)	3.93	3.82	4.57	4.52	-0.03
Employment at the entrant's site	100.00	100.32	99.89	100.10	0.00
Productivity increase (%)	3.01	3.01	3.01	3.01	0.00

Notes: Medians across 55 events. Employment is per 100 entrant jobs, the original promised jobs. Income is per $100 of the full model's additional payroll at the entrant's site, using the same denominator in all four specifications. Every specification receives the full model's productivity increase. Welfare is annual dollars per baseline resident of each specification. Each column subtracts its own no-entry baseline. The interaction is computed within events before taking its median; differences between column medians need not equal median paired differences. Appendix D.3 defines the contrast and reports its distribution.

Imposing the restriction that all shopping must originate from home tracts lowers the service gain by about 2.5 jobs per 100 entrant jobs, a fifth of the response, but leaves net jobs and welfare almost unchanged. This is because the service jobs it removes also free workers and floor space that other tradable producers can use. Fixing inbound commuting changes service jobs, net jobs, and welfare more substantially. It lowers the service gain by about 4.3 and net jobs by about 16 per 100 entrant jobs, yet resident income rises from about $37 to $41 per $100 of payroll and welfare per resident from $3.9 to $4.6. Under this restriction, the entrant must now fill its jobs from inside the CZ either from prior residents or from people that migrate into the CZ. Existing or new residents take the jobs that commuters would have filled, so more of the payroll stays with them as earnings. Labor supply to the CZ is less elastic, so wages rise more and other tradable producers shed more jobs. A labor market that cannot draw workers from outside therefore creates fewer jobs but leaves its residents better off.

The two margins overlap. Removing both lowers the service gain by about 1.3 jobs less than the sum of removing each alone. This is because inbound commuters are also many of the workers who shop near the entrant's site. Commuting from outside the CZ adds more new workers to the labor force than the migration or nonemployed worker channels. However, unlike movers and nonemployed workers, its workers are not residents. A majority of the service demand from inbound commuters leaves the CZ. This measure of labor market openness raises employment but lowers what residents gain.

6.2 Where do the gains land within the CZ?

I start by comparing the county in which the firm enters to the entire CZ as a way to illustrate the spatial distribution. At the median, the host county holds 38 percent of the CZ's residents and receives 49 percent of the gain in resident income, but only 28 percent of the CZ's service-job gain.[25] Its service employment falls in 23 of the 55 events. This is because income follows the entrant's workers home. Many hired residents live near the entrant, but service jobs accrue in the locations where people shop. By raising wages and rents, the entrant increases factor costs and prices for service firms, so shoppers take more of their spending away from the entrant.

Figure 5 shows the pattern in one market of median size. Panel A maps resident disposable-income changes and Panel B service-employment changes. The host county contains 26 percent of baseline residents and receives 33 percent of the income gain, but its service employment falls, by an amount equal to 26 percent of the CZ's net service-job gain. The rest of the CZ gains 126 percent of it. An exact accounting of the host county's service jobs shows why. Higher spending by residents and workers adds about 405 service jobs there, but the shift of shopping toward destinations elsewhere removes about 414, and fewer jobs per dollar of revenue remove about 66 more (Appendix Table D.6).

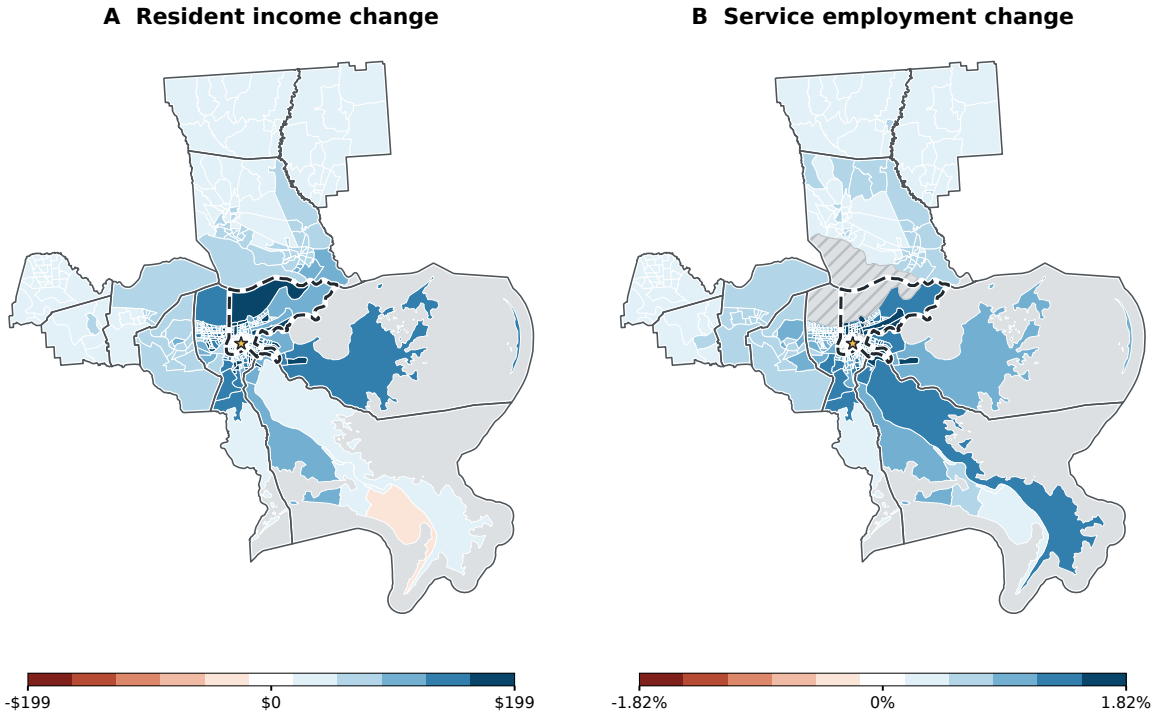
It is worth noting that income gains are more concentrated than service gains. The tenth of tracts with the largest income gains receives about 28 percent of the CZ's income gain but 21 percent of its service-job gain.

To explain the variation across tracts, I examine the home and workplace shopping networks. Workers travel to consume services either by originating their trips from their home tract or their workplace tract. To see which origin predicts where service jobs grow, I regress each tract's change in service employment, per 100 entrant jobs, on its exposure to trip origin, leaving out the entrant's own tract. Home exposure is the dollars of each \$100 of entrant payroll that reach the tract through the shopping of individuals from their residences, and work exposure is the dollars that reach it through purchases made from their workplace. I measure the exposure from the baseline payroll-weighted commuting shares, service expenditure shares, and shopping probabilities, so it is fixed before entry. Each specification includes event-by-county fixed effects, and the controls are the tract's baseline service employment, service wage, and residential rent.

Both routes raise service employment. Each dollar of home exposure adds about 0.53 service jobs per 100 entrant jobs and each dollar of work exposure about 0.43. Home exposure alone explains 72 percent of the variation within counties (column 2). The entrant's own tract behaves differently. There the productivity increase competes with service firms for workers and commercial floor space, so including it turns the work-exposure coefficient to -5.59 and lowers the share explained to 36 percent (Appendix Table D.4). Where the entrant's workers live therefore predicts most of where service gains land, and their workplace shopping adds a similar amount per dollar.

[25] The service share is the signed ratio of the host county's service-job change to the CZ's, taken over the 52 events with a positive CZ change. Shares of income and residents use all 55 events.

Figure 5: Spatial incidence of an expansion: income and service employment



Area	Baseline population (% of CZ)	Income change (% of CZ change)	Service-job change (% of CZ change)
Host county	**25.6%**	**32.8%**	**-26.0%**
Rest of the CZ	74.4%	67.2%	126.0%

Notes: The star marks the expansion tract; solid lines mark county boundaries and the dashed line the host county. Income is in annual 2023 dollars per baseline resident; employment is the percentage change from baseline service jobs. Scales are centered on zero with five magnitude classes on each side, defined separately for each outcome. Gray or hatched tracts have undefined measures. Table shares use changes in levels; negative and above-100 shares retain their signs.

6.3 Why do multipliers differ across commuting zones?

So far, I have presented the results mainly as the median change across firm entry events. This masks substantial heterogeneity in the response to the entry shock across CZs. Between the 10th and 90th percentiles of events, the service multiplier ranges from about 4 to 21 jobs per 100 entrant jobs. Net jobs range from about 37 to 65, and welfare per resident from about \$1 to \$14, a factor of more than ten. A natural first explanation is that the events differ greatly in size, from 150 to 12,000 promised jobs. To control for this, I recalibrate the productivity shock at each of the 55 observed sites, in the entrant's observed sector, so that employment there rises by 500 jobs. Removing differences in absolute size reduces the cross-event variance of the service multiplier by only 0.2 percent. I then standardize the relative size of the shock, so that employment at the site rises by 0.5 percent of the CZ's baseline employed residents. This raises the variance explanation to 4 percent. The remaining differences reflect the characteristics of the local market, mainly its reliance on workers who commute in from outside the CZ, and the industry of the entrant. Even together, the market characteristics I measure explain only about 20 percent of the variation within

Table 8: Commuting and shopping exposure and tract service-job changes

	(1)	(2)	(3)	(4)
Home-route exposure		0.822***	0.566***	0.528***
		(0.080)	(0.088)	(0.093)
Work-route exposure			0.408***	0.433***
			(0.147)	(0.143)
Log(1 + baseline service jobs)	0.042***			0.010***
	(0.005)			(0.002)
Log baseline service wage	0.009			-0.069***
	(0.010)			(0.012)
Log baseline residential rent	0.006			0.002
	(0.004)			(0.002)
Event × county effects	Yes	Yes	Yes	Yes
Within-county R^2 (%)	13.4	72.3	74.0	75.1
Tract–event observations	30,787	30,787	30,787	30,787
CZ clusters	38	38	38	38

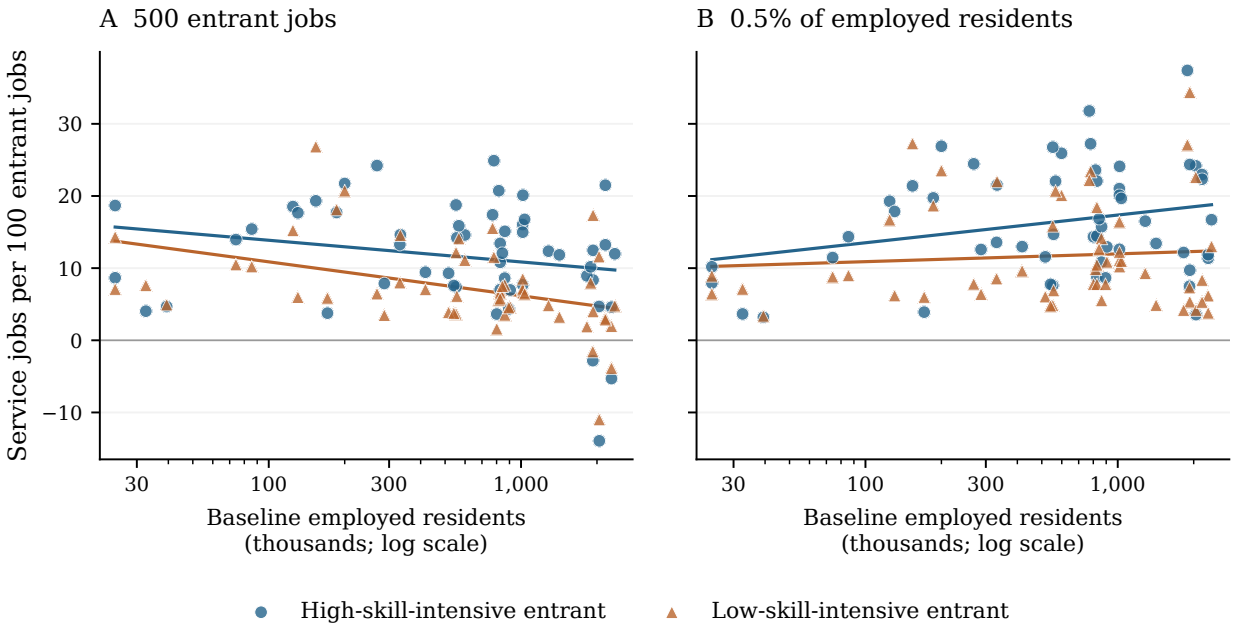
Notes: The dependent variable is the tract's service-job change per 100 entrant jobs. Exposures are dollars reaching the tract per $100 of entrant payroll through the indicated route, including its home/work budget weight. All columns use the same 30,787 tract–event observations with defined controls, from 55 events in 38 CZs, and exclude each entrant's own tract. Observations receive equal weight. Parentheses report standard errors clustered by CZ; fit is measured after removing event-by-county effects. Stars indicate $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

industries (Table 9), so much of each market's response depends on the particular tract the firm enters and the commuting and shopping flows from that tract.

Table 9 relates the proportional-shock multiplier to market characteristics, controlling for the entrant's industry. Market size explains 3 percent of the variation within industries. Adding inbound commuting, shopping leakage, the residential supply elasticity, and the calibrated service spending share raises this to 19.2 percent in column (5). Column (6) adds the shopping elasticity $\sigma_{N,c}$ and the commuting elasticity χ_{ce} used in each event's calibration. These govern substitution across shopping destinations and residence–workplace choices. They raise the explained share only to 19.7 percent. The share of residents who cross the CZ border to work is the most significant factor because markets that already rely on commuters remove more income from the CZ. In summary, these results show that most of the variation across markets is not explained by characteristics that are easy to observe. This illustrates that a multiplier estimated in one economy transfers poorly to another.

I then compare across industries. Under the 500-job shock, a high-skill-intensive entrant adds a median 12 service jobs per 100 entrant jobs and a low-skill-intensive entrant adds 7 jobs. The ratio is about 1.9, much lower than the ratio of roughly three in Moretti (2010) (4.9 local service jobs per high-technology job and 1.6 per manufacturing job). Part of the difference is pay. High-skill-intensive jobs pay about $102,000 a year against $85,000. To hold pay fixed, I simulate the entry of a high- and a low-skill-intensive entrant that both match the same $50 million of payroll. In this situation, the high-skill-intensive entrant adds about 57 service jobs and the low-skill-intensive entrant about 40. Holding payroll fixed lowers the high-skill-intensive entrant's advantage from a ratio of about 1.9 to about 1.4, so higher pay accounts for about 49 percent of the gap. The rest of

Figure 6: Market size and service multipliers under standardized entry



Notes: Each point is an event–industry experiment at the observed site. Both panels contain the same 55 event calibrations in 38 CZs and both tradable industries. Lines are separate linear fits against log baseline employed residents. Axes are common across panels. The experiments hold job gains at the entrant's site fixed at the absolute or proportional size; productivity and payroll adjust to meet that target.

Table 9: Market characteristics and service multipliers across commuting zones

	(1)	(2)	(3)	(4)	(5)	(6)
Log baseline employed residents	1.28	-0.28	-0.69	1.43	0.25	1.26
	(0.79)	(0.73)	(1.05)	(0.93)	(1.25)	(1.58)
Inbound employment share		-3.03***			-3.38***	-3.39***
		(0.71)			(1.20)	(1.23)
Shopping leakage			-2.95***		0.21	0.50
			(0.91)		(1.23)	(1.16)
Residential supply elasticity				0.21	0.56	0.54
				(1.11)	(1.18)	(1.16)
Service expenditure share					1.25*	1.30*
					(0.71)	(0.77)
Shopping elasticity, $\sigma_{N,c}$						-0.89
						(1.64)
Commuting elasticity, χ_{ce}						-0.13
						(1.23)
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Within-industry R^2 (%)	3.1	15.8	12.2	3.1	19.2	19.7
Event–industry observations	110	110	110	110	110	110
CZ clusters	38	38	38	38	38	38

Notes: The outcome is service jobs per 100 entrant jobs under the 0.5-percent employment shock. Each coefficient is the change associated with one standard deviation of the predictor after removing industry effects. Shopping leakage weights baseline destination shares by home- and workplace-origin service budgets; housing elasticity is weighted by baseline residents. Column (6) adds the saved shopping and commuting elasticities. Reported R^2 measures fit beyond industry effects.

the difference comes from the size of the incumbent tradable sector, and the difference in residential location between high- and low-skill workers.

7 Counterfactuals

This section asks what the same firm entry would deliver if it located elsewhere in the same CZ, holding fixed the jobs it promises and the subsidy it receives. The answer matters most to the local government that pays part of the subsidy, because it determines how cost-effective its subsidy is. Subsidy packages for large firms usually combine state incentives with a local share, most often a property-tax abatement granted by the county where the firm locates. Because local property-tax deals need the county's approval, the county also has a say in where the firm builds (Appendix A.1). This is the choice a local government faces once a firm has chosen its market, and it is where the commuting and shopping networks matter most. Moving an entry within its own county changes what residents gain by about as much as moving it to a different market. The same entry delivers fewer employment spillovers where economic activity is already concentrated and where its workforce commutes in from outside the market. It is larger in tracts where it must hire from the surrounding neighborhoods. Section 7.1 first sets these gains against what the subsidy costs, Section 7.2 moves each entry across locations in its county and asks why the outcomes differ and how much of the gain the county keeps, and Section 7.3 swaps the entrant's industry. Each event keeps its recorded subsidy in 2023 dollars, spread over ten annual payments with a median of about $1.6 million, split equally between the host county and the state.

7.1 What does a job cost, and is the subsidy worth paying?

If the subsidy brought the firm, residents gain more than it costs. At the median, CZ residents gain about $1.85 a year for each dollar of the annual payment, counting both the county's and the state's halves, and the gain exceeds the payment in 37 of the 55 events. From the county's perspective, the added property tax that the entry generates covers the county's half of the payment in 45 of the 55 events. In the other 10, CZ residents gain about $1.07 per net county dollar and host-county residents about $0.31. In only 5 of the 55 events does the recorded subsidy exceed the payment at which residents' gain would just equal the county's net cost. In those events, residents of the paying county capture much less of the gain than the CZ as a whole. These ratios are fiscal accounting in the spirit of the marginal value of public funds (Hendren and Sprung-Keyser, 2020). Table 10 collects the measures.

The county can afford its half because the entry enlarges its property tax base. Entry adds floor space and raises rents on residential and commercial property in the county, which raises its annual receipts by a median of about 3.7 times its share of the annual payment, enough to cover it in 45 of the 55 events. The model sends tax receipts out of the economy, so residents' welfare does include the tax dollar gain, while the county's accounts credit them against its payrment of the subsidy.

Table 10: What the subsidy costs and what residents gain

Measure	Median	P10	P90
Annual subsidy ($)	1,630,000	134,000	16,700,000
Subsidy per promised job ($)	22,400	2,060	133,000
Subsidy per net CZ job ($)	43,600	4,660	322,000
CZ welfare / annual subsidy	1.85	0.296	23.1
CZ welfare / net county outlay	1.07	0.489	4.25
Host welfare / net county outlay	0.306	-0.0501	1.62

Notes: Event-weighted summaries at observed locations. 2023 dollars. Subsidy per job uses the full ten-year commitment; welfare and county outlays are annual. Ratios with net county outlay as the denominator use only the 10 events with positive outlay. The other 45 events have additional host-county tax receipts at least as large as half the annual subsidy. Welfare is residential-opportunity EV, not an additive cohort measure. The CZ and county welfare indices are evaluated separately and are not spatially additive, so the difference between the two county-outlay ratios is not the welfare of the rest of the CZ. Appendix E defines each measure.

Policymakers more often quote the cost per job. The recorded subsidies come to about $22,000 per promised job at the median. Because an entry adds only about half the jobs it promises (Section 6.1), the subsidy per net CZ job is about $44,000, about twice as much, with a 10th–90th percentile range of about $4,700 to $320,000. Every observed event adds net jobs. Appendix E reports the annual-payment version, which equals the subsidy per job-year if the jobs last the ten years of payments.

For comparison, the megadeals in Slattery and Zidar (2020) averaged $178 million for 1,500 promised jobs between 2002 and 2017, about $120,000 per promised job, so the subsidies here are modest by that standard. Program evaluations instead measure the jobs an incentive causes (Criscuolo et al., 2019; Siegloch et al., 2025; Freedman et al., 2023). For the California Competes Tax Credit, Freedman et al. (2023) estimate about $3,960 per job created in recipient tracts, against $9,900 per pledged job.[26] Even if every subsidy brought its firm, the median cost per net job here is an order of magnitude above that estimate per job created, and the gap widens with every subsidy that did not change a firm's location decision.

How often subsidies change firms' location decisions also decides whether the subsidy is worth paying. A subsidy creates the gain only when the firm would not have come without it. If the subsidy changes the location decision in, say, a quarter of cases, residents can expect a quarter of the gain. Bartik (2018) places plausible rates for typical incentives between 2 and 25 percent.[27] At 25 percent, the median cost per job the subsidy created rises to about $170,000, residents gain about $0.46 per public dollar, and 19 of the 55 subsidies exceed the county's break-even. At 2 percent, the cost is about $2.2 million per job, residents gain about $0.04 per public dollar, and 47 subsidies exceed the break-even (Appendix Table E.1). The median subsidy is worth its full public cost only if it is pivotal for at least 54 percent of decisions, about twice Bartik's upper bound. Overall, the subsidies pay for themselves only if they cause most of the entries they subsidize, and

[26]These figures keep each study's price basis and are not converted to 2023 dollars. Differences in geography, duration, and additionality also limit direct comparison.

[27]The lower end reflects the median of the estimates without an obvious bias, 3.4 percent, and the upper end is close to the 23.5 percent median of estimates biased upward (Bartik, 2018, pp. 1–2).

the evidence on incentives suggests that they do not.

7.2 How much would the same subsidy buy at another location?

The location a firm chooses is usually not the one that would serve residents best. In the typical event, about 70 percent of the other locations I evaluate in the host county would give residents a larger gain for the same jobs and subsidy. Moving the entry to the county's median evaluated location would raise residents' gain by about 18 percent with the same subsidy, and moving it to a location better than three quarters of the candidates would raise it by about 57 percent. These location effects are of the same order as the differences across markets. Within a county, the best tenth of locations delivers about 2.1 times the gain of the worst tenth, and within the whole CZ about 2.7 times. Across observed events, residents' gain per entrant job varies by a factor of about 3.6 between the 10th and 90th percentiles, and by about 3.4 when every event receives the same 500-job expansion. Where a county puts an entry therefore matters on the same order as which market the entry enters.[28]

For each event, I move the entry to at least ten tracts in its host county with baseline employment in the entrant's industry, and recalibrate productivity so that employment in that industry rises by the same promised jobs, with the same subsidy. The median host-county comparison contains 11 locations including the observed one, and the whole-CZ comparison uses the original sample of up to 20 alternatives per CZ.

Table 11: The same job target and subsidy at alternative locations

Measure	Host county	Whole CZ
Observed-location welfare percentile	30.0	40.0
Welfare P90/P10 across locations	2.12	2.71
Welfare change at median location (%)	17.6	6.6
Welfare change at P75 location (%)	56.9	46.7
Net jobs–welfare rank correlation	0.78	0.68
Cost reduction at lowest-cost location (%)	26.1	28.9
Additional jobs at lowest-cost location	148.8	204.1
Welfare difference there ($/resident)	1.98	2.60

Notes: Medians of event-level comparisons, including the observed location in each candidate set. Both columns use all 55 events. The host-county column includes the added county alternatives; the CZ column retains the original CZ draw. Welfare refers to the entire CZ in both columns. The observed-location rank runs from zero for the lowest to 100 for the highest evaluated welfare. Median- and P75-location welfare changes compare within-event quantiles with the observed outcome. Cost reductions, additional jobs, and welfare differences are paired observed-versus-lowest-cost comparisons. Welfare is annual 2023 dollars per baseline adult resident. The best location is the best among those evaluated, and its advantage grows with the number of locations tested.

Locations whose workers already live in the CZ tend to deliver more to its residents. Table 12 groups each event's evaluated locations by two baseline characteristics, the share of the location's workers who commute from outside the CZ and the share of their work-route spending that stays

[28]The within-event figures are medians of each event's 90th-to-10th percentile ratio across its evaluated locations, and the across-event figures compare events directly. Welfare per public dollar is not comparable across events, because annual subsidies themselves vary by a factor of about 125 between the 10th and 90th percentiles.

in it. Locations with few inbound commuters deliver about 15 and 7 percent more welfare than the event's median location, and locations with many deliver about 10 percent less, whatever their shopping retention. Net jobs differ by at most about 5 percent across the four groups. Within events, the rank correlation between a location's inbound share and residents' gain is −0.43 at the median, against 0.03 for work-route retention. The same subsidy could therefore benefit residents at a location where a majority of the jobs are filled by locals.

Table 12: Which locations deliver more to residents?

	Work-route spending kept in the CZ	
	Below median	Above median
Residents' gain relative to the event's median location (%)		
Few inbound commuters	+14.7	+7.4
Many inbound commuters	−10.2	−10.5
Net CZ jobs relative to the event's median location (%)		
Few inbound commuters	+2.8	0
Many inbound commuters	0	−5.1
Evaluated locations		
Few inbound commuters	275	323
Many inbound commuters	323	225

Notes: All locations in the original CZ-wide sample for the 55 events, including the observed location. Within each event, locations are split at the median of two baseline characteristics of the entrant's industry at that location, the share of its workers who commute from outside the CZ and the share of its workers' workplace-origin service spending that stays in the CZ. Entries are medians of each location's percentage difference from the event's median evaluated location, holding the promised jobs and the subsidy fixed. Welfare is annual CZ residential-opportunity EV.

These baseline results motivate repeating the commuting and shopping parameter restrictions across locations. Within host counties, fixing inbound commuting reduces the standard deviation of net-job gains across locations by about 23 percent at the median, home-only shopping by 5 percent, and both by 25 percent. Fixing inbound commuting changes the net-job-maximizing location in 24 of 55 events, although the median rank correlation with the full model remains 0.92.

Welfare differences across locations remain largely intact. Under all three restrictions, the within-event standard deviation of welfare gains rises by about one to three percent at the median, and median rank correlations are at least 0.99. The original CZ draw and the size-cap sensitivity give the same qualitative pattern (Appendix E.3). These comparisons hold productivity fixed, so actual job gains in the entrant's tract and sector are 100.21, 99.94, and 100.10 per 100 promised jobs under home-only shopping, fixed inbound commuting, and both, respectively, at the median of the within-event location averages. Table E.7 reports these employment responses alongside the outcomes.

The location that adds the most jobs is usually not the one residents value most, even though the two rankings are closely related. Within host counties, the rank correlation between net jobs and residents' gain is 0.78 at the median, but the same location maximizes both in only 22 of 55 comparisons. The evaluated location with the most net jobs cuts the subsidy per net job by about

26 percent and adds about 149 net jobs at the median, and searching the whole CZ raises these to 29 percent and 204 jobs.

Part of what separates locations is scale. Creating the same promised jobs takes more payroll at some locations than at others, and more payroll means more income to spend. To separate scale from location, I re-solve each event at different locations in its county this time matching the additional payroll at the observed location rather than its jobs, with the same industry and subsidy.[29] Matching payroll changes how many jobs a location creates, from about 0.25 to 1.6 times the promised jobs. Equalizing payroll narrows the range of service-job gains across locations by about 34 percent and the range of welfare gains by about 29 percent at the median, while the range of net-job gains barely changes. Roughly a third of the differences between locations in service jobs and welfare therefore come from how much payroll it takes to create the promised jobs, and the rest from who receives that income, where they spend it, and how the local economy adjusts.

Location still matters at equal payroll (Table 13). The highest-welfare location raises residents' gain by about 43 percent at the median, against 88 percent at equal jobs, and the location with the most net jobs lowers the subsidy per net job by about 14 percent, against 22 percent at equal jobs on the same locations. Even with scale removed, where an entry locates still changes what the subsidy buys.

Table 13: Relocating the same expansion at equal jobs or equal payroll

Measure	Fixed jobs	Fixed payroll
Range across locations, percent of the fixed-job range		
Service-job gains		66.2
Welfare gains		70.6
Net-job gains		100.1
Best evaluated location relative to the observed location		
Location with most net jobs: cut in subsidy per net job (%)	22.2	13.7
Location with most net jobs: additional net jobs	128.7	50.2
Highest-welfare location: welfare gain (%)	87.5	43.2

Notes: The same 51 events and 381 host-county alternatives from the original CZ-wide sample in both columns, with each event's observed location included once. The fixed-job column re-reports the equal-jobs relocation on these locations, which are fewer than in Table 11, so its entries differ from that table. All outcomes refer to the entire CZ. The first panel reports the median within-event ratio of the maximum-minus-minimum outcome range at equal payroll to the same range at equal jobs, times 100. The remaining rows are medians of paired improvements over the observed location. The location with the most net jobs has the lowest subsidy per net job. Each event retains its own industry and recorded subsidy. Equal payroll means the same annual incremental tract–industry payroll as at the observed location, in 2023 dollars. Equal payroll creates between 0.25 and 1.6 times the promised jobs, and service jobs fall at 13 of the 381 locations. These comparisons cover evaluated locations, not all feasible county locations.

The differences across locations follow the existing geography of jobs and workers around each location. Across each event's host-county locations, residents' gain falls with the service employment in the location's tract (median rank correlation −0.83, negative in 55 of 55 events), with the jobs

[29] Payroll is the change in the wage bill of the entrant's industry at the location, including wage changes for workers already there. The comparison covers the 51 events with an alternative in the original draw (Appendix E.4).

per resident there (−0.74), with existing employment in the entrant's industry (−0.67), and with the share of the location's workers who live outside the CZ (−0.48). It rises with payroll per entrant job (0.62). Where the entrant's industry is already large, the same job target takes a smaller productivity increase, so wages and payroll rise less per entrant job, and a workforce drawn from outside the CZ takes much of its pay home. How concentrated a location's commuting and shopping flows are matters much less (Appendix Table E.9). These are associations across locations rather than separate effects, but they fit the results above. Equalizing payroll removes only about a third of the spread across locations, and shutting off the commuting and shopping responses barely changes the welfare ranking, so what separates locations is the geography already in place around them.

The county and the CZ largely agree on where within the county the entry should go, and the county's choice does little to change how much of the gain it keeps. The location best for host-county residents is also the CZ's best host-county location in 33 of 55 events, and in 43 once the county counts its added property tax, and the two rankings have a median rank correlation of 0.92. Locations within a county differ by a median of only about 17 percentage points in the share of their workers who live in the county. A more local catchment raises the county's share of the service-job gain only slightly (median rank correlation 0.19 across the 40 events with enough locations) and lowers its share of the welfare gain (−0.43 for home-origin shopping in the county), consistent with the spending it brings into the county also raising local rents and service prices that residents who are not hired pay. How much of the gain a county keeps is set mainly by its boundary and its market rather than by where in the county the entry goes.

7.3 Would a different type of entrant deliver more per dollar?

A high-skill-intensive entrant delivers more for the same subsidy, mostly in welfare rather than jobs. For each event, I keep the observed location, the promised jobs, and the subsidy, and swap only the entrant's tradable industry. The subsidy per net job is about the same for the two, about $49,000 and $50,000 at the median, but residents gain about $1.66 per public dollar from a high-skill-intensive entrant against $1.40 from a low-skill-intensive one. The high-skill-intensive entrant delivers more welfare per dollar in 76 percent of events and costs less per net job in 71 percent. Its jobs pay about 19 percent more, which accounts for part of its advantage, but Section 6.3 shows that its service advantage persists at equal payroll, so the rest comes from who fills its jobs and where they spend. Overall, the type of entrant changes what residents gain from a subsidy more than it changes the cost per job.

Panel B of Appendix Table D.7 repeats the comparison at 500 jobs, at 0.5 percent of baseline employed residents, and at $50 million of added payroll.

Taken together, the subsidy is worth paying only if it changes location decisions, and when it does, the locations and entrants whose jobs go to residents deliver the most for each public dollar.

Table 14: Entrant industry at the same observed location, job target, and subsidy

Measure	High-skill industry	Low-skill industry
Job increase at the entrant's location	979.0	979.0
Net CZ job increase	450.9	376.4
Service job increase	106.8	57.0
Incremental payroll ($m)	101.75	81.64
Subsidy per net job ($)	48,683	50,080
CZ welfare ($/resident)	4.13	2.35
CZ welfare / annual subsidy	1.66	1.40

Notes: Medians over the same 55 completed event pairs in both columns. Each event retains its original promised job increase at the entrant's location and its recorded subsidy. Welfare and incremental payroll are annual 2023 dollars; subsidy per net job uses the full ten-year commitment. Payroll includes wage changes for existing workers in the entrant's tract and industry. No subsidy-financing equilibrium is introduced.

8 Conclusion

State and local governments spend about $20 billion a year on firm-specific subsidies, justified by the direct jobs these firms create and the indirect jobs their payroll supports. This paper asks, once a large tradable firm has entered, how many local service jobs its payroll supports and where they appear. It follows the payroll from the firm to its workers, from the workers to the neighborhoods where they live and shop, and from those neighborhoods to local service firms. It uses linked worker histories to see who is hired and a tract-level model of an open commuting zone to add up the pieces in a local multiplier.

The data show why the indirect jobs are fewer than the payroll suggests. About half of an entrant's new employees were not previously attached to the local labor market, and many of the rest leave other local jobs, so only about $32 of every $100 of entrant payroll becomes new income for residents. Residents spend that income close to home, and service employment grows in the neighborhoods where the entrant's workers lived before entry.

The model turns these pieces into a net effect. An entry adds about half of the jobs it promises, because other tradable producers lose workers and floor space to it, and 100 entrant jobs support about 12 service jobs at the median, fewer than most existing estimates. Residents' welfare rises by about $4,000 a year per entrant job, small relative to the wage because each hire gains only the new wage net of their outside option. The gains spread across the market. The host county holds 38 percent of the commuting zone's residents but receives only 28 percent of its service-job gain. For entries of the same relative size, the multiplier is about 3.7 times as large at the 90th percentile event as at the 10th, and the aggregate characteristics of the local economy explain only about 20 percent of the variation within industries. Much of the rest reflects the commuting and shopping flows around the entrant's site.

These results put a price on the jobs. The recorded awards come to about $22,000 per promised job but about $44,000 per net job, because an entry adds only half the jobs it promises. In 45 of the 55 events, the added property tax from the entry covers the host county's share of the award, and assuming the firm would not have entered otherwise, residents gain about $1.85 for each public

dollar. The median award pays for itself, however, only if it changes the location decision in at least 54 percent of cases, about twice the upper end of the evidence on incentives. The same award would also buy more at sites whose jobs go to people who already live in the market.

The model takes the firm's entry as given, so it measures what an award buys once the firm has come, not whether the award changed its location decision. Within that limit, the lesson is that what an entry is worth to a place depends less on how many jobs the firm promises than on who fills them, where they live, and where they shop.

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Appendix

Appendix roadmap. Appendix A presents the data construction and additional empirical results. Appendix B derives the model from individual preferences, including employment and location choices, consumption, production, and equilibrium accounts. Appendix C describes quantification and equilibrium computation. Appendix D supports the results in Section 6, and Appendix E supports the counterfactuals in Section 7.

A Data and Additional Empirics

A.1 Award data and event samples

Award records and site verification. The award data begin with three public sources. State economic development agencies publish registers of the awards they approve, Good Jobs First's Subsidy Tracker compiles company-specific awards from state and local disclosures (Tarczynska, 2022), and the Slattery series records the announced packages of large incentive competitions, which often bundle state and local incentives (Slattery, 2024). Where supporting documents exist, I reconcile each recorded amount with agency reports, executed agreements, legislative materials, and local authorizing documents, matching on the recipient, the program, the project location, the award or agreement date, and the contract identifier when one is reported. Because an award names a firm rather than an establishment, I also match each award to its establishment in Data Axle and check the site against a document, preferring primary incentive documents to agency announcements and news reports, and against 2020 tract boundaries and available parcel records. I drop 36 awards that do not attach to a single employer's site, such as grants to prepare industrial land, film credits, and tax increment financing for retail developers, and 11 more whose address cannot be placed in a 2020 tract (Table A.1).

From award records to events. One project often appears in several award records. A state credit, a training grant, and a local abatement may each finance it, and Good Jobs First and a state register may list the same award. I therefore consolidate records into firm-site events. After normalizing firm names, which merges four sites where one firm appears under two spellings, I group the awards to the same firm in the same tract and start a new episode when at least five years separate two awards. Only one site has awards that far apart. When every award at a site promises the same number of jobs, the awards form one event, the jobs count once, and the subsidy is the sum across instruments. When the promises differ, I split the awards by award year and combine the instruments within each year, which turns six sites into nine events. The one exception is a project whose three annual awards report slightly different totals for the same plant, which I keep as a single event. Each event is dated to its first award year, and its promised jobs are the largest promise among its awards. Overlapping valuations of the same award count once, so when a package from the Slattery series already contains a state program's award, I keep the package

figure and do not add the program. These steps turn 299 award records into 226 events involving 212 firms at 201 tracts in 32 states, and 179 of the events rest on a single award.

Realized jobs and payments. Promised jobs come from the award itself, but realized jobs and payments are observed only where a program requires reports and the agency publishes them. Realized employment is observed for 95 events, most of them in Texas, Virginia, North Carolina, and New Jersey, which publish performance reports for their largest programs. The administering agency verified or certified 53 of these counts. The rest are recipients' self-reports republished by the agency, figures carried in the award record, or press reports. When an event has more than one count, I prefer verified or certified counts, then a count attached to a documented post-performance review, then the most recent count, and finally the larger one. The 64 events with a count at least four years after the award, when most of the promised hiring should have occurred, form the descriptive attainment sample in Figure A.1, which is not used in the regressions or the model.

An approved amount is not the amount paid, so I collect payment and clawback records separately where they are public. In Indiana, I match awards to the 2017 Economic Incentives and Compliance Report by recipient, funding program, and exact contract amount, and record the paid or certified amount with its reporting date. In Virginia, I match the Commonwealth's Development Opportunity Fund awards to the state's post-performance reports by grant amount, locality, and performance-agreement date, keep the latest project status among the editions examined, and separate clawback requests from repayments received where the report allows. This yields paid or certified amounts for 22 events and clawback records for five. These records give cumulative payments or recoveries as of a reporting date rather than spending over the completed life of an award.

State and local support. Separating the local share starts with whether a site received any local property-tax relief. Among the 226 events, I establish the local relief for 86, either as a dollar amount (43) or as a documented absence of any abatement (43). For another 13, an abatement is documented but its amount is not published, and the remaining 127 are unresolved. For 33 events, I can document both the state and the local share of the award. Fifteen receive state support only through state programs, fourteen are large packages whose state side I rebuild from the individual programs named in the granting documents, and four have a local deduction whose dollar value I compute from the exempted assessed value. I build the state amount from documented state instruments, rather than as the residual left after subtracting an announced local contribution from the package total. Each component is classified by the revenue source it reduces, so a local property-tax exemption that the state administers remains a local cost. I also set apart spending that does not accrue to the firm, such as public infrastructure, transfers to third parties, and bond face values, and I separate generally available tax preferences from negotiated, firm-specific incentives.

Property-tax relief is researched facility by facility. I first identify the parcel, its owner of record, and the taxing jurisdictions, including the municipality and school district, allowing for

subsidiaries, successor firms, landlords, and industrial development authorities that hold title to the property the recipient occupies. The sources are assessor and auditor records, state abatement registers, industrial development board filings, executed agreements, ordinances, council materials, and project disclosures in Annual Comprehensive Financial Reports. Government-wide abatement totals are not assigned to a firm without project-level evidence. When a source reports an abated assessed value rather than a dollar saving, annual relief equals the jurisdiction's tax rate per dollar of assessed value times the exempted value, and market values are converted only when the assessment basis is known. Under a payment in lieu of taxes, relief is the tax that would have been owed less the required payment. Relief repeated across several state award records counts once, while concessions from separate taxing authorities are added.

Observed annual savings, approval-stage estimates, and projected full-term values are kept separately. When the observation window is incomplete, the full-term value follows the contract's term and schedule, including any recorded cancellation, and otherwise extends the most recent observed annual saving. Cases that require an assumed horizon or a declining deduction schedule are flagged, and all amounts are in 2023 dollars. A search that finds nothing is recorded as missing rather than as zero support, and a documented absence of support is distinguished from an instrument whose amount is unknown. Public-ownership exemptions and substitute cash grants are recorded separately from conventional abatements. Because disclosure and historical coverage vary across jurisdictions, and realized abatements depend on the investment the assessor eventually values, these comparisons describe only the projects whose components can be documented, and their full-term values are estimates rather than verified spending.

Table A.1: From all-state award records to firm-site events

Stage	Change	Count
Raw all-state award records	—	202,070
Package > $1 million or promised jobs > 100	−184,903	17,167
Exactly one usable announced location	−2,587	14,580
Exclude loans	−974	13,606
Append Slattery awards	+385	13,991
Geographic and establishment matching	−5,857	8,134
Award years 2004–2017	−3,503	4,631
Tradable industry	−2,687	1,944
Promised jobs > 500 or > 0.50% of baseline CZ employment	−1,598	346
Restrict to single-employer, site-specific awards	−36	310
Require substantiated address and 2020 tract	−11	299
Consolidate award records into firm-site events	—	**226**

Notes: Counts refer to award records until the final row. CZ employment is measured in the year before the award. Multiple awards for the same firm-site project are combined into one event, dated to the first award year; distinct project commitments and episodes separated by at least five years are kept separate.

The award universe. Table A.2 describes the awards of at least 100 promised jobs in the 29 states with usable records, 2002–2024. Tax credits and rebates are half of the awards but a fifth of the value. Property tax abatements are 7 percent of the awards but a quarter of the value, and they

are local rather than state revenue. The 2 percent of awards that bundle several instruments carry 46 percent of all value, so mean-based statistics are driven by the tail. Most programs require an application before the siting decision, the largest awards are discretionary, and local property-tax deals need a second approval from the county, school board, or industrial development board.

Table A.2: Instrument mix of awards with at least 100 promised jobs, 2002–2024

Instrument	Share of awards	Value ($bn)	Share of value
Tax credit or rebate	52.40%	15.50	21.40%
Grant (incl. loan hybrids)	37.90%	5.70	7.80%
Property tax abatement	6.70%	18.00	24.80%
Megadeal (mixed package)	2.30%	33.20	45.80%
Other (infrastructure, reimbursement, assistance)	0.80%	0.10	0.20%
Total	100%	72.50	100%

Notes: Author's award file, 29 states, 2002–2024, all awards of at least 100 promised jobs. Values are reported package values.

Table A.3 codes the tax base of the 17 programs that carry 99 percent of the credit and rebate dollars. Three fifths are payroll or withholding based, so the firm keeps part of the income tax withheld from its own employees' wages. Only 6 percent are conventional corporate income tax credits. Most "corporate tax credits" in this universe are therefore wage subsidies financed by employees' own withholding. Awards are nominally conditional on job counts, wage floors, and investment minimums, but enforcement ranges from post-performance payment on verified jobs to self-reported counts with weak audit, and reported jobs are usually projections.

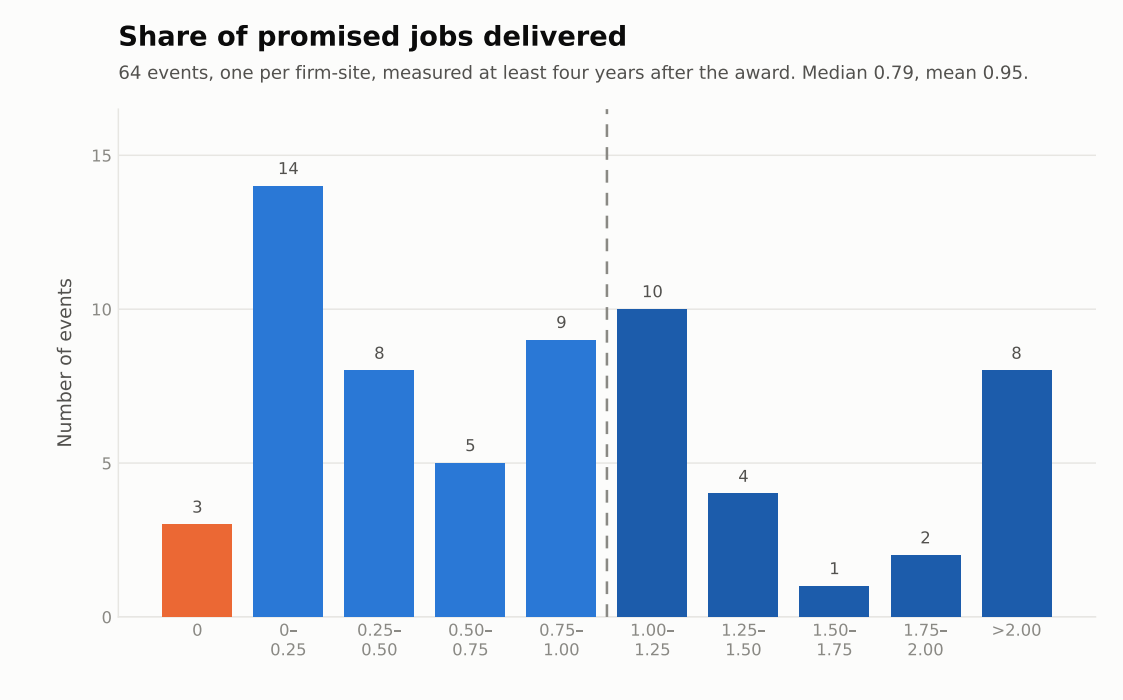
Table A.3: Tax base of credit and rebate awards

Base	Share of credit and rebate value
Payroll or withholding	61.30%
Sales or property, in a state without an income tax	8.10%
Corporate income, conventional	5.90%
Mixed income, sales, and withholding	3.00%
Local or unclassified	3.20%

Notes: Shares of the $15.50 billion coded as tax credit or rebate in Table A.2, after hand-coding the 17 programs that carry 99 percent of it. Two New Jersey programs (25.20 percent of the total) are split between the first and third rows.

Promised and delivered jobs. Figure A.1 uses the 64 firm-site events with reported employment at least four years after award. The median delivered 0.79 of the jobs promised and the mean 0.95. 39 fell short, 17 delivered under a quarter of the promise, 3 recorded zero or negative job gains, and 8 more than doubled the promise. Total realized jobs divided by total promised jobs is 1.06. Because disclosure is selective, these are descriptive statistics rather than estimates of the effect of subsidies.

Figure A.1: Share of promised jobs delivered, firm-site events observed at least four years after award



Notes: 64 events from the consolidated award registry, with employment observed at least four years after award. Each bar counts firm-site events by realized-to-promised employment; the first includes zero and negative gains. The dashed line marks delivery of exactly the promised count. Median 0.79, mean 0.95, pooled ratio 1.06.

The samples used in the paper. The paper draws on several samples of entries, and Table A.4 relates them. The award registry provides the subsidy events. The LEHD entrant sample contains all large tradable openings and expansions in the 29 states with restricted data, whether or not they were subsidized, and supports Facts 1 and 3. The public registry links awards to an entry year, a 2020 tract, and a commuting zone; a candidate enters it only if LODES employment in the entry tract and industry rises by more than 500 jobs relative to baseline in at least one of event years 1 through 5, and when two events occupy the same tract within two years the earlier is kept. The model is quantified for the 55 registry events with complete model inputs. Because the public registry conditions on a realized employment increase, it describes what follows a large entry that occurred and does not identify whether the award caused the firm to locate where it did.

A.2 Public data sources

Employment and commuting. Workplace employment by tract comes from LODES Workplace Area Characteristics, resident workers from Residence Area Characteristics, and tract-to-tract commuting flows from the Origin–Destination files. All tract identifiers are harmonized to 2020 geography. The public service-employment outcome is raw annual workplace employment in NAICS 44–45, 53, 61, 62, 72, and 81, available for 97.48% of tract-years. In the public exposure designs,

Table A.4: Event samples

Sample	Screen	Events	CZs	Used in
Award registry (Table A.1)	Tradable; $>$ 500 jobs or $>$ 0.5% of CZ employment; 2004–2017	226	—	Section 2
with realized jobs $\geq$ 4 years after award	Reported employment	64	—	Figure A.1
LEHD entrant sample	Same size and industry screen; 29 states	400	—	Tables 1–2, Fact 1
pre-entry-residence stack	CZs with one observed event	—	40	Fact 3, Figure 3
new-income stack	CZs with one observed event	—	30	Figure A.7
continuous exposure	All events and tracts	—	100	Figure 4
incumbent-industry event study	Matched control CZs	—	80	Appendix A.4
Public registry	LODES tract-industry employment +500 in years 1–5	124	69	Appendix A.6
boundary comparison	All verified award sites, +5 years	180	86	Section 2, Table A.11
Model sample	Registry events with complete inputs	55	38	Sections 5–7

Notes: Entrant and event counts from restricted data are disclosure-rounded. A dash means the count is not released or not applicable. The LEHD entrant sample includes the subsidized events that meet its screen. The award registry's CZ count is not separately tabulated.

entry tracts and their adjacent tracts receive no exposure, so that exposure is never assigned to employment at the plant itself.

Shopping visits. Shopping flows come from 2019 Veraset smartphone visit records (Veraset, 2022). Explicit home and work records are not counted as visits. Remaining visits to service establishments are aggregated from 2010 block groups to 2020 tracts with the official crosswalk. Each device receives total weight one within a month and industry group, so the annual measure is a device-weighted distribution of visits from each home tract to each destination tract. Visits are counts of trips, not dollars. The visit data cover retail, selected real estate, education excluding colleges, food services, and other services; they contain no health care (NAICS 62) bucket, so that sector enters the employment outcome but not the shopping allocations. Because the visits are observed in 2019, after the events, they describe where service demand is located rather than a predetermined treatment map.

Travel times. Driving times between tract centroids use the 2020 OpenStreetMap road network (OSRM). Same-tract travel is set to five minutes. Gravity comparisons use tract pairs with finite driving time of at most 180 minutes.

Wages. Public workplace wages combine QWI county earnings by education with LODES tract employment counts by education. Because LODES publishes education and industry as separate margins, education categories are allocated to the tradable sector using the tract-year tradable

employment share. This is a proxy for workplace wages by tract, not a joint tabulation.

Property. CoreLogic deed and assessor records over 2004–2020 provide residential and commercial repeat-sales and hedonic price indexes, building square footage, land area, and property counts by 2020 tract. The repeat-sales estimator follows a three-stage Case–Shiller design, and the hedonic regressions include physical characteristics, property type, quality, condition, and missingness indicators. Commercial transactions are sparse in some tracts, and tract-years without support are left missing. Property values are converted to annual rent equivalents in the quantification through the price-to-rent assumptions of Appendix C.

Restricted records. Worker histories and establishment records come from the restricted LEHD and business register (Abowd and Vilhuber, 2011; U.S. Census Bureau, 2025). Entrant hires are observed from 12 quarters before to 12 quarters after hire. A worker's primary employer in a quarter is the employer paying the highest earnings. All restricted statistics in the paper have passed Census Bureau disclosure review.

A.3 Fact 1: Hires, residence, and earnings

Residence and prior workplace. Table A.5 expands Table 3 into ten joint residence–workplace categories for the 551,850 hires in the released residence tabulation, pooling entrant types and CZ sizes. Residence is measured one year before and one year after hire. Let C_e be the event CZ, r_i^- and r_i^{+1} residence before and after hire, and w_i^- the prior workplace. Continuing residents satisfy $r_i^- = r_i^{+1} = C_e$; movers satisfy $r_i^- \neq C_e$ and $r_i^{+1} = C_e$, including those who commuted into the CZ before moving; outside residents satisfy $r_i^{+1} \neq C_e$. Residence and prior workplace carry different information. Continuing residents who previously worked outside the CZ are 9.31 percent of hires, and movers who previously commuted to a job in the CZ are 2.17 percent. Among continuing residents, 9.56 percent have no observed prior job. For these workers I treat no observed job as nonemployment, which includes nonparticipation. For workers who lived outside the CZ before hire, a missing prior job may instead be a job in a state outside the data, so their prior earnings are treated as missing rather than zero.

Market size. Table A.6 splits hires by the released CZ-size class. Continuing residents are 62.27 percent of hires in small CZs, 60.03 percent in medium CZs, and 54.64 percent in large CZs, while the share living outside the CZ after hire rises to 31.26 percent in large markets from about 25 percent in small and medium ones. Movers are 13.09, 15.42, and 14.10 percent of hires, so migration does not rise with market size. These comparisons pool entrant types and do not separate market size from entrant composition.

Prior industry. Table A.7 reports hires' prior industries. Prior jobs in tradables account for 35.41 percent of hires and prior jobs in nontradables for 17.92 percent, of which 10.60 percent of

Table A.5: Residence transitions and prior workplace of entrant hires

Residence before hire	Residence after hire	Prior workplace	Hires	Share of all hires (%)
Outside CZ	Outside CZ	Outside CZ	83,400	15.11
Outside CZ	Outside CZ	Inside CZ	27,700	5.02
Outside CZ	Outside CZ	No observed job	35,600	6.45
Outside CZ	Inside CZ	Outside CZ	37,700	6.83
Outside CZ	Inside CZ	Inside CZ	12,000	2.17
Outside CZ	Inside CZ	No observed job	29,300	5.31
Inside CZ	Outside CZ	Unresolved	7,250	1.31
Inside CZ	Inside CZ	Outside CZ	51,400	9.31
Inside CZ	Inside CZ	Inside CZ	237,000	42.95
Inside CZ	Inside CZ	No observed job	30,500	5.53
Total			551,850	100.00
Among continuing CZ residents (318,900 hires)				
Share with no observed prior job (%)				9.56

Notes: Low- and high-skill entrants and all CZ-size classes are pooled. Residence is measured one year before and one year after hire; workplace refers to the prior job. The ten joint categories are mutually exclusive and exhaustive. Counts are disclosure-rounded, and shares are calculated from those counts, so they may not sum to 100. The final line divides the 30,500 hires with no observed prior job by the 318,900 continuing CZ residents. No observed prior job does not distinguish unemployment from nonparticipation.

Table A.6: Recruitment and residence by CZ size

CZ size	Hires	Continuing CZ residents (%)	Moved into CZ (%)	Lives outside CZ after hire (%)
Small	106,150	62.27	13.09	24.63
Medium	171,900	60.03	15.42	24.55
Large	273,800	54.64	14.10	31.26

Notes: Entrant types are pooled within each CZ-size class. Each percentage uses all hires in its row as the denominator. Continuing residents live in the event CZ one year before and one year after hire; movers live outside it before and inside it after; the final category lives outside it after hire. Counts are disclosure-rounded. Small, medium, and large are the released size labels; population cutoffs are not released.

all hires came from nontradable employers inside the event CZ. A further 17.45 percent have no observed prior job.

Earnings. For released group g with n_g hires, let q_g^+ and q_g^- be the after-hire and prior annual earnings quasi-medians, and let $\mathcal{O}(\mathcal{G})$ be the groups in a set $\mathcal{G}$ with observed prior earnings. The

Table A.7: Prior industry and workplace of entrant hires

Prior industry / job status	All hires (%)	Prior job inside CZ (%)	Prior job outside CZ (%)
Low-skill nontradables	13.06	7.32	5.74
High-skill nontradables	4.87	3.28	1.59
Low-skill tradables	15.42	10.40	5.01
High-skill tradables	19.99	12.78	7.21
Other observed industries	29.22	17.31	11.91
No observed prior job	17.45	—	—

Notes: Entrant types and worker skills are pooled. Each percentage uses all 550,700 hires in the prior-industry tabulation as its denominator; this sample is separate from the 551,850 hires in the residence tabulation. Low and high skill refer to the prior employer's industry. The inside- and outside-CZ columns split each observed-industry row geographically. Counts are disclosure-rounded.

reported after-hire earnings and earnings increase are

$$Q_{\mathcal{G}}^{+} = \frac{\sum_{g\in\mathcal{G}} n_g q_g^{+}}{\sum_{g\in\mathcal{G}} n_g}, \qquad D_{\mathcal{G}} = \frac{\sum_{g\in\mathcal{O}(\mathcal{G})} n_g (q_g^{+} - q_g^{-})}{\sum_{g\in\mathcal{O}(\mathcal{G})} n_g}. \tag{A.1}$$

These are count-weighted averages of group quasi-medians, not medians of individual earnings changes. Among the 454,600 hires with observed prior earnings, after-hire earnings average $85,500 and the increase $22,500. The increase is $18,400 for hires whose prior job was in the CZ and $29,300 for hires whose prior job was outside it (Table A.8). The 96,100 hires with no observed prior job earn $72,200 after hire and have no reported increase. The increases can reflect hours as well as pay rates.

Table A.8: Earnings after hire and observed earnings increases

Hire group	Hires	2023 $ thousands After hire	Increase
All hires with observed prior earnings	454,600	85.5	22.5
Prior workplace inside CZ	281,350	84.9	18.4
Prior workplace outside CZ	173,250	86.4	29.3
Low-skill workers, observed prior earnings	299,600	61.7	18.3
High-skill workers, observed prior earnings	155,000	131.4	30.7
No observed prior job	96,100	72.2	—

Notes: Entrant types are pooled. Earnings are annual, in thousands of 2023 dollars, and are count-weighted averages of released group quasi-medians; increases are the corresponding averages of after-minus-before quasi-medians. The prior-workplace rows and the worker-skill rows are alternative splits of the same 454,600 hires. Missing prior earnings are not treated as zero. Counts are disclosure-rounded; the earnings sample contains 550,700 hires.

Payroll by residence. Table A.9 splits the payroll shares in Table 3 by worker skill. Of the $69.035 billion in entrant payroll, 71.69 percent is paid to workers living inside the event CZ one year after hire. The inside-CZ total combines continuing residents and movers.

Table A.9: Entrant payroll by post-hire residence and worker skill

	Annual payroll (2023 $ billions)			
Residence one year after hire	Low-skill workers	High-skill workers	Total payroll	Share of total payroll (%)
Inside the CZ	23.852	25.639	49.491	71.69
Outside the CZ	9.284	10.260	19.544	28.31
Total	**33.136**	**35.899**	**69.035**	**100.00**

Notes: Low and high skill refer to individual workers; entrant types are pooled. Residence is measured one year after hire. Payroll is annual, in billions of 2023 dollars, from released earnings sums. These are gross payments by post-hire residence, not additional resident income.

The resident-income accounting. Fact 1 allocates each $100 of entrant payroll among workers living outside the CZ, the prior earnings of continuing residents, and new resident income. This paragraph states the object being approximated and the assumptions the approximation uses. For entrant e, let H be its hire cohort and $P^H_{e,0}$ the entrant's payroll to that cohort in a reference year. For horizon h, let $\mathcal{W}^-_h$ and $\mathcal{W}^+_h$ be annual earnings windows before and after hire, r^-_h and r^+_h the residence reference quarters, $Y^{\text{cov}}_{i,\mathcal{W}}$ worker i's covered earnings from all employers in window $\mathcal{W}$, and $Y^e_{i,\mathcal{W}}$ earnings paid by the entrant. The change in resident earnings and the entrant payroll paid to outside residents are

$$\Delta Y^{H,R}_{e,h} \equiv \sum_{i \in H} \left[\mathbf{1}\{R_{i,r^+_h} = C_e\} Y^{\text{cov}}_{i,\mathcal{W}^+_h} - \mathbf{1}\{R_{i,r^-_h} = C_e\} Y^{\text{cov}}_{i,\mathcal{W}^-_h} \right], \tag{A.2}$$

$$P^{H,C}_{e,h} \equiv \sum_{i \in H} \mathbf{1}\{R_{i,r^+_h} \neq C_e\}\, Y^e_{i,\mathcal{W}^+_h}, \tag{A.3}$$

and the resident-earnings ratio is $\iota^{\text{earn}}_{e,h} \equiv \Delta Y^{H,R}_{e,h} / P^H_{e,0}$. Equation (A.2) requires earnings classified jointly by residence before and after hire. The released tabulations classify residence (Table A.5) and earnings (Table A.8) separately, so the $100 allocation in Fact 1 approximates it under three assumptions. First, each residence–workplace category's earnings equal the released quasi-median for its prior workplace, pooling prior industries and worker skills within entrant type. Second, continuing residents with no observed prior job had zero prior earnings. Third, prior and after-hire allocations are each scaled to match the released payroll totals by entrant type and post-hire residence. The approximation omits replacement hiring at the employers the hires left and any wage and employment response elsewhere in the CZ, which is why Fact 1 presents its $32 as a lower bound on the resident income an entry creates.

No double counting. The accounting above enters the model exactly once. In Section 4, entrant jobs enter tradable labor demand, the resulting wages accrue to the workers who fill them, and their earnings pass once through the housing and service shares and the 70/30 home/work shopping split, with inbound commuters' home-origin spending and outbound commuters' workplace-origin spending leaving the local account. Adding the Fact 1 allocation as a second spending shock would count the same payroll twice. The model's recruitment of entrant workers is an equilibrium outcome rather than an imposed share.

Education and reasons for moving. The higher migrant share at high-skill entrants in Section 2 is consistent with public evidence on why workers move. The CPS ASEC for 2007–2019 records each respondent's main reason for moving. For respondents aged 25–64, I estimate survey-weighted linear probability models of five mobility outcomes on an indicator for a bachelor's degree, controlling for survey year, five-year age group, sex, race and ethnicity, and marital status, and report average standardized predictions by education group with confidence intervals from the 160 CPS replicate weights (Table A.10). College graduates are no more likely to move in a given year (11.44 versus 11.35 percent), but they are 2.07 times as likely to move for a new job or job transfer (1.87 versus 0.90 percent), and among movers 16.41 percent of graduates cite a new job against 8.01 percent of others. Education is associated not with whether people move but with whether the move is tied to a job.

Table A.10: Educational attainment and reasons for residential mobility

Outcome	Below BA (%)	BA+ (%)	Difference pp [95% CI]	Rate ratio [95% CI]
Any residential move	11.35	11.44	0.10 [−0.07, 0.26]	1.01 [0.99, 1.02]
Employment-related move	2.05	3.24	1.18 [1.09, 1.28]	1.58 [1.53, 1.63]
New job or job transfer	0.90	1.87	0.97 [0.90, 1.03]	2.07 [1.98, 2.17]
Employment reason, among movers	18.20	28.38	10.18 [9.41, 10.94]	1.56 [1.51, 1.61]
New job or transfer, among movers	8.01	16.41	8.40 [7.82, 8.98]	2.05 [1.96, 2.14]

Notes: Public-use CPS ASEC, 2007–2019, respondents aged 25–64. BA+ denotes a bachelor's degree or higher; associate degrees are in the below-BA group. The first three rows use all respondents as the denominator; the last two condition on having moved. Rates are average standardized predictions from survey-weighted linear probability models controlling for year, age group, sex, race and ethnicity, and marital status. Confidence intervals use CPS successive-difference replicate weights.

A.4 Incumbent adjustment

Source establishments. For hires who previously worked in the CZ, the source establishment is the most recent primary job with real quarterly earnings above the earnings threshold; a source spell may bridge up to two quarters of nonemployment but not an intervening job. An episode (e, s) pairs an entrant e with a source s that lost $D_{es} \geq 1$ workers to it. Let τ_{es} be the first departure year and $k = t - \tau_{es}$ event time. Exposure is the share of baseline employment lost, $x_{es} = D_{es}/E_{s,\tau_{es}-2}$, and the figures compare the highest-exposure group (Q3) with establishments that lost no workers

to an entrant. For group q with episodes $\mathcal{G}_q$, let $\mathcal{A}_q(k)$ be the episodes still active at k and $P_{s,t}$ annual payroll. The indices, each equal to 100 in year -1, are

$$I_q(k) = 100\,\frac{\sum_{\mathcal{G}_q} E_{s,\tau_{es}+k}}{\sum_{\mathcal{G}_q} E_{s,\tau_{es}-1}}, \qquad I_q^A(k) = 100\,\frac{\sum_{\mathcal{A}_q(k)} E_{s,\tau_{es}+k}}{\sum_{\mathcal{A}_q(k)} E_{s,\tau_{es}-1}},$$
$$W_q^A(k) = 100\,\frac{\sum_{\mathcal{A}_q(k)} P_{s,\tau_{es}+k} / \sum_{\mathcal{A}_q(k)} E_{s,\tau_{es}+k}}{\sum_{\mathcal{A}_q(k)} P_{s,\tau_{es}-1} / \sum_{\mathcal{A}_q(k)} E_{s,\tau_{es}-1}}. \tag{A.4}$$

I_q includes establishments that close, which then contribute zero; I_q^A and W_q^A follow establishments that remain open.

Every establishment in the sample exists at year 0, so I_q rises mechanically before year 0 as establishments open and falls after it as they close, for exposed and unexposed establishments alike. The informative comparison is therefore between the exposed and unexposed paths at the same event time, not the level of either. At year seven, the full-sample index for the highest-exposure tradable sources is 66.97, close to the unexposed path, while their active-establishment employment is 99.72 and average pay 114.10 (Figure 1). For nontradable sources the three indices are 78.45, 104.80, and 114.10, and for other sectors 65.08, 97.20, and 133.20 (Figure A.2). Establishments that lost workers to an entrant and stayed open kept their headcount and paid more. These are descriptive paths without confidence intervals, and establishments that lost workers may differ from the unexposed group in ways other than exposure.

Incumbent industries. The industry event study asks whether incumbent industries that are closely tied to the entrant change differently from other industries in the same CZ. For entrant e with industry n_e, realized jobs J_e, event year τ_e, and CZ c_e, the sites are $c \in \{c_e\} \cup \mathcal{M}(e)$, where $\mathcal{M}(e)$ are matched control CZs with no sample entry within ± 5 years, and $T_{ec} = \mathbf{1}\{c = c_e\}$. Incumbent industries s (three-digit NAICS) are those present at $k = -2$, excluding the entrant's parent, with baseline employment $E_{c_e s,b}$ in the treated CZ. Let $\phi_{s\to d}$ be the national share of job-to-job movers leaving industry s who enter d, averaged over the three pre-entry years, and ι_{sn} the sum of supplier and customer shares between s and n in the 2017 BEA use table. Labor and input–output relatedness and the two treatment intensities are

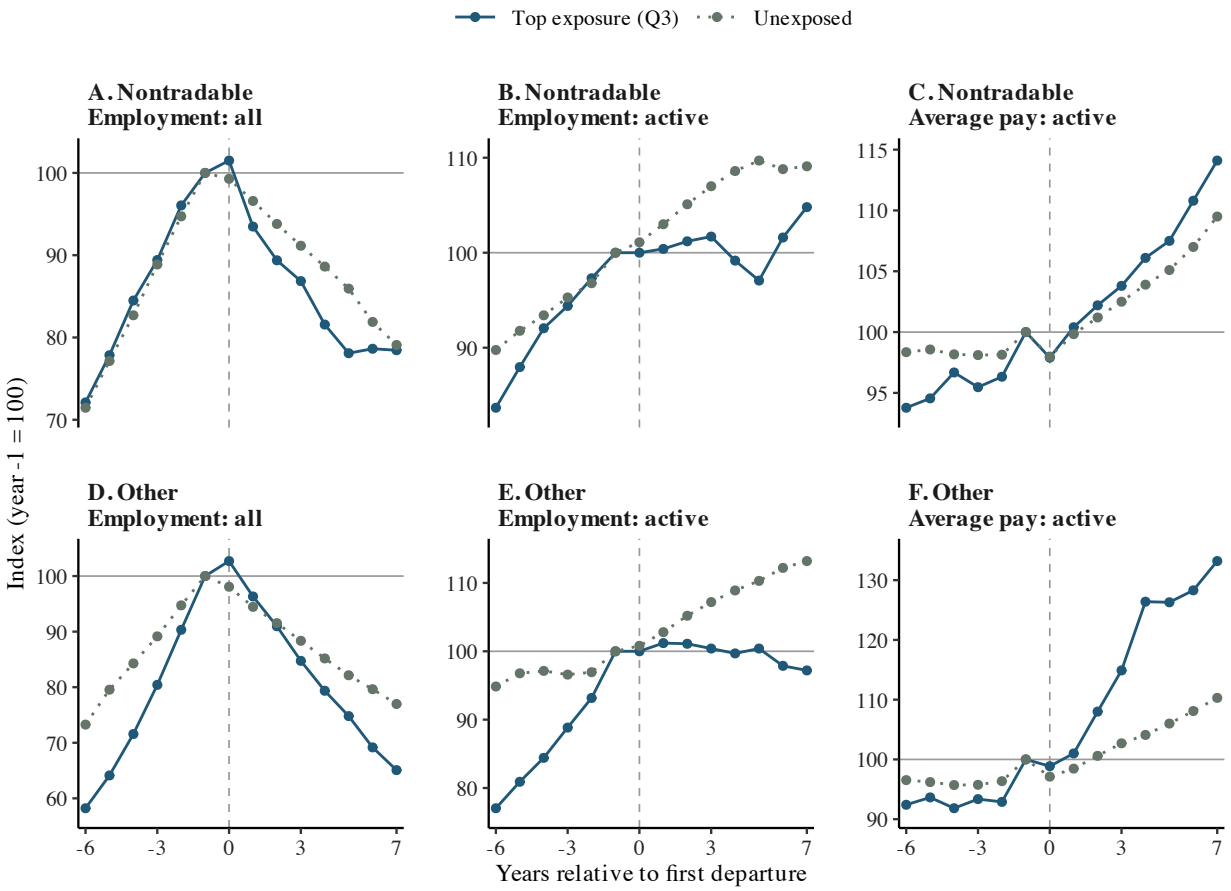
$$r_{se}^L = \phi_{s\to n_e} + \phi_{n_e\to s}, \quad r_{se}^{IO} = \iota_{sn_e}, \qquad \mathcal{D}_{es}^m = \frac{J_e\, r_{se}^m}{E_{c_e s,b}}, \quad \tilde{\mathcal{D}}_{es}^m = \frac{\mathcal{D}_{es}^m}{\mathrm{IQR}(\mathcal{D}^m)}, \quad m \in \{L, IO\}, \tag{A.5}$$

with the interquartile range taken across treated industries $s \neq n_e$. For log employment, log payroll, and log earnings per worker Y_{ecst},

$$Y_{ecst} = \alpha_{ecs} + \lambda_{ect} + \mu_{est} + \sum_{k\neq -2} \mathbf{1}\{t - \tau_e = k\}\, T_{ec}\big(\beta_k^L \tilde{\mathcal{D}}_{es}^L + \beta_k^{IO} \tilde{\mathcal{D}}_{es}^{IO}\big) + \varepsilon_{ecst}. \tag{A.6}$$

The CZ-by-year effects λ_{ect} absorb shocks common to all industries in a CZ, and the industry-by-year effects μ_{est} absorb each industry's national path through the control CZs, so β_k^L is the

Figure A.2: Nontradable and other source-establishment trajectories by exposure

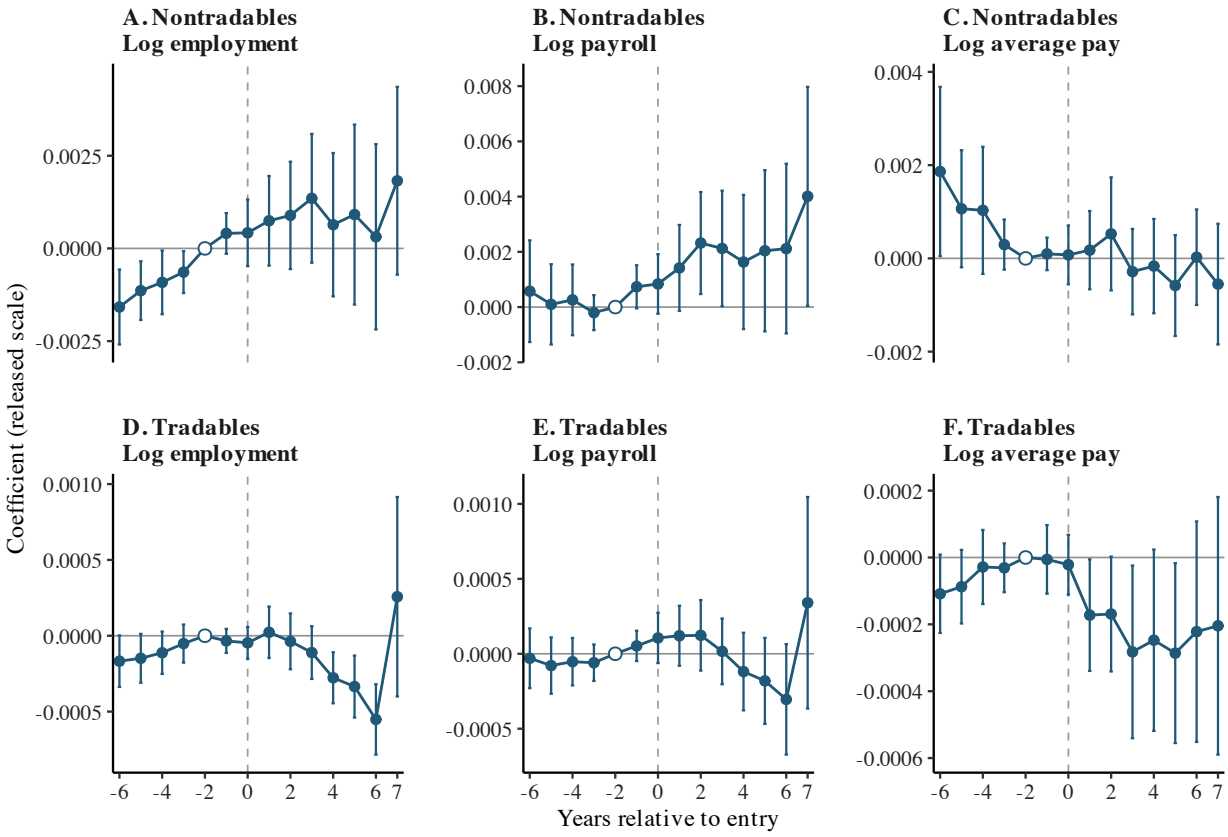


Notes: Released paths for nontradable (top row) and other sectors (bottom row). Groups, indices, and display conventions follow Figure 1.

differential change at event time k for an incumbent industry one interquartile range more labor-related to the entrant, holding input–output relatedness fixed, and β_k^{IO} the converse. The regression excludes $s = n_e$, weights by $\omega_{ec}E_{c_e s,b}$ with $\omega_{ec} = 1$ for the treated CZ and $1/|\mathcal{M}(e)|$ for each control, and clusters by event CZ and incumbent industry. The release covers 80 CZs.

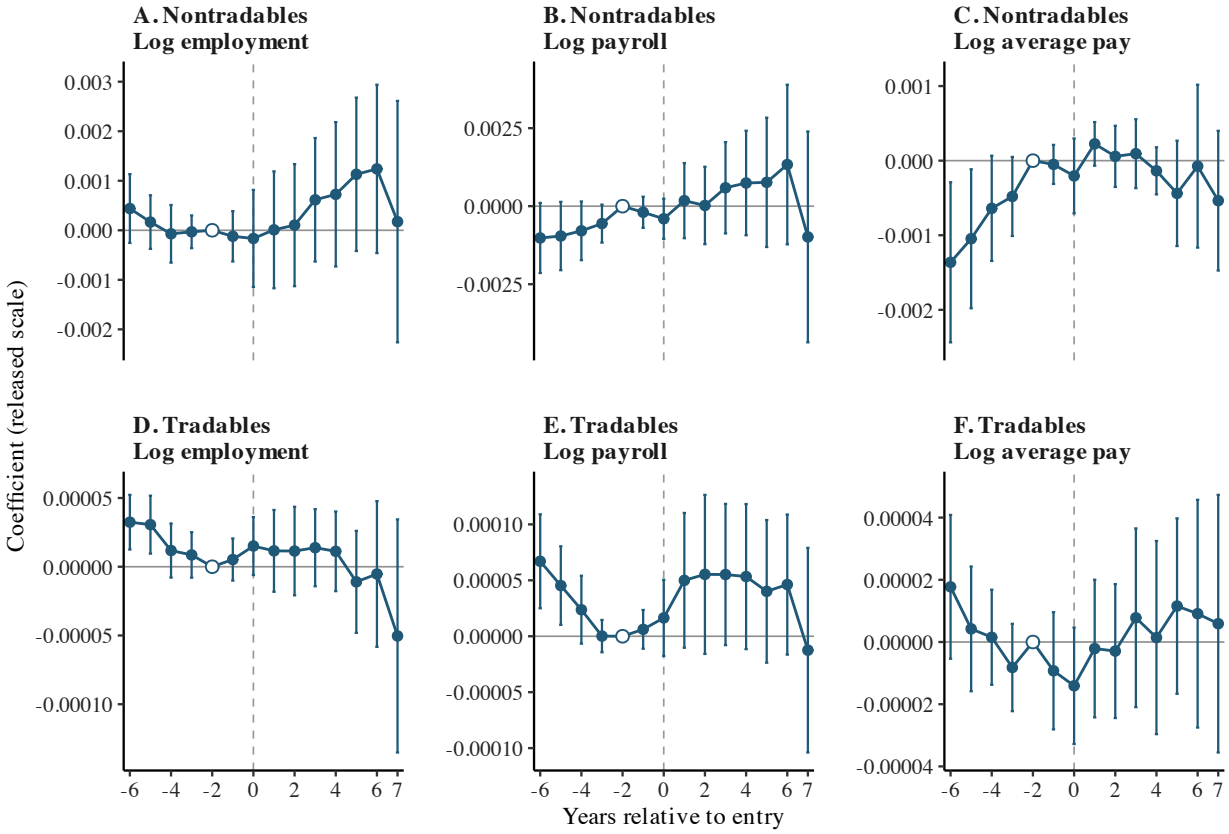
Figures A.3 and A.4 plot the labor and input–output coefficients from the joint specification. For tradable industries, the labor-exposure employment coefficients are negative in years four through six, with confidence intervals below zero, and the year-seven interval includes zero. Nontradable payroll's labor-exposure coefficient is positive in years two, three, and seven. Several pre-entry intervals also exclude zero, in particular for nontradable labor exposure and tradable input–output exposure, so these paths are suggestive rather than clean event studies. They are within-CZ contrasts across industries and do not measure the total change in incumbent employment.

Figure A.3: Joint industry exposure: labor component



Notes: Labor coefficients β_k^L from equation (A.6), controlling for input–output exposure. Outcomes are log employment, log payroll, and log average pay for nontradable and tradable incumbent industries. Points and bars are released coefficients and 95 percent confidence intervals on the interquartile-range exposure scale. Year −2 is omitted; the dashed line marks entry; each panel has its own scale. 80 CZs.

Figure A.4: Joint industry exposure: input–output component



Notes: Input–output coefficients β_k^{IO} from equation (A.6), controlling for labor exposure. Conventions follow Figure A.3.

A.5 Fact 2: Shopping geography and openness

Service expenditure shares. The 2024 BEA PCE Bridge maps each NIPA consumption category to input–output commodities and reports producers' values, transportation costs, and wholesale and retail margins. I define service expenditure N as household consumption of educational services (NAICS 61), health care and social assistance (62), accommodation and food services (72), and other services (81), each at producers' value, plus the retail margin on goods sold by retailers (44–45). Using the margin rather than the full purchase price keeps the manufactured value of goods out of retail. This gives $N = \$9.04$ trillion in 2024, or $N/Y^d = 0.41$ of disposable personal income Y^d, which was \$21.92 trillion (NIPA Table 2.1). Housing $H = \$3.15$ trillion is the PCE housing-services category, including imputed rent on owner-occupied housing; a \$3.29 billion overlap with group housing is removed. The share of after-housing income spent on services is

$$s_N^{AH} = \frac{N - 0.00329}{Y^d - H} = \frac{9.04}{21.92 - 3.15} = 0.48,$$

or 49.34% if utilities are also treated as housing. These are average national shares of total household consumption, including services purchased on households' behalf by employers and governments, not marginal propensities, and the sectors are locally oriented rather than perfectly nontraded.

Openness at county and CZ boundaries. Figure A.5 compares, for each county and CZ, the share of residents' primary jobs located inside the area with the share of residents' service visits that end inside it. The commuting share uses 2019 LODES resident workers; the visit share uses raw 2019 Veraset visits in retail, selected real estate, education excluding colleges, food services, and other services. Neither is a dollar share.

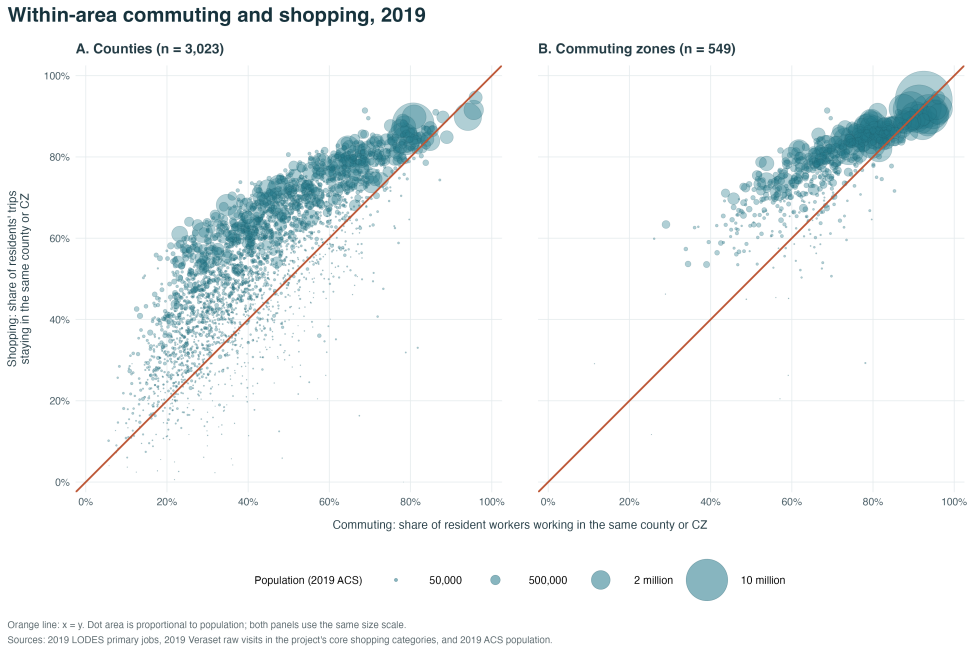
The entrant-level comparison in Section 2 uses the 180 verified award episodes at 167 workplace tracts in 86 CZs, with 4 episodes more than 100 km from their CZ centroid excluded. Let F_{ie,t_e+5} be the number of primary-job workers living in tract i and employed in the event tract e five years after the award. The share living inside boundary B is

$$S_e(B) = \frac{\sum_{i \in B} F_{ie,t_e+5}}{\sum_i F_{ie,t_e+5}}, \tag{A.7}$$

for the event county, the CZ, and the tracts within 60 driving minutes of the event tract. These are post-award levels for all workers in the event tract, not the entrant's own workforce. Entrant industries are classified as high skill when their four-digit industry's college share in the 2000 Census is at or above the national median of 20.66%, giving 132 high-skill and 46 low-skill episodes. To separate accessible population from administrative geography, I also compute the 2019 population within 40 driving minutes of the event tract,

$$P_e^{40} = \sum_i \text{Pop}_{i,2019}\ \mathbb{1}\{\tau_{ie} \leq 40 \text{ minutes}\}, \tag{A.8}$$

Figure A.5: Within-area commuting and shopping, 2019



Notes: Each dot is a county (3,023) or CZ (549). The horizontal axis is the share of resident primary-job workers in LODES 2019 who work inside the area; the vertical axis is the share of residents' raw Veraset visits in NAICS 44, 45, selected 53, 61 (excluding 6113), 722, and 81 that end inside it. Cross-county flows within a CZ count as internal. Dot area is proportional to 2019 population; the line is equality.

and split episodes at common terciles (448,246 and 1,735,422 residents). Table A.11 reports the shares. In the lower and middle terciles, high-skill sites keep a larger share of their workers inside the county than low-skill sites, the reverse of the unconditional comparison, so the aggregate skill contrast partly reflects where each type of entrant locates. The upper tercile has only 2 low-skill events.

Where within-CZ visits go. Table 4 in Section 3 splits within-CZ visits into three mutually exclusive destinations. A same-tract visit ends in the device's home tract; an adjacent-tract visit ends in a different 2020 tract sharing a boundary point with the home tract; a nonlocal hub visit ends in a shopping-hub footprint outside the home tract and its neighbors. For the core sectors (NAICS 44–45, 722, 81), 10.60 percent of within-CZ visits stay in the home tract, 23.81 percent go to an adjacent tract, and 29.25 percent to a nonlocal hub. Hubs of all kinds receive 43.40% of within-CZ visits; 14.10% of within-CZ visits are to a hub that is also in the home tract or an adjacent tract, so 67.50% of hub visits are away from home. Across all destinations, 84.80% of core-sector visits stay in the home CZ. The core combines NAICS 44, 45, selected NAICS 53, restricted NAICS 61, NAICS 722, and NAICS 81. Selected NAICS 53 consists of 5311 (Lessors of Real Estate), 5312 (Offices of Real Estate Agents and Brokers), 5313 (Activities Related to Real Estate), 5321 (Automotive Equipment Rental and Leasing), 5322 (Consumer Goods Rental), and 5323 (General Rental Centers). Restricted NAICS 61 includes Educational Services other than 6113

Table A.11: Share of workers living inside the boundary, by nearby population and entrant industry

Entrant industry	Events	County	CZ	60-minute area
		Mean share in percent (SD in percentage points)		
All retained events				
High-skill	132	44.60 (18.50)	78.50 (11.40)	83.50 (7.50)
Low-skill	46	48.50 (15.20)	75.10 (12.20)	77.70 (9.90)
Lower tercile: at most 448,246 residents				
High-skill	25	57.60 (12.90)	79.60 (8.80)	81.40 (7.70)
Low-skill	34	50.50 (15.70)	77.00 (10.10)	76.10 (10.10)
Middle tercile: 448,246–1,735,422 residents				
High-skill	49	45.60 (19.00)	81.20 (9.50)	81.80 (8.50)
Low-skill	10	42.20 (13.00)	70.80 (15.20)	81.00 (8.60)
Upper tercile: more than 1,735,422 residents				
High-skill	58	38.30 (17.20)	75.70 (13.20)	85.90 (5.70)
Low-skill	2	45.90 (16.70)	63.70 (25.50)	87.80 (4.20)

Notes: Each event receives equal weight. The denominator is all LODES primary jobs in the event workplace tract in award year +5, not the entrant workforce. Columns change only the residential boundary. Parentheses contain cross-event standard deviations, not standard errors.
Nearby population is 2019 ACS population summed across 2020 tracts whose reference points are at most 40 driving minutes *to* the event tract, irrespective of county or CZ membership. Whole-tract populations are included by reference-point membership. This is an accessible-population measure, not population density or a pre-entry covariate.
Common tercile cutoffs use all 180 retained award episodes, including 2 with unresolved industry skill classifications. Those two episodes are omitted from the displayed high/low rows. Ties stay together; displayed cutoffs are rounded to the nearest resident. The upper-tercile low-skill group has only 2 events.
High skill is at or above the nationwide median four-digit-industry college share in Census 2000; pooled industry matches are flagged in the source data. See Appendix A.5 for the sample and coverage checks.

(Colleges, Universities, and Professional Schools). Mobility records for NAICS 62 (Health Care and Social Assistance), including ambulatory health-care providers, hospitals, nursing and residential-care facilities, and social-assistance providers, are excluded for privacy reasons. NAICS 721 is not included in the core or reported separately. Table A.12 reports the cumulative driving-time distribution of all visits, including those that cross CZ boundaries.

Shopping hubs. Hubs are built from destination activity, adapting the employment-subcenter method of Giuliano and Small (1991) and McMillen (2001) to visits. Let V_j be all 2019 device-weighted core-sector visits received by tract j, and $\widetilde{V}_j = V_j + \sum_{k \in \mathcal{N}(j),\, c(k)=c(j)} V_k$ its sum with its same-CZ neighbors. A tract is a candidate peak when $\widetilde{V}_j$ is at least as large as that of every adjacent tract in its CZ. Peaks are processed from strongest to weakest, and a weaker peak is merged into a stronger one when the mean driving time between them is at most ten minutes. A hub's footprint $\mathcal{F}_g$ is the union of its peak tracts and their adjacent tracts, with shared tracts

Table A.12: Cumulative driving time of U.S. service visits, 2019

Sector	≤ 5	≤ 10	≤ 15	≤ 20	≤ 30	≤ 40	≤ 60	> 60
NAICS 44	18.99	38.11	52.51	62.61	74.86	81.16	86.82	13.18
NAICS 45	14.86	34.60	50.95	62.16	75.23	81.72	87.43	12.57
NAICS 722	16.25	34.75	49.25	59.60	72.23	78.79	84.80	15.20
NAICS 81	19.20	36.28	50.92	61.99	75.72	82.59	88.45	11.55
Core	*16.86*	*35.52*	*50.48*	*61.11*	*73.97*	*80.52*	*86.38*	*13.62*
Selected NAICS 53	9.26	22.44	35.35	45.79	59.50	66.84	73.76	26.24
Restricted NAICS 61	11.28	26.24	41.35	53.74	69.77	78.22	85.99	14.01

Notes: Percentages of annual device-weighted visits, including visits that cross CZ boundaries. Columns are cumulative minutes of tract-center driving time on the 2020 road network; same-tract visits are assigned five minutes. Pairs more than 100 straight-line miles apart, routed times above 60 minutes, and unreachable pairs are in the > 60 column. The core combines NAICS 44, 45, selected NAICS 53, restricted NAICS 61, NAICS 722, and NAICS 81. Selected NAICS 53 consists of 5311 (Lessors of Real Estate), 5312 (Offices of Real Estate Agents and Brokers), 5313 (Activities Related to Real Estate), 5321 (Automotive Equipment Rental and Leasing), 5322 (Consumer Goods Rental), and 5323 (General Rental Centers). Restricted NAICS 61 includes Educational Services other than 6113 (Colleges, Universities, and Professional Schools). Mobility records for NAICS 62 (Health Care and Social Assistance), including ambulatory healthcare providers, hospitals, nursing and residential-care facilities, and social-assistance providers, are excluded for privacy reasons. NAICS 721 is not included in the core or reported separately.

assigned to the stronger hub, and its share of destination activity is $H_g = \sum_{j \in \mathcal{F}_g} V_j / \sum_{j:c(j)=c(g)} V_j$. A hub qualifies under the 2 percent definition when $H_g \geq 0.02$; the 1 and 5 percent definitions use 0.01 and 0.05. The same footprints are used for every sector in Table 4.

Hubs and market size. Across the 446 CZs with 2019 population of at least 50,000, the number of hubs rises with population. A log-link count model of the hub count on log population gives an elasticity of 0.59 under the 1 percent definition, 0.43 under the 2 percent definition, and 0.10 under the 5 percent definition, which mechanically caps how many hubs can qualify (Table A.13).

Table A.13: Scaling of within-CZ shopping hubs with CZ population

Minimum hub share	Preferred model	Intercept	Robust SE	Elasticity	Robust SE	95% CI
1%	Negative binomial	-5.95	(0.30)	0.59	(0.02)	[0.54, 0.63]
2%	Negative binomial	-3.99	(0.32)	0.43	(0.03)	[0.38, 0.48]
5%	Poisson	-0.20	(0.30)	0.10	(0.02)	[0.06, 0.15]

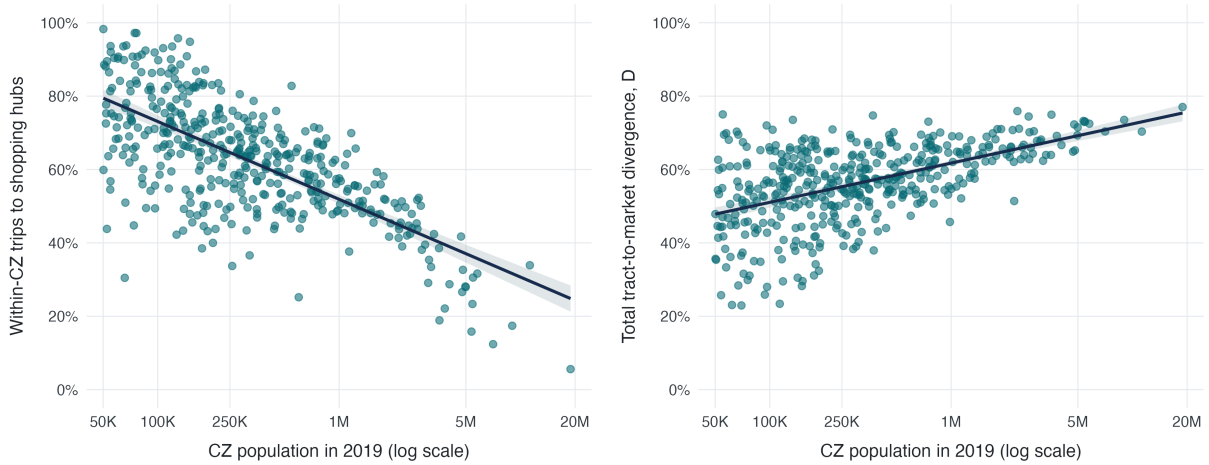
Notes: Each row reports the valid AIC-preferred count model for one hub-mass threshold. The sample contains 446 CZs with 2019 population of at least 50,000. The dependent variable is the number of within-CZ hubs. Standard errors are heteroskedasticity-robust HC1 estimates. At the 5% threshold, the negative-binomial dispersion collapses to the Poisson boundary and is excluded from model comparison.

A tract's shopping versus its market's. Fact 3 assigns exposure to the home tract and its neighbors rather than through the observed shopping network, on the grounds that the home-and-adjacent component is the part of shopping that differs across tracts. This paragraph measures that. For home tract i in CZ c, let p_{ij} be the share of its device-weighted within-CZ visits ending in tract j, and let

$$q_{-i,j} = \frac{\sum_{k \in c,\, k \neq i} v_{kj}}{\sum_{k \in c,\, k \neq i} \sum_{\ell \in c} v_{k\ell}} \tag{A.9}$$

be the destination shares of all other residents of the CZ. The total-variation distance $D_i = \sum_j \max\{p_{ij} - q_{-i,j}, 0\}$ is the share of tract i's visits that would have to be reassigned to make its pattern match its market's. Across 438 CZs with population of at least 50,000, 34.40% of within-CZ visits end in the home tract or an adjacent tract, against 3.20% under the other residents' pattern, and the home-and-adjacent set accounts for 48 percent of the pooled positive deviation $\sum_i V_i D_i$ (weighting origins by their visits V_i). Figure A.6 shows how the two parts of this vary with market size. The share of within-CZ visits going to 2 percent hubs falls from three quarters in the smallest tract-count decile to two fifths in the largest (correlation -0.70 with log population), while the visit-weighted mean of D_i rises from about one half to two thirds (correlation 0.52). Small markets send most visits to a few shared hubs, so tracts look like their markets; large markets spread visits over many destinations, and tracts differentiate.

Figure A.6: Shopping hubs and total tract-to-market divergence by market size



Notes: Each dot is one of the 438 CZs with 2019 population of at least 50,000 and positive within-CZ shopping support. Left: share of residents' within-CZ device-weighted visits received by 2 percent hub footprints. Right: visit-weighted mean of D_i across the CZ's home tracts. Lines are OLS fits in log population with pointwise 95 percent HC1 bands.

Shopping and commuting gravity. This paragraph documents the comparison of travel-time decay behind Figure 2 and equation (1). It is distinct from the sector-by-CZ shopping gravity used to quantify the model in Section 5, which adds same-tract and same-county terms. For shopping, F^S_{ijc} is the number of 2019 device-weighted visits from residents of i to establishments in j, pooling

NAICS 44–45, 5311–5313, 5321–5323, 61 (excluding 6113), 722, and 81. For commuting, F^C_{ijc} is the 2019 LODES count of workers living in i and working in j. For each mode m and CZ c, Poisson pseudo-maximum likelihood solves

$$(\widehat{\alpha}^m_c, \widehat{\delta}^m_c, \widehat{\beta}^m_c) = \arg\max_{\alpha,\delta,\beta} \sum_{(i,j)\in\mathcal{J}_c} \left[F^m_{ijc} \log \mu^m_{ijc} - \mu^m_{ijc}\right], \qquad \mu^m_{ijc} = \exp\left(\alpha^m_{ic} + \delta^m_{jc} + \beta^m_c t_{ijc}\right), \quad \text{(A.10)}$$

with standard errors two-way clustered by origin and destination tract. PPML accommodates the many tract pairs with zero flows, which carry information about decay, and avoids the retransformation problems of log-linear OLS (Silva and Tenreyro, 2006). Both modes use the same tract-pair support: all non-water pairs within a CZ with finite driving time of at most 180 minutes, with same-tract pairs at five minutes and absent flows entered as zero. A CZ needs at least ten candidate pairs and ten positive-flow pairs. The shopping estimator converges in 542 CZs and the commuting estimator in 530; the comparison uses the 526 CZs with both. Because travel time enters in levels, I report the exact percentage decline per minute, $d^m_c = 100[1 - \exp(\widehat{\beta}^m_c)]$. Table A.14 reports the coefficient moments; the mean declines are 8.26% for shopping and 5.92% for commuting, and the cross-CZ correlation of the two coefficients is 0.86.

Table A.14: CZ-level shopping and commuting decay estimates

Flow	Converged CZ fits	Mean $\widehat{\beta}_c$	Median $\widehat{\beta}_c$	Median decline per minute
Shopping	542	−0.09	−0.08	8.08%
Commuting	530	−0.06	−0.06	6.10%

Notes: Coefficient moments and exact declines are computed on the 526-CZ matched comparison sample; the converged-fit column counts estimates available before matching the two modes.

The estimates are in line with the literature. Ahlfeldt et al. (2015) estimate a commuting semi-elasticity of about 0.07 per minute in Berlin, against the median 6.10% here, and Monte et al. (2018) estimate a county-to-county distance elasticity of −4.43. For consumption trips, Miyauchi et al. (2021) estimate travel-time elasticities of −1.82 from home and −1.81 from work in Tokyo, and Oh and Seo (2023) find that ten additional minutes reduce service-trip flows by about 80 percent in Seoul, against 57.00% here. The comparison is descriptive. Identification of β^m_c requires that travel time be uncorrelated with omitted pair-specific determinants of flows conditional on the fixed effects, and the visit data count trips rather than dollars. In the model the two coefficients map to different structural objects, $\beta^S_c = (1 - \sigma_N)\kappa$ and $\beta^C_c = -\chi\kappa$, which is why the comparison is made on the reduced-form declines rather than on the parameters.

A.6 Fact 3: Residential exposure designs

Fact 3 compares service employment in the tracts most and least exposed to entrant workers' spending. This section gives the alternative exposure measures and specifications estimated on the restricted data, the public analog built from predicted residences, and the reasons the observed

shopping network is not used as exposure.

The home-and-adjacent kernel. Every exposure measure in this section places each hire's spending at a home tract i and spreads it over $\mathcal{N}(i)$, the home tract and all 2020 tracts sharing a boundary point with it, with weights K_{ij} that assign 30 percent to the home tract and share 70 percent equally among the neighbors,

$$K_{ij} \geq 0, \qquad \sum_{j \in \mathcal{N}(i)} K_{ij} = 1, \qquad K_{ij} = 0 \text{ if } j \notin \mathcal{N}(i), \tag{A.11}$$

so each hire is counted once across tracts. The 30/70 split is the share of visits to the home tract among visits that end in the home tract or an adjacent tract (Appendix A.6). The entrant's workplace tract is excluded from every outcome sample.

Stacked new-income exposure. The first alternative ranks tracts by new income rather than by workers. New income is the earnings increase over the previous job for previously employed local hires and the full entrant wage for migrants and hires without a prior job. It is assigned to the hire's residence four quarters after hire, so that G_{ei} is new income received by hires living in tract i, and exposure is

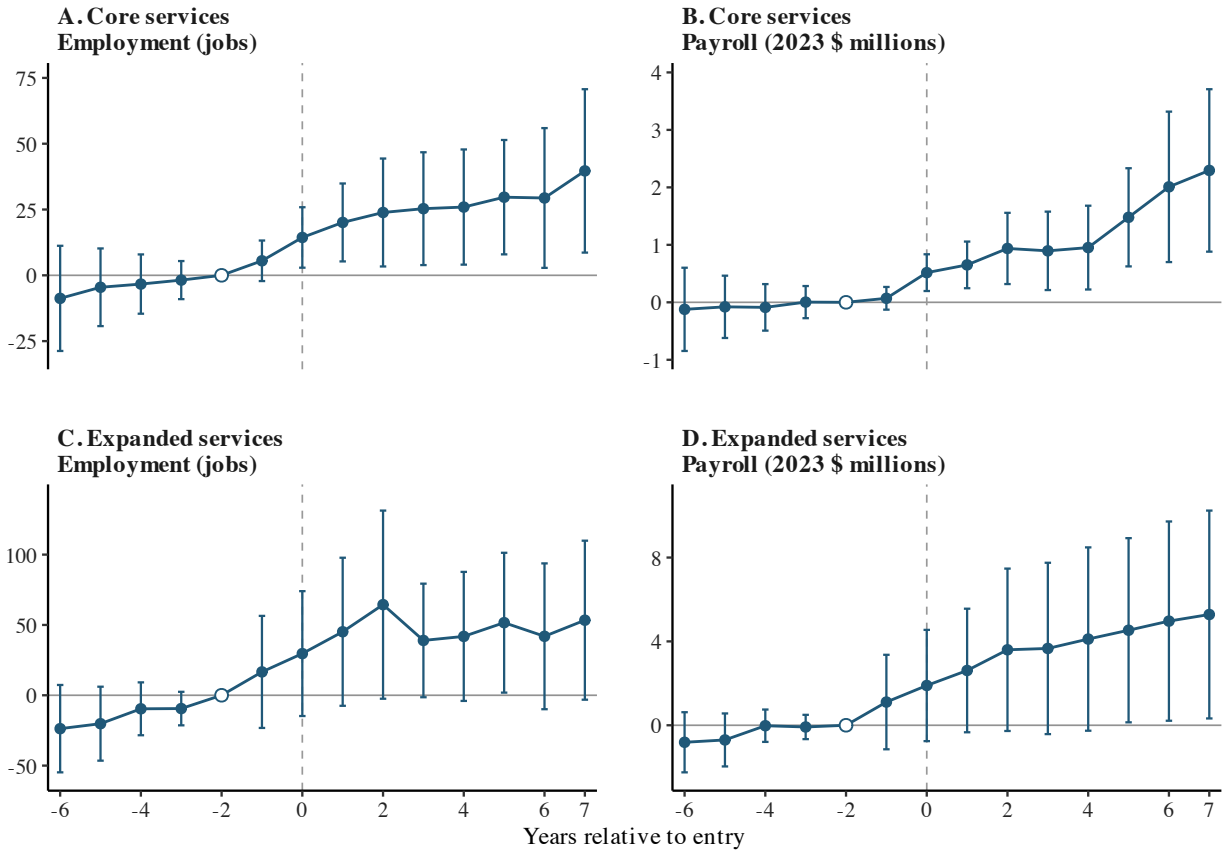
$$A^I_{ej} = \sum_i G_{ei} K_{ij}. \tag{A.12}$$

As in Fact 3, the sample is events in CZs with no other observed event, and I compare the top and bottom quartiles of A^I_{ej} within each event with equation (2). Mean exposure differs between the quartiles by \$1.166 million for the core-service outcomes (NAICS 44, 45, 722, 81) and \$1.164 million for the expanded outcomes that add 61 and 62; the release covers 30 CZs. By year seven, the core-service contrasts are 39.67 jobs and \$2.295 million in annual payroll, with both intervals above zero (Figure A.7). The expanded outcomes give 53.37 jobs and \$5.283 million, with the employment interval including zero. Pre-entry intervals include zero in all four panels. Because this measure uses post-hire residences, it can reflect moves after hiring as well as income gains at existing addresses.

Continuous exposure. The continuous specification uses all events and all tracts, including CZs with several events and the middle of the exposure distribution. For event e, let H_{ei} be the number of hires living in tract i of the event CZ four quarters after hire, with $A^W_{ej} = \sum_i H_{ei} K_{ij}$ the neighborhood allocation. Calendar-year exposure is $X^W_{jt} = 100^{-1} \sum_{e:\tau_e = t} A^W_{ej}$ for workers and $X^I_{jt} = 10^{-6} \sum_{e:\tau_e = t} A^I_{ej}$ for income. For either measure X and outcome o,

$$Y^o_{jt} = \alpha^o_j + \delta^o_{c(j)t} + \sum_{k \in \{-6,\ldots,7\} \setminus \{-2\}} \beta^o_k X_{j,t-k} + \gamma^o_- X^{\text{old}}_{jt} + \gamma^o_+ X^{\text{future}}_{jt} + \varepsilon^o_{jt}, \tag{A.13}$$

Figure A.7: Stacked service responses to neighborhood new-income exposure



Notes: Top-minus-bottom-quartile contrasts in new-income exposure from equation (2), in CZs with no other observed events. Panels A–B cover NAICS 44, 45, 722, and 81; panels C–D add 61 and 62. Income is assigned to residences four quarters after hire and spread with the 30/70 kernel. Bars are 95 percent confidence intervals clustered by CZ; year −2 is normalized to zero. Employment is in jobs and payroll in millions of 2023 dollars. 30 CZs.

with tract and CZ-by-year fixed effects, no weights, CZ-clustered standard errors, and stocks of exposure from events outside the window carried in X^{old} and X^{future}. Worker-exposure outcomes cover NAICS 44, 45, 722, 81, 61, and 62; income-exposure outcomes cover 44, 45, 722, and 81. Figure 4 in Section 3 reports the worker-exposure estimates. Stacked and continuous coefficients answer different questions: the stacked contrast is the change in the gap between the top and bottom quartiles in single-event CZs, while the continuous coefficient is a slope per 100 workers across the full distribution in all events, so their magnitudes are not directly comparable.

The public analog. The public design replaces identified hires with predicted residences and LEHD outcomes with LODES employment. It covers the 124 registry events dated 2011–2017 in 69 CZs.

Residences are predicted in two steps. First, for each event, the pre-entry LODES origin–destination counts $C_{hw_e,t}$ from residence tract h to the workplace tract w_e are averaged over event

years -4 through -2 and normalized within the CZ,

$$\overline{C}_{eh} = \frac{1}{3}\sum_{\ell=2}^{4} C_{hw_e,\tau_e-\ell}, \qquad p^C_{eh} = \frac{\overline{C}_{eh}}{\sum_{r\in c_e}\overline{C}_{er}}. \tag{A.14}$$

These are the residences of all workers in the tract, not of the entrant's hires. Second, because movers may settle outside the existing commuting catchment, a mover distribution p^M_{eh} is the share of the CZ's recent in-migrants aged 18–64 living in tract h, from the ACS five-year table B07001 ending in $\tau_e - 2$,

$$p^M_{eh} = \frac{M_{h,\tau_e-2}}{\sum_{r\in c_e} M_{r,\tau_e-2}}. \tag{A.15}$$

The predicted residence distribution mixes the two with a fixed weight,

$$p_{eh} = \omega\, p^C_{eh} + (1-\omega)\, p^M_{eh}, \qquad \omega = 0.84, \tag{A.16}$$

where ω is the share of CZ-resident hires who were previously employed or nonemployed in the CZ, and $1-\omega$ the share who migrated in, from the source shares behind Fact 1. The weights were fixed before any outcome was examined. Predicted residences are then scaled to the event's exposure mass N_e, the promised jobs allocated to CZ tracts by pre-entry commuting shares, on the support S_e that excludes the workplace tract and its neighbors,

$$R_{eh} = N_e\, \frac{p_{eh}\, \mathbf{1}\{h \in S_e\}}{\sum_{r\in S_e} p_{er}}. \tag{A.17}$$

Exposure then follows the home-and-adjacent rule. Let $\mathcal{A}_e(h)$ be the tracts in c_e adjacent to h. The 2019 visit data assign 0.09 of all service visits to the home tract and 0.20 to adjacent tracts; conditional on these local destinations, the home share is 0.31 and the adjacent share 0.69, which are the 30/70 weights used throughout. Destination exposure is

$$q^{HT}_{ehj} = 0.31\, \mathbf{1}\{j = h\} + \frac{0.69}{|\mathcal{A}_e(h)|}\, \mathbf{1}\{j \in \mathcal{A}_e(h)\}, \qquad E_{ej} = N_e\, \frac{\sum_h R_{eh}\, q^{HT}_{ehj}}{\sum_m \sum_h R_{eh}\, q^{HT}_{ehm}}, \tag{A.18}$$

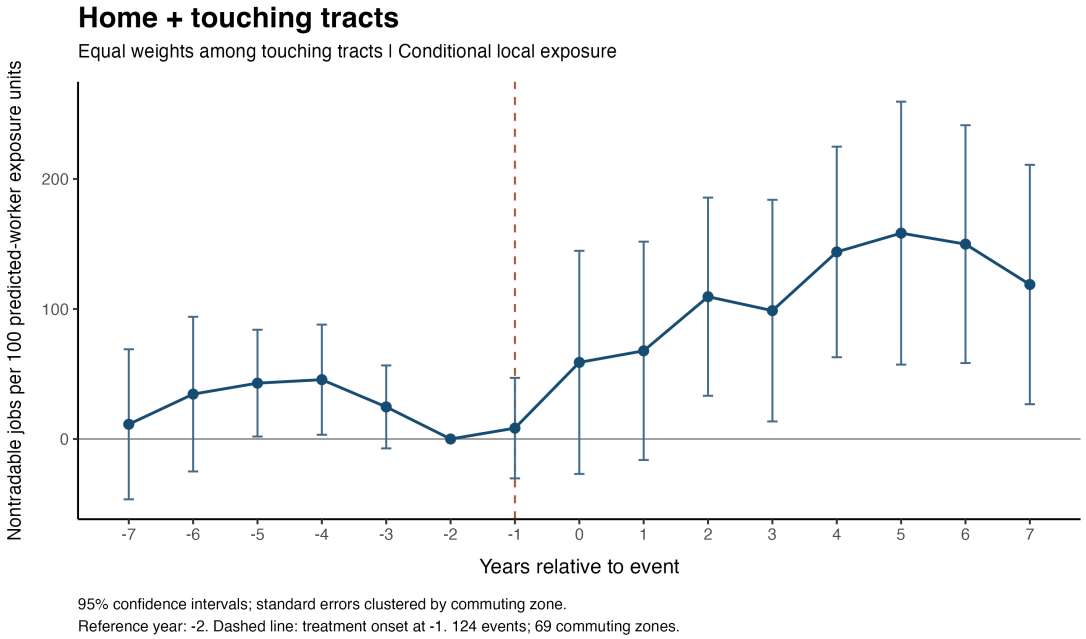
so each event's exposure mass is preserved. With $X^k_{jt} = \sum_e (E_{ej}/100)\, \mathbf{1}\{t-\tau_e = k\}$, the specification is

$$Y_{jt} = \alpha_j + \lambda_{c(j)t} + \sum_{k=-7,\, k\neq -2}^{7} \beta_k X^k_{jt} + \gamma_O O_{jt} + \gamma_F F_{jt} + \varepsilon_{jt}, \tag{A.19}$$

where Y_{jt} is LODES private service employment, α_j and $\lambda_{c(j)t}$ are tract and CZ-by-year fixed effects, and O_{jt} and F_{jt} carry exposure to events more than seven years in the past and future. The regression is unweighted and clustered by CZ, with 772,144 tract-years on 36,791 tracts over 2002–2022. The coefficient β_k is the differential employment response per 100 predicted-worker exposure units relative to year -2. Figure A.8 plots the estimates. The mean over years $+2$ through $+7$ is 129.90 jobs per 100 units (SE 41.00), and the joint test that the leads from -7 through -3 are zero

has $p = 0.06$, so the pre-entry evidence warrants caution. Like Fact 3, this is a within-CZ spatial response, not net job creation.

Figure A.8: Public home-plus-adjacent exposure and service employment



Notes: Coefficients β_k from equation (A.19), in private service jobs per 100 predicted-worker exposure units, with 95 percent confidence intervals clustered by CZ. Year -2 is normalized to zero; the dashed line marks treatment onset at -1. 124 events in 69 CZs.

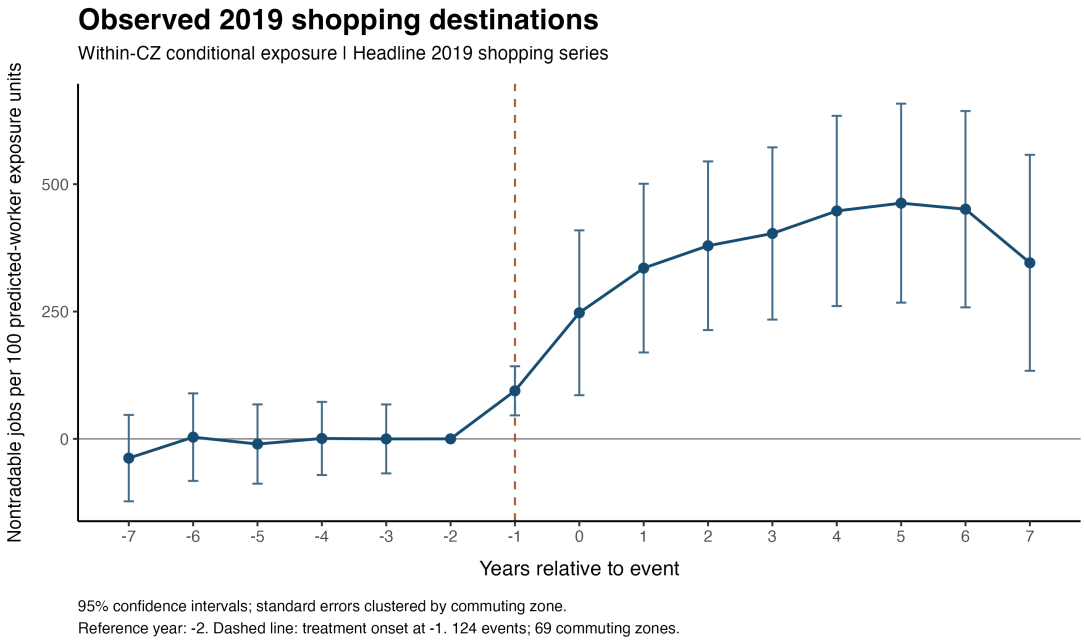
Why the observed shopping network is not used as exposure. The same predicted residences can instead be routed to destinations through the observed 2019 within-CZ visit shares, $S_{ej} = \sum_h R_{eh} s_{ehj}$, or through a gravity network built from pre-entry destination employment and travel time. Both alternatives have the same outcome, specification, and sample as the home-and-adjacent design. Two features rule them out as identifying variation. First, residents of different neighborhoods in a CZ shop at the same hubs, so exposure routed through the network lands on nearly the same destination tracts for every event in a CZ. Among the 85 entries in CZs with more than one entry, destination fixed effects alone explain 76.60% of the variation in network exposure. Second, the 2019 network is measured after the 2011–2017 events and may reflect their effects.

Figure A.9 shows what the observed network produces anyway. The mean coefficient over years +2 through +7 is 414.90 jobs per 100 predicted-worker exposure units (SE 89.20), several times the home-and-adjacent estimate, with a joint pretrend p-value of 0.37. The log and stacked versions of the same design reject the pretrend test ($p = 0.00$ and $p = 0.01$). Placebo networks that shuffle destination exposure within event, size-decile, and travel-time groups give a mean post-entry coefficient of 68.70 against the observed 414.90.[30] The coefficient measures employment growth at

[30]The placebo distribution uses 199 shuffles that preserve each event's exposure mass and destination support; a

the shared shopping destinations of each market in the years after entries, which is where entry-linked spending lands, but it cannot attribute that growth to a particular entrant. The residential design remains the identifying one.

Figure A.9: Observed 2019 shopping-destination exposure and service employment



Notes: Coefficients from equation (A.19) with exposure assigned through observed 2019 within-CZ shopping destination shares, in private service jobs per 100 predicted-worker exposure units, with 95 percent confidence intervals clustered by CZ. Year -2 is normalized to zero; the dashed line marks treatment onset at -1. 124 events in 69 CZs. Because the 2019 network postdates the events, the series is descriptive.

A.7 Aggregate and median housing shares across commuting zones

The aggregate housing share measures the fraction of a CZ's total household resources devoted to housing. The median household share describes the housing expenditure burden of the middle household. Figure A.10 compares these moments using a common sample and measurement procedure across 583 CZs in the 50 states and the District of Columbia. This national comparison uses a common 2008–2017 period; the event calibration in equation (C.18) targets the household median in each CZ's pre-entry sample.

Sample and geographic assignment. The data are the Census Bureau's national ACS five-year Public Use Microdata Samples for 2008–2012 and 2013–2017, linking housing records to their adult person records. Each release receives one half of its published survey weight. Pooling these nonoverlapping releases produces a ten-year household distribution and average population stocks. The housing sample contains 12,113,453 occupied households with at least one adult aged 18 or

restricted wild-cluster bootstrap with 999 CZ-level draws gives $p = 0.00$.

older and usable housing-service measures. Rents and incomes use the ACS adjustment factors followed by the CPI conversion to annual 2023 dollars.

Public Use Microdata Areas (PUMAs) are allocated fractionally to CZs using 2010 census population in their intersections. The allocation uses 2000 PUMAs for 2008–2011 interviews and 2010 PUMAs for 2012–2017 interviews. The former crosswalk is constructed from the Missouri Census Data Center's Geocorr PUMA–county intersections; the latter uses Census tract–PUMA relationships and tract populations. Historical county codes and merged PUMAs are reconciled before aggregation. For household m in PUMA p, let a_{pc} be the fraction allocated to CZ c. Its estimation weight is $w_{mc} = \frac{1}{2}\text{WGTP}_m a_{pc}$.

Housing services and household resources. Paying renters contribute their annual contract rent. Owner-occupied and rent-free dwellings contribute imputed rental services from a national, survey-weighted regression estimated on 3,322,577 paying renter households:

$$\log E_m^{\text{rent}} = X_m'\beta + \lambda_{p(m),t(m)} + u_m. \tag{A.20}$$

Here X_m contains log rooms and its square, bedrooms and its square, structure-type indicators, and structure-type interactions with log rooms. The fixed effects identify state, PUMA vintage, PUMA, and interview year. Imputed annual service expenditure is $E_m = \exp(X_m'\widehat{\beta} + \widehat{\lambda}_{p(m),t(m)})\widehat{s}_{b(m)}$, where $\widehat{s}_b$ is the survey-weighted mean of $\exp(\widehat{u}_m)$ among renters of structure type b. This retransformation incorporates the residual distribution when converting predicted log rents to levels.

Household resources sum income and housing-portfolio dividends over adults. An employed adult receives 95 percent of observed wage and salary income. A nonworker receives the CZ–skill mean of public assistance, Supplemental Security Income, and Social Security income among nonworkers, estimated with person weights. High skill denotes a bachelor's degree or higher. The CZ housing portfolio equals total housing-service expenditure divided by 1.28. Its skill shares are based on reported owner-occupied housing values, split equally between the adult householder and spouse or unmarried partner when both are present. Each skill's share of the portfolio is distributed equally among its represented adults in the eligible sample. Thus

$$Y_{mc} = Y_{mc}^{\text{inc}} + \sum_{k\in\{H,L\}} n_{mk} D_{ck}, \qquad D_{ck} = \frac{\omega_{ck}}{N_{ck}} \frac{\sum_m w_{mc} E_m}{1.28}, \tag{A.21}$$

where n_{mk} is the household's adult count of skill k, $N_{ck} = \sum_m w_{mc} n_{mk}$, and ω_{ck} is that skill's ownership-value share. Property tax revenue leaves the modeled economy, so residents receive no rebate. These resources implement the model's individual budget convention.

The two expenditure shares. Both statistics use the same eligible households, service bills, and resources. With $s_{mc} = E_m/Y_{mc}$, define

$$A_c = \frac{\sum_m w_{mc} E_m}{\sum_m w_{mc} Y_{mc}} = \sum_m \frac{w_{mc} Y_{mc}}{\sum_a w_{ac} Y_{ac}} s_{mc}, \tag{A.22}$$

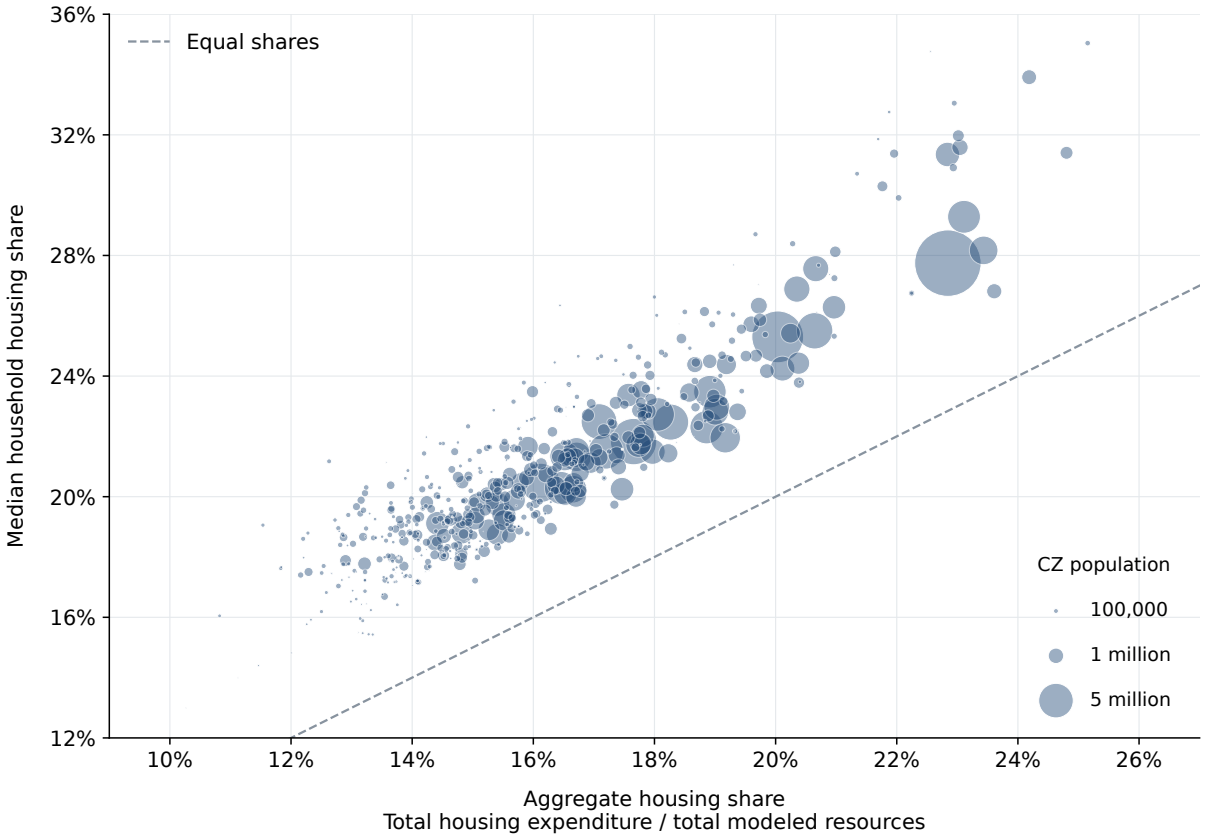
$$M_c = \inf \left\{ s : \frac{\sum_m w_{mc} \mathbf{1}\{s_{mc} \le s\}}{\sum_m w_{mc}} \ge \frac{1}{2} \right\}. \tag{A.23}$$

The aggregate A_c weights household shares by resources as well as survey weights; M_c is the household-weighted median of the pooled observations. The aggregate share describes housing's claim on total resources, while the median describes a typical household's expenditure burden. Greater resource weight on households with lower housing shares can therefore produce a lower aggregate share.

Across CZs, the median of A_c is 15.40 percent and the median of M_c is 20.20 percent. The median within-CZ gap is 4.70 percentage points, and the household median exceeds the aggregate in all 583 CZs. Each dot's area is proportional to average 2008–2017 population, obtained from all-age person records, including group quarters, with the same half-weight pooling and geographic allocation. The cross-CZ summaries give each CZ equal weight.

The comparison isolates the choice of expenditure-share statistic under common measurement. Both moments depend on the rental-service imputation and resource convention. The model targets the median household share; its baseline portfolio and housing quantities are determined jointly by the budget and market-clearing equations. A common Cobb–Douglas share generally cannot reproduce both the observed median and aggregate share. Fractional geographic assignment assumes that households in a PUMA represent its constituent CZ portions; small CZs sharing a PUMA can consequently have similar estimates. The plotted values are point estimates conditional on these imputations and geographic allocations.

Figure A.10: Aggregate and median household housing shares across U.S. commuting zones



Notes: ACS 2008–2012 and 2013–2017 five-year microdata, pooled with one half of each release's survey weights. The horizontal axis is total housing-service expenditure divided by total modeled household resources; the vertical axis is the household-weighted median expenditure share. Both statistics use identical eligible households and measurement rules. Housing services are observed contract rents or imputed rental equivalents. Resources include modeled income and housing-portfolio dividends. Dot area represents average all-age CZ population during 2008–2017. The dashed line denotes equal shares. All 583 CZs are shown; the axes have different ranges. PUMA-to-CZ locations are assigned using population weights.

B Model

This appendix derives individual demand, employment and location choices, firms' input demands, and the equilibrium accounts from the preferences and technologies in Section 4. A superscript 0 denotes the no-entry equilibrium. Appendix C describes the quantification and solution of these conditions.

B.1 Individual preferences

An individual ω of skill k chooses consumption and a joint residence and employment choice ℓ. For workers, $\ell = (i, j, g)$ specifies residence, workplace, and sector. For nonworkers it specifies residence and sector. The housing share is $\alpha_\ell = \alpha_H$ for $i \in \mathcal{J}$ and $\alpha_\ell = \overline{\alpha}_H^o$ for $i = o$. Write $\gamma_\ell = (1 - \alpha_\ell)s_N$

and $\beta_\ell = (1-\alpha_\ell)(1-s_N)$, so that $\alpha_\ell + \gamma_\ell + \beta_\ell = 1$. The utility in (3) can be written

$$U_\ell(\omega) = z_\ell(\omega)\mathcal{B}_\ell h^{\alpha_\ell} n^{\gamma_\ell} x^{\beta_\ell}, \qquad \mathcal{B}_\ell = \begin{cases} B^R_{ik} B^W_{jgk}/d_{ij}, & i,j \in \mathcal{J},\ e = W, \\ \overline{B}^{\text{out}}_{igk}, & i \in \mathcal{J},\ j = o,\ e = W, \\ \overline{B}^{\text{in}}_{jgk}, & i = o,\ j \in \mathcal{J},\ e = W, \\ \overline{B}^{\text{oo}}_{gk}, & i = j = o,\ e = W, \\ B^R_{ik} B^U_{ik}, & e = U. \end{cases} \tag{B.1}$$

Here $e = W$ denotes work and $e = U$ nonemployment. For internal commuters, $d_{ij} = \exp(\kappa t_{ij})$. The boundary amenity composites include fixed commuting costs and the attractiveness of access to outside jobs or residences. Each has at most one location index within the CZ; they preserve the separately calibrated inbound and outbound flows and the outside worker stock. Their construction is given in Appendix C.6.1. Internal commuting retains separate residence and workplace factors, with workplace amenities varying by sector. Housing and the two nonhousing goods have constant expenditure shares. After optimizing consumption and absorbing fixed expenditure-share constants into participation preference parameters, the systematic value is $\Phi_\ell = \mathcal{B}_\ell y_\ell / [(q^R_i)^{\alpha_\ell} \mathcal{P}^{\gamma_\ell}_\ell]$. The next subsection derives the employment and location probabilities; Appendix B.3 derives income, consumption demands, and this normalization.

B.2 Nested tastes and employment and location choices

Individuals choose jointly among residence and employment choices. The nested Fréchet distribution allows their tastes to be more similar among location pairs than across participation choices, sectors, and areas. Draws are independent across individuals and are held fixed across equilibria. For a nest v containing choices $a \in \mathcal{C}(v)$, let $\overline{B}_{va,k}$ be preference parameters and θ_v its shape parameter. The value index and conditional choice probability are

$$V_{vk} = \left[\sum_{a \in \mathcal{C}(v)} \overline{B}_{va,k} V^{\theta_v}_{ak} \right]^{1/\theta_v}, \tag{B.2}$$

$$\pi_{a|v,k} = \frac{\overline{B}_{va,k} V^{\theta_v}_{ak}}{V^{\theta_v}_{vk}}. \tag{B.3}$$

For a specific residence and employment choice, $V_{ak} = \Phi_\ell$ and $\overline{B}_{va,k} = 1$: fixed location attractiveness is contained in the amenities in Φ_ℓ. The $\overline{B}$'s describe preferences over broader choices, whose values summarize the residence and employment opportunities they include. A nest with one potential choice has a preference parameter equal to one. The shape parameters satisfy

$$0 < \mu \leq \nu \leq \eta \leq \psi \leq \chi. \tag{B.4}$$

The migration, sector, and participation parameters μ, ν, and η are common across skills and CZs. The residence–workplace parameter χ is estimated from commuting gravity for each CZ, and ψ varies across events, so the ordering holds event by event. Their calibration is reported in Appendix C.1.

B.2.1 Value indices and conditional shares

The region consists of two areas, the CZ and the accessible outside area. Let $r \in \{I, O\}$ denote the area of residence, so that $r = I$ for CZ residents and $r = O$ for outside-area residents. Let $g \in \mathcal{G} = \mathcal{G}_T \cup \mathcal{G}_N$ be a sector, with $\mathcal{G}_T = \{T_H, T_L\}$ and $\mathcal{G}_N = \{N_H, N_L\}$ (for a nonworker, the sector in which they would work), and $e \in \{W, U\}$ work or nonemployment. A location is a CZ tract or the outside area, and a location pair ℓ is a residence and workplace location, or a residence location for a nonworker. CZ residents choose directly among location pairs with workplaces inside and outside the CZ. Outside-area residents who work first choose a job location $s \in \{I, O\}$, inside or outside the CZ, and then a location pair within it. Each nest's index equals the value index in (B.2) raised to the nest's shape parameter. The working, nonworking, sector, and area indices are defined in (B.8) and (B.9) below, and applying (B.3) at each margin gives the conditional shares that follow them. At the top, residence in the region and in the rest of the country form one nest,

$$(V_k^{\text{all}})^\mu = \overline{B}_k^{\text{nat}} (V_k^{\text{nat}})^\mu + \overline{B}_k^{\text{reg}} (V_k^{\text{reg}})^\mu, \qquad (V_k^{\text{reg}})^\mu = \sum_{r \in \{I,O\}} \overline{B}_{rk}^R \mathcal{Q}_{rk}^{\mu/\nu}, \tag{B.5}$$

where the national value V_k^{nat} is fixed for national population $\mathcal{N}_k$. Because migration and the choice between the two areas share μ, the choice to live in the region and the choice of area combine into one upper nest.

For a group of location pairs $\mathcal{D}$ with shape χ,

$$\pi_{\ell|\mathcal{D},k} = \frac{\Phi_\ell^\chi}{\sum_{a \in \mathcal{D}} \Phi_a^\chi}. \tag{B.6}$$

For outside-area workers, multiplying this location-pair share by the job-location share yields (B.17). CZ residents and nonworkers have no job-location choice, and nonworkers choose among residence locations with shape χ.

B.2.2 Expanded utilities and population allocations

This subsection writes the indices and allocations explicitly for both areas and then derives the CZ resident notation used in the main text. The preference parameters are $\overline{B}_{rk}^R$ for area r, $\overline{B}_{rgk}^S$ for sector g, $\overline{B}_{rgk}^E$ and $\overline{B}_{rgk}^U$ for work and nonemployment, and $\overline{B}_{gsk}^O$ for job location s, a choice only outside-area residents make. Write $\mathcal{L}_{Igk}^W$ for the working location pairs available to CZ residents, $\mathcal{L}_{Ogsk}^W$ for those available to outside-area residents with job location s, with $\mathcal{L}_{Ogk}^W = \mathcal{L}_{OgIk}^W \cup \mathcal{L}_{OgOk}^W$, and $\mathcal{L}_{rgk}^U$ for the nonworking residence locations. Let i_ℓ denote the residence location in pair ℓ and,

for workers, j_ℓ its workplace location. The residence location determines the area, with $i_\ell \in \mathcal{I}_I = \mathcal{J}$ for CZ residents and $i_\ell \in \mathcal{I}_O = \{o\}$ for outside-area residents. Nonworking choices retain sector g, but their systematic utility and conditional residence probabilities are identical across sectors.

Working and nonworking utility. Using the housing share α_ℓ defined in Appendix B.1, systematic utility is

$$\Phi_\ell = \begin{cases} \dfrac{\mathcal{B}_\ell y_\ell}{(q^R_{i_\ell})^{\alpha_\ell}(P^{\rho}_{i_\ell} P^{1-\rho}_{j_\ell})^{(1-\alpha_\ell)s_N}}, & \text{if working,} \\ \dfrac{\mathcal{B}_\ell y_\ell}{(q^R_{i_\ell})^{\alpha_\ell} P^{(1-\alpha_\ell)s_N}_{i_\ell}}, & \text{if nonemployed.} \end{cases} \tag{B.7}$$

Disposable income y_ℓ is given by (B.33). Outside income and prices are fixed except that an outside resident working inside the CZ receives the endogenous wage at that workplace. The values use the expenditure-share normalization derived in Appendix B.3.

Working, nonworking, and sector indices. The index $\mathcal{W}_{Igk}$ summarizes CZ residents' working location pairs and $\mathcal{W}_{Ogsk}$ those of outside-area residents with job location s. The index $\widetilde{\mathcal{W}}_{Ogk}$ combines the two job locations, and $\mathcal{U}_{rgk}$ summarizes nonworking residence locations,

$$\begin{aligned} \mathcal{W}_{Igk} &= \sum_{\ell \in \mathcal{L}^W_{Igk}} \Phi^{\chi}_\ell, & \mathcal{W}_{Ogsk} &= \sum_{\ell \in \mathcal{L}^W_{Ogsk}} \Phi^{\chi}_\ell, \\ \widetilde{\mathcal{W}}_{Ogk} &= \sum_{s \in \{I,O\}} \overline{B}^O_{gsk} \mathcal{W}^{\psi/\chi}_{Ogsk}, & \mathcal{U}_{rgk} &= \sum_{\ell \in \mathcal{L}^U_{rgk}} \Phi^{\chi}_\ell, \end{aligned} \tag{B.8}$$

with Φ_ℓ from (B.7). Working and nonworking opportunities combine within sector, and sectors combine within each area,

$$\begin{aligned} \mathcal{S}_{Igk} &= \overline{B}^E_{Igk} \mathcal{W}^{\eta/\chi}_{Igk} + \overline{B}^U_{Igk} \mathcal{U}^{\eta/\chi}_{Igk}, \\ \mathcal{S}_{Ogk} &= \overline{B}^E_{Ogk} \widetilde{\mathcal{W}}^{\eta/\psi}_{Ogk} + \overline{B}^U_{Ogk} \mathcal{U}^{\eta/\chi}_{Ogk}, \\ \mathcal{Q}_{rk} &= \sum_{g \in \mathcal{G}} \overline{B}^S_{rgk} \mathcal{S}^{\nu/\eta}_{rgk}. \end{aligned} \tag{B.9}$$

Because CZ residents have no job-location choice, ψ does not enter their indices.

CZ resident notation. Dropping the area subscript for CZ residents gives the main-text notation,

$$\begin{aligned} \mathcal{W}_{gk} &= \mathcal{W}_{Igk}, & \mathcal{U}_{gk} &= \mathcal{U}_{Igk}, & \mathcal{S}_{gk} &= \mathcal{S}_{Igk}, \\ \overline{B}^{E}_{gk} &= \overline{B}^{E}_{Igk}, & \overline{B}^{U}_{gk} &= \overline{B}^{U}_{Igk}, & \overline{B}^{S}_{gk} &= \overline{B}^{S}_{Igk}, \\ N_k &= N_{Ik}, & N_{gk} &= N_{Igk}, & E_{gk} &= E_{Igk}, \end{aligned} \tag{B.10}$$

with choice sets $\mathcal{L}^{W}_{gk} = \mathcal{L}^{W}_{Igk}$ and $\mathcal{L}^{U}_{gk} = \mathcal{L}^{U}_{Igk}$. The first line of (B.9) is then the main-text sector index (7).

Regional migration and residence. The fixed national contribution is $\overline{\mathcal{O}}_k = \overline{B}^{\text{nat}}_k (V^{\text{nat}}_k)^{\mu}$, and V^{reg}_k is given in (B.5). The regional share and the share living in each area are

$$\begin{aligned} b^{\text{reg}}_k &= \frac{\overline{B}^{\text{reg}}_k \sum_{r\in\{I,O\}} \overline{B}^{R}_{rk} \mathcal{Q}^{\mu/\nu}_{rk}}{\overline{\mathcal{O}}_k + \overline{B}^{\text{reg}}_k \sum_{r\in\{I,O\}} \overline{B}^{R}_{rk} \mathcal{Q}^{\mu/\nu}_{rk}}, \\ \pi(r \mid \text{reg}, k) &= \frac{\overline{B}^{R}_{rk} \mathcal{Q}^{\mu/\nu}_{rk}}{\sum_{r'\in\{I,O\}} \overline{B}^{R}_{r'k} \mathcal{Q}^{\mu/\nu}_{r'k}}. \end{aligned} \tag{B.11}$$

Multiplying these shares gives

$$\begin{aligned} N_{rk} &= \mathcal{N}_k b^{\text{reg}}_k \pi(r \mid \text{reg}, k) = \mathcal{N}_k \frac{\overline{B}^{\text{reg}}_k \overline{B}^{R}_{rk} \mathcal{Q}^{\mu/\nu}_{rk}}{\overline{\mathcal{O}}_k + \overline{B}^{\text{reg}}_k \sum_{r'\in\{I,O\}} \overline{B}^{R}_{r'k} \mathcal{Q}^{\mu/\nu}_{r'k}}, \\ N^{\text{reg}}_k &= \sum_r N_{rk} = \mathcal{N}_k b^{\text{reg}}_k, \qquad N^{\text{nat}}_k = \mathcal{N}_k (1 - b^{\text{reg}}_k). \end{aligned} \tag{B.12}$$

Both migration and the choice between the two areas use μ. For the main-text CZ population equation, define

$$\begin{aligned} \overline{B}^{R}_{k} &= \overline{B}^{\text{reg}}_k \overline{B}^{R}_{Ik}, \\ \mathcal{O}_k &= \overline{\mathcal{O}}_k + \overline{B}^{\text{reg}}_k \overline{B}^{R}_{Ok} \mathcal{Q}^{\mu/\nu}_{Ok}. \end{aligned} \tag{B.13}$$

The term $\mathcal{O}_k$ includes both the fixed national contribution and living in the outside area. It is endogenous because outside-area residents can work at CZ workplaces. Setting $r = I$ in (B.12) and substituting $\mathcal{Q}_{Ik} = \sum_g \overline{B}^{S}_{gk} \mathcal{S}^{\nu/\eta}_{gk}$ gives (8).

Sector allocation. The share of area-r residents choosing sector g is $\overline{B}^{S}_{rgk} \mathcal{S}^{\nu/\eta}_{rgk} / \mathcal{Q}_{rk}$, so

$$N_{rgk} = N_{rk} \frac{\overline{B}^{S}_{rgk} \mathcal{S}^{\nu/\eta}_{rgk}}{\mathcal{Q}_{rk}}. \tag{B.14}$$

For $r = I$, this gives (9). The allocation includes both employed and nonemployed individuals. Nonemployment income, prices, and amenities are identical across sectors at each residence location. The baseline allocation and its normalization are derived below.

Labor-force participation. The employed share of those choosing sector g is the working term of $\mathcal{S}_{rgk}$ relative to its total,

$$\frac{E_{Igk}}{N_{Igk}} = \frac{\overline{B}^E_{Igk}\mathcal{W}^{\eta/\chi}_{Igk}}{\mathcal{S}_{Igk}}, \qquad \frac{E_{Ogk}}{N_{Ogk}} = \frac{\overline{B}^E_{Ogk}\widetilde{\mathcal{W}}^{\eta/\psi}_{Ogk}}{\mathcal{S}_{Ogk}}. \tag{B.15}$$

The first gives (10). The nonemployed population in sector g is $N_{rgk} - E_{rgk}$.

Job location and location pairs. An outside-area resident who works in sector g chooses job location s and then a location pair $\ell \in \mathcal{L}^W_{Ogsk}$, with shares

$$\pi(s \mid O, g, W, k) = \frac{\overline{B}^O_{gsk}\mathcal{W}^{\psi/\chi}_{Ogsk}}{\widetilde{\mathcal{W}}_{Ogk}}, \qquad \pi_{\ell|O,g,W,s,k} = \frac{\Phi^\chi_\ell}{\mathcal{W}_{Ogsk}}. \tag{B.16}$$

A CZ resident chooses a location pair directly. The conditional location-pair probabilities of workers are therefore

$$\pi_{\ell|I,g,W,k} = \frac{\Phi^\chi_\ell}{\mathcal{W}_{Igk}}, \qquad \pi_{\ell|O,g,W,k} = \frac{\overline{B}^O_{gsk}\mathcal{W}^{\psi/\chi}_{Ogsk}}{\widetilde{\mathcal{W}}_{Ogk}}\frac{\Phi^\chi_\ell}{\mathcal{W}_{Ogsk}}, \tag{B.17}$$

and the main-text probability abbreviates $\pi_{\ell k} = \pi_{\ell|I,g,W,k}$. Multiplying by the area, sector, and employment allocations gives the working population, for CZ residents and outside-area residents respectively,

$$\begin{aligned} F_{ijgk} &= N_{Ik} \underbrace{\frac{\overline{B}^S_{Igk}\mathcal{S}^{\nu/\eta}_{Igk}}{\mathcal{Q}_{Ik}}}_{\text{sector}} \underbrace{\frac{\overline{B}^E_{Igk}\mathcal{W}^{\eta/\chi}_{Igk}}{\mathcal{S}_{Igk}}}_{\text{work}} \underbrace{\frac{\Phi^\chi_\ell}{\mathcal{W}_{Igk}}}_{\text{location pair}}, \qquad i \in \mathcal{J}, \\ F_{ojgk} &= N_{Ok} \underbrace{\frac{\overline{B}^S_{Ogk}\mathcal{S}^{\nu/\eta}_{Ogk}}{\mathcal{Q}_{Ok}}}_{\text{sector}} \underbrace{\frac{\overline{B}^E_{Ogk}\widetilde{\mathcal{W}}^{\eta/\psi}_{Ogk}}{\mathcal{S}_{Ogk}}}_{\text{work}} \\ &\quad \times \underbrace{\frac{\overline{B}^O_{gsk}\mathcal{W}^{\psi/\chi}_{Ogsk}}{\widetilde{\mathcal{W}}_{Ogk}}}_{\text{job location}} \underbrace{\frac{\Phi^\chi_\ell}{\mathcal{W}_{Ogsk}}}_{\text{location pair}}. \end{aligned} \tag{B.18}$$

The first line gives (11). Substituting (B.12) for N_{rk} gives the unconditional number choosing each pair.

Nonemployment and population accounting. For a nonworking choice $\ell \in \mathcal{L}^U_{rgk}$ with residence location i and sector g, the nonemployed population is

$$U_{igk} = (N_{rgk} - E_{rgk})\frac{\Phi^{\chi}_{\ell}}{\mathcal{U}_{rgk}}, \qquad U_{ik} = \sum_{g\in\mathcal{G}} U_{igk}. \tag{B.19}$$

An unavailable nonworking choice has $U_{igk} = 0$. Thus $\mathcal{U}_{rgk}$ is a utility index, while U_{igk} and U_{ik} count individuals. The allocations satisfy

$$\begin{aligned} \sum_{g} N_{rgk} &= N_{rk}, & \sum_{\ell\in\mathcal{L}^W_{rgk}} F_{i_\ell j_\ell gk} &= E_{rgk}, \\ \sum_{i\in\mathcal{I}_r} U_{ik} &= \sum_{g}(N_{rgk} - E_{rgk}), & \sum_{r} N_{rk} + N^{\text{nat}}_{k} &= \mathcal{N}_k. \end{aligned} \tag{B.20}$$

Conditional shares apply to nests with positive value; an empty nest has zero population. Write $\pi^{\text{all}}_{\ell k}$ for the unconditional probability, distinguished from the conditional location-pair probability $\pi_{\ell k}$ in the main text. For a regional choice,

$$\pi^{\text{all}}_{\ell k} = b^{\text{reg}}_{k}\pi(r \mid \text{reg}, k)\pi(g \mid r, k)\pi(e \mid r, g, k)\pi_{\ell|r,g,e,k}.$$

For workers the last factor is (B.17); for nonworkers it is the residence-location share in (B.19). The national choice has probability $1 - b^{\text{reg}}_{k}$. For the full feasible choice set $\mathcal{A}_k$,

$$n_{\ell k} = \mathcal{N}_k \pi^{\text{all}}_{\ell k}, \qquad \sum_{\ell\in\mathcal{A}_k} \pi^{\text{all}}_{\ell k} = 1. \tag{B.21}$$

These probabilities determine equilibrium populations. Individual transitions across equilibria are derived in Appendix B.12.

B.2.3 Derivation from the taste distribution

For any positive vector of choice values x, let $\mathcal{V}_k(x)$ denote the all-location value obtained from the indices above, with each location-pair value Φ_ℓ replaced by x_ℓ and the national value replaced by x_{nat}. Define $G_k(x) = \mathcal{V}_k(x)^{\mu}$. The nested Fréchet joint distribution in (5) is therefore

$$\Pr\{z_\ell \le t_\ell \text{ for every } \ell\} = \exp\left\{-\left[\mathcal{V}_k\big((t_\ell^{-1})_\ell\big)\right]^{\mu}\right\}. \tag{B.22}$$

The ordering in (B.4) ensures that this is a valid nested extreme-value distribution. Since each value index is homogeneous of degree one, maximized utility has distribution

$$\Pr\{\max_{\ell} z_\ell\Phi_\ell \le u\} = \exp\left[-\left(\frac{V^{\text{all}}_{k}}{u}\right)^{\mu}\right]. \tag{B.23}$$

Let $\partial_\ell G_k$ denote the derivative of G_k with respect to x_ℓ. The probability that ℓ maximizes utility integrates the density that its utility equals u while every other choice gives at most u:

$$\begin{aligned}\pi^{\text{all}}_{\ell k} &= \int_0^\infty \Phi_\ell \, \partial_\ell G_k(\Phi) \, u^{-\mu-1} \exp[-G_k(\Phi)u^{-\mu}] \, du \\ &= \frac{\Phi_\ell \, \partial_\ell G_k(\Phi)}{\mu G_k(\Phi)} = \frac{\partial \log V^{\text{all}}_k}{\partial \log \Phi_\ell}.\end{aligned} \tag{B.24}$$

At any nest,

$$\frac{\partial \log V_{vk}}{\partial \log V_{ak}} = \frac{\overline{B}_{va,k} V_{ak}^{\theta_v}}{\sum_{u \in \mathcal{C}(v)} \overline{B}_{vu,k} V_{uk}^{\theta_v}} = \pi_{a|v,k}. \tag{B.25}$$

The chain rule therefore gives the product of conditional shares in (B.21) and the preceding probability expression. For two choices a and d in the same nest, $\pi_{a|v,k}/\pi_{d|v,k} = (\overline{B}_{va,k}/\overline{B}_{vd,k})(V_{ak}/V_{dk})^{\theta_v}$. Hence θ_v is the elasticity of conditional odds with respect to relative value. Holding other values fixed, the own-share elasticity is $\theta_v(1 - \pi_{a|v,k})$. In particular,

$$\frac{\partial \log N^{\text{reg}}_k}{\partial \log V^{\text{reg}}_k} = \mu(1 - b^{\text{reg}}_k). \tag{B.26}$$

B.2.4 Baseline normalization and sectors

Let $\pi^0_{a|v,k}$ denote the model's baseline conditional probability. At the location-pair margin, it is predicted by the calibrated amenities, commuting costs, and baseline economic conditions:

$$\pi^0_{\ell|\mathcal{D},k} = \frac{(\Phi^0_\ell)^\chi}{\sum_{m \in \mathcal{D}^0} (\Phi^0_m)^\chi}, \qquad \ell \in \mathcal{D}^0, \tag{B.27}$$

where $\mathcal{D}^0$ contains the choices available at baseline. Internal commuting probabilities come from fitted gravity flows, so an observed zero on a feasible link need not imply a zero probability. Broader conditional probabilities reproduce the corresponding calibrated population, sector, and participation totals. They are distinct from amenities and from preference parameters in levels.

At a broader nest, the preference parameters satisfy

$$\overline{B}_{va,k} = \pi^0_{a|v,k} \left(\frac{V^0_{vk}}{V^0_{ak}} \right)^{\theta_v}. \tag{B.28}$$

At the location-pair margin, $\overline{B}_{va,k} = 1$ and this identity follows from (B.27). Writing $\widehat{V}_{ak} = V_{ak}/V^0_{ak}$ yields, on the baseline choice set,

$$\widehat{V}_{vk} = \left[\sum_a \pi^0_{a|v,k} \widehat{V}^{\theta_v}_{ak} \right]^{1/\theta_v}, \qquad \pi_{a|v,k} = \frac{\pi^0_{a|v,k} \widehat{V}^{\theta_v}_{ak}}{\sum_u \pi^0_{u|v,k} \widehat{V}^{\theta_v}_{uk}}. \tag{B.29}$$

In particular, when the feasible bilateral choice set is unchanged,

$$\pi^1_{\ell|\mathcal{D},k} = \frac{\pi^0_{\ell|\mathcal{D},k}(\Phi^1_\ell/\Phi^0_\ell)^\chi}{\sum_{m\in\mathcal{D}^0}\pi^0_{m|\mathcal{D},k}(\Phi^1_m/\Phi^0_m)^\chi}. \tag{B.30}$$

The fixed amenities and commuting costs cancel from the utility ratios and remain embodied in the baseline predicted probabilities. No additional bilateral preference parameter enters this update. Baseline counts and amenity calibration are described in Appendix C.6.1.

For area r, let E^0_{rgk} be employment in sector g, $E^0_{rk} = \sum_g E^0_{rgk}$, and U^0_{rk} nonemployment. The sector allocation of nonworkers is

$$a^0_{rgk} = E^0_{rgk}/E^0_{rk}, \qquad U^0_{igk} = a^0_{rgk}U^0_{ik}, \qquad i \in r. \tag{B.31}$$

All sectors at a nonworking residence have the same income, prices, and amenities. Summing over sectors recovers U^0_{ik}, and the baseline work probability within every sector is $\pi^0_{rk} = E^0_{rk}/(E^0_{rk}+U^0_{rk})$. With hats denoting ratios to baseline values of the indices in (B.8) and (B.9),

$$\widehat{\mathcal{S}}_{rgk} = \pi^0_{rk}\widehat{\widetilde{\mathcal{W}}}^{\eta/\psi}_{rgk} + (1-\pi^0_{rk})\widehat{\mathcal{U}}^{\eta/\chi}_{rgk}, \qquad \widehat{\mathcal{Q}}_{rk} = \sum_g a^0_{rgk}\widehat{\mathcal{S}}^{\nu/\eta}_{rgk},$$

$$\pi(g \mid r,k) = \frac{a^0_{rgk}\widehat{\mathcal{S}}^{\nu/\eta}_{rgk}}{\widehat{\mathcal{Q}}_{rk}}, \qquad \pi(W \mid r,g,k) = \frac{\pi^0_{rk}\widehat{\widetilde{\mathcal{W}}}^{\eta/\psi}_{rgk}}{\widehat{\mathcal{S}}_{rgk}}. \tag{B.32}$$

The parameter ν therefore governs sector choice, including the associated value of nonemployment; η governs work versus nonwork within a sector. An unconditional employed-sector share also reflects participation and location responses. When $\eta = \nu$, these two value indices combine into a single employment and sector index after aggregating the identical nonemployment choices. The national choice is not divided across sectors.

B.3 Disposable income and consumption demand

Disposable income is the wage or nonemployment income plus the residential dividend,

$$y_\ell = \begin{cases} w^g_{jk}, & \text{if working,} \\ y^U_k, & \text{if nonemployed,} \end{cases} + \mathbf{1}\{i \in \mathcal{J}\}D_k + \mathbf{1}\{i = o\}\overline{D}_{ok}. \tag{B.33}$$

Nonemployment income is externally financed. For CZ residents, this reduces to (12). Outside dividends are fixed. An outside resident employed inside the CZ receives that workplace's endogenous wage. A feasible consumption choice requires $y_\ell > 0$.

B.3.1 Expenditure shares and home and work shopping

A worker's service composite is $n = n_i^{\rho} n_j^{1-\rho}$. For fixed locations, the consumption problem is

$$\begin{aligned} \max_{h,n_i,n_j,x>0} \quad & \alpha_\ell \log h + \gamma_\ell \rho \log n_i + \gamma_\ell (1-\rho) \log n_j + \beta_\ell \log x, \\ \text{subject to} \quad & q_i^R h + P_i n_i + P_j n_j + x \le y_\ell. \end{aligned} \tag{B.34}$$

The amenity and taste terms are constant within this problem. The first-order conditions equate each good's expenditure to its utility exponent divided by the marginal utility of income. Summing these conditions and using the budget constraint shows that the marginal utility of income is $1/y_\ell$. Consequently,

$$\begin{aligned} q_i^R h_\ell &= \alpha_\ell y_\ell, & P_i n_i &= \rho \gamma_\ell y_\ell, \\ P_j n_j &= (1-\rho)\gamma_\ell y_\ell, & x_\ell &= \beta_\ell y_\ell. \end{aligned} \tag{B.35}$$

Housing demand is

$$h_\ell = \frac{\alpha_\ell y_\ell}{q_i^R}, \tag{B.36}$$

and spending on housing, services, and the tradable good is $\alpha_\ell y_\ell$, $\gamma_\ell y_\ell$, and $\beta_\ell y_\ell$. For CZ residents, $\alpha_\ell = \alpha_H$ and $\gamma_\ell = \gamma = (1-\alpha_H)s_N$, giving (13). A nonworker buys services only from home, so $P_i n_i = \gamma_\ell y_\ell$, with the same housing and tradable demands. Thus a dollar of additional disposable income raises service expenditure by γ_ℓ, allocated between home and work according to employment status.

B.3.2 Indirect utility and the expenditure-share normalization

Define $R_W = \rho^\rho (1-\rho)^{1-\rho}$ and $R_U = 1$, with endpoint values interpreted by continuity. Substitution of the demands into (B.1) gives

$$\begin{aligned} U_\ell^*(\omega) &= z_\ell(\omega) K_\ell \frac{\mathcal{B}_\ell y_\ell}{(q_i^R)^{\alpha_\ell} \mathcal{P}_\ell^{\gamma_\ell}}, & K_\ell &= \alpha_\ell^{\alpha_\ell} \gamma_\ell^{\gamma_\ell} \beta_\ell^{\beta_\ell} R_e^{\gamma_\ell}, \\ \mathcal{P}_\ell &= \begin{cases} P_i^\rho P_j^{1-\rho}, & e = W, \\ P_i, & e = U. \end{cases} \end{aligned} \tag{B.37}$$

The housing share, nonhousing shares, and shopping allocation are fixed within each entry comparison. The constant K_ℓ depends only on area r and employment status e, so write it as K_{re}. It is common across location pairs within each such group, although it can differ between workers and nonworkers and across areas. Homogeneity of the working and nonworking indices allows these constants to be absorbed into the participation shifters:

$$\overline{B}_{rgk}^E = \overline{B}_{rgk}^{E,\text{raw}} K_{rW}^\eta, \qquad \overline{B}_{rgk}^U = \overline{B}_{rgk}^{U,\text{raw}} K_{rU}^\eta. \tag{B.38}$$

Thus the normalized systematic value in (4) produces the same probabilities and inclusive values as the raw indirect utility when the participation shifters are expressed in the corresponding units. This fixed taste normalization preserves the original Cobb–Douglas preferences and expenditure shares without a bilateral preference parameter. Each shopping counterfactual exercise uses the same normalization in its entry and no-entry equilibria.

B.4 Population, employment, and labor supply

Multiplying the conditional shares yields a regional choice probability $\pi^{\text{all}}_{\ell k} = b^{\text{reg}}_k \pi(r \mid \text{reg}, k)\pi(g \mid r, k)\pi(e \mid r, g, k)\pi_{\ell|r,g,e,k}$. For outside-area workers, the last probability includes the job-location choice. The national share is $1 - b^{\text{reg}}_k$. Conditional shares sum to one at every margin, so $\sum_\ell \pi^{\text{all}}_{\ell k} = 1$ and $n_{\ell k} = \mathcal{N}_k \pi^{\text{all}}_{\ell k}$ exhausts the fixed national population.

Let F_{ijgk} denote working counts and U_{igk} nonworking counts by sector. The relevant employment and population totals are

$$\begin{aligned} U_{ik} &= \sum_g U_{igk}, & L^g_{jk} &= \sum_i F_{ijgk}, \\ N_k &= \sum_{i \in \mathcal{J}} \left(\sum_{j,g} F_{ijgk} + U_{ik} \right), & N &= \sum_k N_k. \end{aligned} \tag{B.39}$$

Workplace employment includes inbound commuters; resident employment includes outbound commuters. These sums distinguish changes in local jobs from changes in residents' employment. In particular, entry can raise workplace employment through commuting without changing the number of employed CZ residents.

B.5 Individual values and choice changes

Let $y_{\ell k}$ be disposable income in choice ℓ and $q_i = (1+t)r^R_i$ the residential rent for a local resident. Housing has the Cobb–Douglas expenditure share α_ℓ. For an employed individual living in i and working in j, the change in the systematic utility is

$$d_{\ell k} = \log \frac{y_{\ell k}}{y^0_{\ell k}} - \alpha_\ell \log \frac{q_i}{q^0_i} - (1 - \alpha_\ell) s_N \left[\rho \log \frac{P_i}{P^0_i} + (1-\rho) \log \frac{P_j}{P^0_j} \right]. \tag{B.40}$$

Income includes the current skill dividend alongside wages or nonemployment income. For a nonworker the shopping term is $\log(P_i/P^0_i)$. Outside wages, nonemployment income, dividends, and prices are fixed. An outside resident employed inside the CZ receives the endogenous local wage. Fixed amenities and commuting costs remain embodied in the baseline predicted probabilities.

For each bilateral choice, $d_{\ell k} = \log(\Phi_\ell / \Phi^0_\ell)$.

Set $v_{\ell k} = d_{\ell k}$. Each nest v has log inclusive-value change and conditional probabilities

$$v_{vk} = \frac{1}{\theta_v} \log \sum_{a \in \mathcal{C}(v)} \pi^0_{a|v,k} \exp(\theta_v v_{ak}), \tag{B.41}$$

$$\pi_{a|v,k} = \pi^0_{a|v,k} \exp\{\theta_v (v_{ak} - v_{vk})\}. \tag{B.42}$$

Multiplying the conditional probabilities along the nests containing a choice, and then by $\mathcal{N}_k$, gives its population. The baseline probabilities carry the calibrated amenities and commuting costs, so no additional amenity parameters enter. Unavailable choices have $v_{ak} = -\infty$. A nest with no feasible choices has zero value in levels, log value $-\infty$, and zero choice probability. A choice with nonpositive disposable income is unavailable. The baseline calibration requires positive income in every choice with positive baseline mass.

For two members u and z of the same nest v with positive shares in both equilibria,

$$\log \frac{\pi_{u|v,k}}{\pi_{z|v,k}} - \log \frac{\pi^0_{u|v,k}}{\pi^0_{z|v,k}} = \theta_v (v_{uk} - v_{zk}). \tag{B.43}$$

Thus a scale is a conditional log-odds response to the relative value. For a binary choice with share π, the elasticity of that share to its own value is $\theta_v(1-\pi)$. This distinction applies to participation and inward commuting as well as regional attachment. The two migration choices, between the region and the rest of the country and between residence inside and outside the CZ, share μ and therefore combine into a single nest.

B.6 Service demand and shopping price indices

The service bundle combines varieties from the two service sectors $g \in \mathcal{G}_N = \{N_H, N_L\}$. These sectors differ in skill intensity; each can employ both worker skills k. Appendix C.5.1 gives their sector classifications. Firms form a continuum, each supplying one differentiated variety under monopolistic competition with free entry. The mass m_{jg} is a continuous measure of the number of active firms in sector g and tract j, each charging p_{jg}. All varieties enter the same CES bundle with elasticity $\sigma_N > 1$, so the sector grouping introduces no additional consumption nest. Sector prices and establishment masses are determined separately by production costs and free entry.

Sector-specific shopping demand. Shopping and commuting share the per-minute travel-cost parameter κ. The shopping preference parameters satisfy $\zeta_{ijg} = \zeta_{ij}\xi_{jg}$. The bilateral shifter ζ_{ij} is common to both industries, while ξ_{jg} measures destination–industry appeal. Thus $T_{ijg} = \zeta_{ij}\xi_{jg} \exp\{-(\sigma_N - 1)\kappa t_{ij}\}$.

Each delivered variety price is $p_{jg} \exp(\kappa t_{ij})$; summing the dual price terms over varieties gives $T_{ijg} m_{jg} (p_{jg})^{1-\sigma_N}$, while fixed outside opportunities contribute T_i^O. CES expenditure minimization

at origin i therefore gives

$$P_i = \left[T_i^O + \sum_{j \in \mathcal{J}} \sum_{g \in \mathcal{G}_N} T_{ijg} m_{jg} (p_{jg})^{1-\sigma_N} \right]^{1/(1-\sigma_N)}. \tag{B.44}$$

Shephard's lemma gives the expenditure share received by each industry and destination:

$$\frac{\partial \log P_i}{\partial \log p_{jg}} = \frac{T_{ijg} m_{jg} (p_{jg})^{1-\sigma_N}}{P_i^{1-\sigma_N}} = \pi_{ijg}^I. \tag{B.45}$$

Individuals can also purchase services outside the CZ. The term T_i^O summarizes the contribution of outside varieties, prices, shopping preferences, and travel costs to the service price index at origin i. These outside opportunities are held fixed across equilibria. The outside shopping share and the expenditure adding-up condition are

$$\pi_i^O = \frac{T_i^O}{P_i^{1-\sigma_N}}, \qquad \sum_{j \in \mathcal{J}} \sum_{g \in \mathcal{G}_N} \pi_{ijg}^I + \pi_i^O = 1. \tag{B.46}$$

Although T_i^O is fixed, the outside expenditure share responds to local prices and entry through P_i. Lower local prices or additional local varieties reduce the share spent outside. Expenditure $\mathcal{E}_i \pi_i^O$ accrues to businesses outside the CZ and does not enter local nontradable revenue.

A worker living in i and employed at j allocates $\rho \gamma_\ell y_{ijgk}$ to purchases from home and $(1-\rho)\gamma_\ell y_{ijgk}$ to purchases from work. Summing the first budget over F_{ijgk} and the second over F_{jigk} gives workers' spending from origin i. Nonemployment income and expenditure shares are identical across sectors at a given residence and skill, so nonworkers contribute $\gamma U_{ik}(y_k^U + D_k)$ after summing their counts over sectors. These three contributions give (14). The sums include feasible residence and workplace locations inside and outside the CZ. Inbound commuters contribute spending from work and outbound commuters contribute spending from home. Spending originating in the outside location accrues outside the CZ. Revenue at local destinations equals the spending directed to them from CZ origins.

Aggregation by tract. The main-text shopping equations combine industries within each tract. For a tract with positive variety mass, define total varieties and the CES price index per variety by

$$\begin{aligned} m_j &= \sum_{g \in \mathcal{G}_N} m_{jg}, \\ p_j^N &= \left[\sum_{g \in \mathcal{G}_N} \xi_{jg} \frac{m_{jg}}{m_j} (p_{jg})^{1-\sigma_N} \right]^{1/(1-\sigma_N)}. \end{aligned} \tag{B.47}$$

Let $T_{ij} = \zeta_{ij} \exp\{-(\sigma_N - 1)\kappa t_{ij}\}$. Since $T_{ijg} = T_{ij}\xi_{jg}$, these definitions imply

$$T_{ij} m_j (p_j^N)^{1-\sigma_N} = \sum_{g \in \mathcal{G}_N} T_{ijg} m_{jg} (p_{jg})^{1-\sigma_N}. \tag{B.48}$$

Substituting into (B.44) gives (15). A tract with no varieties contributes zero to the price-index sum. Expenditure shares and revenue aggregate as

$$\pi_{ij}^I = \sum_{g \in \mathcal{G}_N} \pi_{ijg}^I, \qquad R_j^N = \sum_{g \in \mathcal{G}_N} R_{jg}. \tag{B.49}$$

Summing expenditure shares gives (16). Total nontradable revenue in tract j is

$$R_j^N = \sum_{i \in \mathcal{J}} \mathcal{E}_i \pi_{ij}^I. \tag{B.50}$$

All service revenue comes from the spending of individuals in the model. The tract price index reflects both industry prices and the composition of available varieties; the industry prices and entry responses remain distinct.

Changes in shopping shares and revenue. Define destination attraction and the origin denominator in changes by

$$\Omega_{jg} = \widehat{m}_{jg} (\widehat{p}_{jg})^{1-\sigma_N}, \qquad D_i = \sum_j \sum_{g \in \mathcal{G}_N} \pi_{ijg}^{I,0} \Omega_{jg} + \pi_i^{O,0}, \tag{B.51}$$

$$\pi_{ijg}^I = \frac{\pi_{ijg}^{I,0} \Omega_{jg}}{D_i}, \qquad \pi_i^O = \frac{\pi_i^{O,0}}{D_i}, \qquad \widehat{P}_i = D_i^{1/(1-\sigma_N)}. \tag{B.52}$$

Individual NT budgets follow the Cobb–Douglas demands and are assigned to home and work origins using ρ. Nonworkers shop from home. Revenue in each industry and tract is

$$R_{jg} = \sum_{i \in \mathcal{J}} \mathcal{E}_i \pi_{ijg}^I, \tag{B.53}$$

where spending $\mathcal{E}_i$ includes each individual's income and dividends.

B.7 Nontradable production, costs, and entry

The model contains two service sectors, $g \in \mathcal{G}_N = \{N_H, N_L\}$. They differ in skill intensity, but each employs both worker skills k. Productivity, skill shares, and fixed input requirements may differ across tracts and sectors. Wages, prices, and entry are determined separately for each sector. The parameters ϵ_m and σ_N are common to both, and the quantification gives the two sectors the same cost shares.

B.7.1 Production and unit cost

A firm in sector $g \in \mathcal{G}_N$ and tract j produces output Y from labor $\mathcal{L}$, commercial space H, and intermediate inputs X, omitting firm subscripts,

$$Y = A_{jg}\, \mathcal{L}^{\alpha_L^g} H^{\alpha_C^g} X^{\alpha_X^g}, \qquad \alpha_L^g + \alpha_C^g + \alpha_X^g = 1. \tag{B.54}$$

Here A_{jg} is productivity and α_L^g, α_C^g, and α_X^g are the expenditure shares on labor, commercial space, and intermediates. If L_k is the firm's employment of skill k, the two skills form a Cobb–Douglas labor bundle with fixed expenditure shares λ_{jgk}. Minimizing the wage bill required to produce one unit of this bundle gives its unit cost:

$$\mathcal{L} = \prod_k L_k^{\lambda_{jgk}}, \qquad c_{jg}^L = \prod_k \left(\frac{w_{jk}^g}{\lambda_{jgk}} \right)^{\lambda_{jgk}}, \qquad \sum_k \lambda_{jgk} = 1. \tag{B.55}$$

A worker of skill k costs the firm the wage w_{jk}^g. Minimizing the cost of output gives

$$c_{jg} = \frac{(c_{jg}^L)^{\alpha_L^g} (q_j^C)^{\alpha_C^g} (p^X)^{\alpha_X^g}}{A_{jg} (\alpha_L^g)^{\alpha_L^g} (\alpha_C^g)^{\alpha_C^g} (\alpha_X^g)^{\alpha_X^g}}, \qquad p_{jg} = \frac{\sigma_N}{\sigma_N - 1}\, c_{jg}. \tag{B.56}$$

Here $q_j^C = (1+t) r_j^C$ and $p^X = 1$. The constant in the denominator follows from the Cobb–Douglas production normalization. Monopolistic competition under CES demand implies the constant markup shown above. Differentiating unit cost with respect to input prices shows that labor, commercial space, and intermediates receive the constant shares α_L^g, α_C^g, and α_X^g of total cost, whatever the factor prices. Variable production and fixed inputs use the same technology, so these shares apply to their combined expenditure C_{jg}.

B.7.2 Factor demand and free entry

Cost minimization within the Cobb–Douglas labor bundle gives $w_{jk}^g L_k = \lambda_{jgk} c_{jg}^L \mathcal{L}$. Shephard's lemma applied to c_{jg} gives the same constant shares. Aggregating variable and fixed costs across firms within sector g gives the input demands

$$L_{jk}^g = \frac{\alpha_L^g \lambda_{jgk} C_{jg}}{w_{jk}^g}, \qquad H_{jg}^C = \frac{\alpha_C^g C_{jg}}{q_j^C}, \qquad X_{jg} = \frac{\alpha_X^g C_{jg}}{p^X}. \tag{B.57}$$

At the CES markup, variable cost is $(\sigma_N - 1) R_{jg}/\sigma_N$, leaving revenue less variable costs of R_{jg}/σ_N. Each firm also incurs an annual fixed cost f_{jg}. With m_{jg} active firms, free entry drives profits to zero:

$$f_{jg} = \varphi_{jg} c_{jg} m_{jg}^{1/\epsilon_m - 1}, \qquad \frac{R_{jg}}{\sigma_N} = m_{jg} f_{jg}. \tag{B.58}$$

Fixed inputs use the same technology as output, so their quantity per firm is $f_{jg}/c_{jg} = \varphi_{jg} m_{jg}^{1/\epsilon_m - 1}$. The parameter φ_{jg} scales this requirement and is held fixed across equilibria. These inputs are fixed

with respect to an individual firm's output; their cost depends on factor prices and the number of firms, which each firm takes as given. The elasticity of fixed inputs per firm with respect to the number of firms is $1/\epsilon_m - 1$. Thus, when $\epsilon_m < 1$, additional entry raises each firm's fixed input requirement; when $\epsilon_m = 1$, that requirement is constant.

The zero-profit condition implies that revenue covers variable and fixed costs: $C_{jg} = R_{jg}$. Substitution gives firm entry and the employment response within each industry:

$$m_{jg} = \left(\frac{R_{jg}}{\sigma_N \varphi_{jg} c_{jg}} \right)^{\epsilon_m}, \qquad d \log L^g_{jk} = d \log R_{jg} - d \log w^g_{jk}. \tag{B.59}$$

Holding φ_{jg} fixed, ϵ_m is the elasticity of the number of firms with respect to revenue relative to unit cost. At given revenue and unit cost, the elasticity with respect to φ_{jg} is $-\epsilon_m$. Higher fixed input requirements therefore support fewer firms. Skill and cost shares are fixed, so the employment response combines additional revenue with the change in the wage. Payroll in sector g is $\sum_k w^g_{jk} L^g_{jk} = \alpha^g_L R_{jg}$. The calibration of cost shares and fixed input requirements is in Appendix C.6.3.

B.7.3 Aggregation across service sectors

The production and entry equations apply separately to each service sector. Total nontradable employment by skill and commercial-space demand in tract j are

$$L^N_{jk} = \sum_{g \in \mathcal{G}_N} L^g_{jk}, \qquad H^{C,N}_j = \sum_{g \in \mathcal{G}_N} H^C_{jg}. \tag{B.60}$$

These totals enter property-market clearing and the employment measures in Section 4.6. Total varieties are $m_j = \sum_{g \in \mathcal{G}_N} m_{jg}$ and total revenue is $R^N_j = \sum_{g \in \mathcal{G}_N} R_{jg}$. The shopping price p^N_j is the CES price index per variety in (B.47), whereas each p_{jg} is the price charged by firms in sector g. Without a sector index, nontradable employment, commercial-space demand, varieties, and revenue denote the corresponding tract totals. These sums preserve the separate costs and entry responses of the two service sectors.

B.8 Tradable production and the entrant

For a tradable sector $g \in \mathcal{G}_T$ with output Y_{jg}, minimizing cost with the Cobb–Douglas technology gives

$$c_{jg} = \frac{(q^C_j)^{\alpha^g_C} (p^X)^{\alpha^g_X}}{A_{jg}} \prod_k \left(w^g_{jk} \right)^{\alpha^g_L \lambda_{jgk}}. \tag{B.61}$$

Fixed expenditure-share constants are absorbed into A_{jg}, and the intermediate input price is normalized to $p^X = 1$. Differentiating total cost $c_{jg} Y_{jg}$ with respect to each input price gives the labor

demand in (24),

$$L^g_{jk} = \frac{\alpha^g_L \lambda_{jgk} c_{jg} Y_{jg}}{w^g_{jk}}, \tag{B.62}$$

and the remaining input demands,

$$H^C_{jg} = \frac{\alpha^g_C c_{jg} Y_{jg}}{q^C_j}, \qquad X_{jg} = \frac{\alpha^g_X c_{jg} Y_{jg}}{p^X}. \tag{B.63}$$

The skill shares satisfy $\sum_k \lambda_{jgk} = 1$. Because output sells at the national price, the zero-profit condition (23) ties the skill-weighted wage a producer can pay to its productivity and commercial rent,

$$\alpha^g_L \sum_k \lambda_{jgk}\, d\log w^g_{jk} = d\log A_{jg} - \alpha^g_C\, d\log q^C_j. \tag{B.64}$$

Output Y_{jg} then adjusts until employment equals the labor supply at these wages, so a producer shrinks when its workers are offered higher wages elsewhere. Profits are zero in every location and sector.

Entry raises the productivity $A_{j_E g_E}$ of the entrant's location-sector by the factor that raises its employment by J^E in the entry equilibrium. The entrant is not distinguished from incumbent producers in its location-sector. It pays the sector–skill wages in its tract, and its demands for labor, commercial space, and intermediate inputs follow (B.62) and (B.63), so it competes with other producers for local commercial space.

B.9 Property supply, residential dividends, and property taxes

Residential and commercial supply follow (26), independently at each tract. Residential and commercial rents paid by users include the property tax: $q^R_i = (1+t) r^R_i$ and $q^C_j = (1+t) r^C_j$. Housing demand sums the individual demand $h_\ell = \alpha_\ell y_\ell / q^R_i$. Commercial demand sums the tradable and nontradable demands above. Equating each demand to supply gives (27).

Residential rental income $\mathcal{R} = \sum_i r^R_i H^R_i$ is distributed using fixed skill ownership shares. Hence

$$D_k = \frac{\omega^0_k \mathcal{R}}{N_k}, \qquad \sum_k N_k D_k = \mathcal{R}, \qquad d\log D_k = d\log \mathcal{R} - d\log N_k. \tag{B.65}$$

The differential applies to positive ownership shares. Dividends accrue to current residents, including renters and new arrivals; commercial rents accrue to absentee landlords.

The tax applies to pretax rent on all residential and commercial floor space, including the space used by the entrant's location-sector. With commercial rental income $\mathcal{R}^C$,

$$\mathcal{R}^C = \sum_{j \in \mathcal{J}} r^C_j H^C_j, \qquad \mathcal{T} = t\left(\mathcal{R} + \mathcal{R}^C\right). \tag{B.66}$$

Tax revenue leaves the modeled economy. It is not rebated to residents or spent locally. Nonemployment income comes from outside the CZ. The award is a payment to the entrant that does not change its hiring, and the model does not solve how it is financed.

B.10 Equilibrium accounting and feasible choices

For each entrant specification, equilibrium determines

$$\begin{aligned}
&\{w^g_{jk}, r^R_i, r^C_j, p_{jg}, m_{jg}\}_{i,j,k,g},\\
&\{F_{ijgk}, U_{ik}, n_{\ell k}, N^{\text{reg}}_k, N^{\text{nat}}_k, N_k, N\},\\
&\{L^g_{jk}, H^C_{jg}, R_{jg}\},\\
&\{h_\ell, P_i, \pi^I_{ijg}, \pi^O_i, \mathcal{E}_i, D_k\}.
\end{aligned}$$

These quantities satisfy individual choice and consumption, firms' input and entry conditions, labor and property-market clearing, the revenue account, finite-population accounting, and the residential portfolio described in Section 4.4.

In particular, the sector-by-sector labor-market conditions underlying (30) are

$$\sum_i F_{ijgk} = L^g_{jk}, \qquad g \in \mathcal{G},\ j \in \mathcal{J},\ k \in \{H, L\}. \tag{B.67}$$

The tradable conditions include the entrant's location-sector, whose labor demand reflects its higher productivity. The service conditions hold separately for each $g \in \mathcal{G}_N$ and each skill k. Summing commercial-space demands over all four sectors gives (27).

Modeled service destinations are tracts with positive pre-entry nontradable employment. Workplace–sector choices use the observed baseline employment support. The entrant's location-sector has positive baseline employment in every event, so entry opens no new choices. Primitives are calibrated jointly to pre-entry prices, employment, costs, spending, and budgets. Each skill has a positive current resident population, feasible consumption choices, and a housing expenditure share strictly between zero and one. Individual disposable income is positive for every available choice.

B.11 Place-based welfare and employment accounting

For a place a, either a county or the CZ, let V^s_{ak} be the nested inclusive value over choices with residence in a. Amenities, tastes, and preference parameters are common to the entry and no-entry calculations. Let Λ^0_ℓ collect the amenity, commuting, and price factors multiplying y_ℓ in (4), evaluated at no-entry prices. Equivalent variation EV_{ak} solves

$$\begin{aligned}
&V^0_{ak}(\text{EV}_{ak}) = V^1_{ak},\\
&\quad \Phi^0_\ell(x) = \mathbf{1}\{\ell \in \mathcal{A}^0_k\}\Lambda^0_\ell \max\{y^0_\ell + x, 0\}, \qquad i \in a.
\end{aligned} \tag{B.68}$$

Thus EV_{ak} is the uniform transfer, evaluated at no-entry prices and opportunities, that reproduces the entry inclusive value. Here $\mathcal{A}_k^0$ is the structurally available no-entry choice set. A transfer can restore an income-infeasible choice within this set. Place-level equivalent variation weights these skill-specific transfers by no-entry resident populations: $\mathrm{EV}_a = \sum_k N_{ak}^0 \mathrm{EV}_{ak}$ in total dollars and EV_a/N_a^0 per individual. The tables report these aggregates and both skill-specific transfers. This measure values residential opportunities in a place; individual welfare incidence can also be studied using matched transitions.

Workplace employment changes include the entrant's jobs, the adjustment at other tradable producers, and nontradable adjustment. The annual gross fiscal price for place a is $S^E/\Delta L_a$ when $\Delta L_a > 0$. The annual net fiscal measure uses $(\frac{1}{2}S^E - \Delta\mathcal{T})/\Delta L_a$, where the county pays half of the annual award S^E. The full commitment per net job is $\overline{S}^E/\Delta L_a$, where $\overline{S}^E$ is the full award commitment. The government account is CZ-wide, including when the employment denominator is a county.

B.12 Individual transitions and welfare

Equilibrium counts follow the nested shares in (B.21). Individual transitions use common taste draws across the no-entry ($s = 0$) and entry ($s = 1$) equilibria. Let $\mathcal{A}_k^s$ contain the choices with feasible consumption under specification s. Chosen locations and employment, and maximized indirect utility, are

$$\ell_k^s(\omega) = \arg\max_{a \in \mathcal{A}_k^s} z_a(\omega)\Phi_a^s, \qquad v_k^s(\omega) = \max_{a \in \mathcal{A}_k^s} z_a(\omega)\Phi_a^s. \tag{B.69}$$

The taste distribution and potential location–industry choices are fixed across counterfactuals. Feasibility responds to equilibrium prices and income.

The skill-k individuals moving from choice a to choice b number

$$T_{ab,k} = \mathcal{N}_k \Pr\{\ell_k^0 = a,\ \ell_k^1 = b\}, \qquad \sum_b T_{ab,k} = n_{ak}^0, \qquad \sum_a T_{ab,k} = n_{bk}^1. \tag{B.70}$$

The joint probabilities use the same draws in both equilibria. Summing them by sector, workplace, residence, or nonemployment gives movements between these categories. Employment changes are net stock changes; tradable worker transitions are measured at the workplace level.

Welfare comparisons use $v_k^0(\omega)$ and $v_k^1(\omega)$ for the same individuals. Each utility includes the dividend associated with the chosen current residence. Welfare averages among current residents also reflect the changes in composition generated by migration and sorting.

C Quantification and Equilibrium Computation

All monetary flows are annual and expressed in constant 2023 dollars. Superscript 0 denotes the pre-entry equilibrium. A subscript c identifies CZs in the calibration; it is omitted from the single-

CZ model in Section 4 and Appendix B.

This appendix supports Section 5 in the order of its subsections. Appendix C.1 collects the parameters, Appendices C.2–C.4 support the individual choices and budgets in Section 5.1, Appendix C.5 supports Section 5.2, and Appendix C.6 supports the construction of the pre-entry economy in Section 5.3. Appendix C.7 describes the equilibrium computation, and Appendix C.8 the exposure used in the validation of Section 5.4.

C.1 Common and CZ-specific parameters

Table 5 reports the parameters. The tax, travel-cost, technology, establishment-entry, and commercial-supply parameters are common across CZs. Residential supply elasticities vary by tract. Housing expenditure shares, shopping elasticities, ownership shares, and location-specific shifters are calibrated using the corresponding CZ data. Migration, sector, and participation scales are common across skills and CZs: $\mu = 1.38$, $\nu = 1.64$, and $\eta = 3.02$. The boundary scale varies across event baselines according to $\psi_e = q_{\text{size}(e)}\chi_{c(e)}$, where size$(e)$ is the event's baseline CZ-size group and $\chi_{c(e)}$ is the fixed residence–workplace scale of the event's CZ. The ratios for small, medium, and large CZs are $(0.665, 0.811, 1.000)$. Both worker skills face the same boundary scale within an event. Baseline wages, nonemployment rates, and preference parameters may differ by skill.

The nontradable basket comprises private retail, education, health, accommodation and food, and other services (NAICS 44–45, 61, 62, 72, and 81). Appendix C.5.2 derives the production cost shares of the two tradable groups and the nontradable composite, and Appendix C.6.3 describes nontradable fixed input requirements.

C.2 Sector choice, migration, and recruitment

This subsection supports the sector-choice, migration, and participation parameters in Section 5.1.

C.2.1 A public benchmark for the sector-choice elasticity

I discipline the common sector-choice elasticity, ν, using public Quarterly Workforce Indicators (QWI) employment and earnings data for 2012–2019. The preferred pooled employment-stock response is 1.64. The estimation compares tradable and nontradable employment, pooling the model's high- and low-skill industry categories within each broad sector. It provides an external benchmark for the common sector scale; its structural interpretation requires the restrictions discussed below.

Data and specification. Employment is full-quarter jobs (`EmpS`) and earnings are average monthly earnings of full-quarter employees (`EarnS`), for private-sector workers aged 25 and older. Nontradables comprise NAICS 44–45, 61, 62, 72, and 81; tradables comprise the remaining covered private industries. I use published broad-industry totals and sum employment and earnings

bills across counties and industries within each CZ, education group, and sector before calculating average earnings.[31]

For employment L_{cks} and average annualized earnings w_{cks} in CZ c, education group k, and sector $s \in \{T, N\}$, I estimate

$$\Delta \log L_{cks} = \beta^{\text{QWI}} \Delta \log w_{cks} + \alpha_{ck} + \delta_{d(c)s} + \gamma_{ks} + \varepsilon_{cks}, \tag{C.1}$$

where Δ denotes the 2012–2019 change and $d(c)$ is the Census division containing CZ c. The fixed effects absorb CZ-by-education, division-by-sector, and education-by-sector differences in growth. Education distinguishes workers with and without a bachelor's degree; the pooled specification imposes a common coefficient. Each sector receives half its CZ–education group's fixed 2012 benchmark employment weight, held constant across specifications.[32] With equal sector weights within each CZ–education group, the coefficient also equals the IV estimate from the relative-outcome regression

$$\Delta \log\left(\frac{L_{ckN}}{L_{ckT}}\right) = \beta^{\text{QWI}} \Delta \log\left(\frac{w_{ckN}}{w_{ckT}}\right) + \widetilde{\delta}_{d(c)} + \widetilde{\gamma}_k + \varepsilon_{ckN} - \varepsilon_{ckT}, \tag{C.2}$$

where $\widetilde{\delta}_d = \delta_{dN} - \delta_{dT}$ and $\widetilde{\gamma}_k = \gamma_{kN} - \gamma_{kT}$.

Instrument and identification. Because earnings and employment are jointly determined, I estimate equation (C.1) by weighted two-stage least squares. The instrument combines pre-period industry employment shares with subsequent employment growth outside the focal CZ:

$$a_{cksi} = \frac{\overline{L}_{cki}}{\sum_{j \in \mathcal{I}_s} \overline{L}_{ckj}}, \qquad Z_{cks} = \sum_{i \in \mathcal{I}_s} a_{cksi} G_{i,-c}, \tag{C.3}$$

$$G_{i,-c} = \log\left(\frac{L_{i,-c,2019}}{L_{i,-c,2012}}\right), \tag{C.4}$$

where $\overline{L}_{cki}$ is mean industry employment over 2007–2011, $\mathcal{I}_s$ contains the broad industries in sector s, and $L_{i,-c,t}$ pools employment across education groups outside the focal CZ. External growth uses the fixed source sample of 2,365 counties in 30 states, yielding 19 broad-industry shocks. Both stages use the same fixed effects and weights; the instrument for equation (C.2) is $Z_{ckN} - Z_{ckT}$.

Identification requires instrument relevance and orthogonality between residualized exposure and unobserved changes in relative labor supply, conditional on the fixed effects. For a labor-supply interpretation, external growth must affect relative employment through earnings rather than through correlated changes in sector amenities, nonwage compensation, or worker composition, and average earnings must track the opportunities of comparable workers. Predetermined shares and the leave-out construction do not establish these restrictions by themselves (Goldsmith-

[31] A county–industry–education–year observation is retained when at least three quarters have positive, finite employment and earnings with accepted QWI release codes (1, 6, or 10). Annual employment averages those quarters, and the annualized earnings bill averages $12 \times$ `EmpS` $\times$ `EarnS`. Suppressed observations are not imputed.

[32] The benchmark weight is the saved 2012 employment total in NAICS 31–33, 44–45, 51–55, 61, 62, 72, and 81. It is not recalculated when the covered tradable industries are expanded.

Pinkham et al., 2020; Borusyak et al., 2022, 2025). Standard errors are clustered by CZ. Anderson–Rubin confidence intervals address weak instruments under this covariance assumption; they do not account for dependence across CZs arising from shared industry exposures (Adão et al., 2019).

Estimates. The preferred pooled estimate is $\widehat{\beta}^{\mathrm{QWI}} = 1.64$, with a CZ-clustered standard error of 0.51, first-stage $F = 15.03$, and an Anderson–Rubin 95 percent confidence interval of $[0.79, 3.24]$. Table C.1 compares specifications using identical observations, instruments, and weights. Removing regional sector controls yields 0.80 (SE 0.37; $F = 36.68$). Additionally replacing CZ-by-education effects with plain CZ effects yields 1.16 (SE 0.32; $F = 37.05$) and allows differences in growth between education groups within a CZ to contribute to identification. The estimates therefore remain sensitive to the treatment of regional and education-specific growth.

Table C.1: Employment-stock responses across tradable and nontradable industries

Specification	Estimate	SE	First-stage F	AR 95% interval
Preferred controls	1.64^{***}	0.51	15.03	$[0.79, 3.24]$
No regional sector effects	0.80^{**}	0.37	36.68	$[0.05, 1.60]$
CZ and education-by-sector effects	1.16^{***}	0.32	37.05	$[0.57, 1.90]$

Notes: All rows pool education groups and use 2012–2019 changes, 2007–2011 exposure shares, and identical observations, instruments, and weights. Preferred controls are CZ-by-education, division-by-sector, and education-by-sector fixed effects. The second row removes division-by-sector effects; the third also replaces CZ-by-education effects with CZ effects. Standard errors are clustered by CZ. The first-stage statistic is the squared cluster-robust t statistic on the excluded instrument. AR intervals invert the Anderson–Rubin test using the same clustering and an $F(1, C-1)$ reference distribution, where C is the number of CZs. Stars indicate $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$ for two-sided CZ-clustered Wald tests of a zero coefficient, using t_{C-1} with $C = 378$; they do not use the AR tests.

Interpretation and the four-sector model. I use the pooled estimate as an external benchmark for the common sector scale $\nu = 1.64$ in Appendix C.2.2. The regression measures an employment-stock response that can include migration, commuting, participation, and composition. In the four-sector model, ν governs choice among sector opportunity values. Mapping the estimated coefficient into ν therefore requires restrictions connecting those values to observed earnings and accounting for the other adjustment margins. A structural calibration could instead target the model's analogue of equation (C.1), computed with the same aggregation, weights, and horizon.

The four-sector extension yields 0.80 (SE 0.36; first-stage $F = 18.50$; AR 95 percent interval $[0.13, 1.77]$). It uses more detailed industry data and a slightly smaller complete-choice sample, so the comparison also changes data coverage. Pairwise diagnostics vary substantially: the N_H–N_L estimate is -1.46 (SE 0.40; $F = 9.88$), with an AR interval of $[-3.02, -0.80]$. These findings qualify the interpretation of a common elasticity across all four choices. Dynamic worker-transition estimates, such as Artuç et al. (2010) and Caliendo et al. (2019), concern sector transitions governed by moving costs and continuation values. They provide context for sector mobility, but their parameters are not directly comparable to this seven-year employment-stock response.

C.2.2 Recruitment calibration

I hold the sector scale at $\nu = 1.64$ and calibrate five parameters, $\theta = (\mu, \eta, q_S, q_M, q_L)$, using 55 event baselines in 38 commuting zones. Migration and participation are common across events; inward commuting varies with CZ size and the measured residence–workplace scale:

$$\psi_e = q_{\text{size}(e)}\chi_{c(e)}, \qquad 0 < \mu \le \nu \le \eta \le \psi_e \le \chi_{c(e)}, \quad 0 < q_S, q_M, q_L \le 1. \tag{C.5}$$

The measured χ_c are held fixed. Size groups use pre-entry resident employment averaged over years -4 to -2, with cutoffs of 288,152 and 860,806 jobs. These public-event terciles proxy for the source-data size categories; the exact source-data cutoffs are unavailable. The groups remain fixed when selecting an event subset.

Entry is a productivity shock to an existing tract and industry, calibrated to deliver the promised net employment increase. It shares its equilibrium wages with other workers in that tract and industry; there are no entrant-specific joining costs. For each candidate, I rebuild the nested choice probabilities around the observed pre-entry allocation, solve the entry equilibrium, and integrate the joint baseline and post-entry choices under persistent tastes. Gross recruits are workers entering the expanding tract and industry who did not previously work there. Prior workers in that tract and industry are excluded even when they change residence. Both worker skills are pooled.

Let $G_e(\theta)$ denote gross recruits and let $h(e)$ index the six entrant-type–CZ-size cohorts. If cohort h accounts for share H_h/H of observed hires and contains n_h model events, its event count weight is

$$w_e(\theta) = \frac{H_{h(e)}/H}{n_{h(e)}G_e(\theta)}. \tag{C.6}$$

Thus each cohort receives its observed hire share and events within a cohort receive equal hire mass. All six cohorts are represented. For moment s, let A_{se} and B_{se} be its raw event numerator and denominator, and let $e \in s$ index the events that inform it. The pooled ratio, moment errors, and identity-weighted minimum-distance criterion are

$$m_s(\theta) = \frac{\sum_{e\in s} w_e(\theta)A_{se}(\theta)}{\sum_{e\in s} w_e(\theta)B_{se}(\theta)}, \tag{C.7}$$

$$d_s(\theta) = 100\big[m_s(\theta) - \widehat{m}_s\big], \qquad \widehat{\theta} \in \arg\min_{\theta\in\Theta} Q(\theta), \qquad Q(\theta) = d(\theta)' I_5 d(\theta), \tag{C.8}$$

where Θ imposes the nesting restrictions in (C.5). The reported five-moment RMSE is $\sqrt{Q(\widehat{\theta})/5}$, in percentage points.

The first target is the pooled share of gross recruits who previously lived outside the CZ, had no prior job inside it, and live inside after entry: 12.93%. The second is the no-prior-job share among recruits who live inside the CZ both before and after entry: 9.50%. The remaining three targets are new inward commuters, who live outside before and after entry and had no prior CZ job, as a share of all gross recruits within each size group: 20.16%, 19.33%, and 23.16% for small, medium,

and large CZs. Nonemployment uses its own continuing-resident denominator. No-prior-job status is a provisional proxy for nonemployment. National-origin workers have no prior CZ attachment in the model; no prior-employment fraction is imputed to them.

The selected values are $\mu = 1.38$, $\eta = 3.02$, and $(q_S, q_M, q_L) = (0.665, 0.811, 1.000)$. The original calibration has a five-moment RMSE of 1.72 percentage points. Multiple starting values, final refinement, direct perturbations, and conditional reoptimization profiles support local convergence; they do not establish a global optimum. No parameter standard errors are reported because an empirical covariance matrix for the recruitment moments is unavailable. Prior-sector shares and Fact 3 employment and payroll responses are excluded from the calibration objective.

The current specification removes tax receipts from the economy rather than returning them through county fiscal transfers. I retain the selected labor coefficients and recompute the source shares for all 55 events. Table C.2 reports both evaluations. The current five-moment RMSE is 1.73 percentage points. Pooling inward commuting across sizes yields a three-moment RMSE of 1.12, compared with 1.10 in the original calibration; this is a different summary from the five-moment criterion used to fit the parameters. The large-CZ ratio reaches its upper bound, and its current commuting share remains 3.86 percentage points below target. The current evaluation is not a new optimization under the revised fiscal rule.

Table C.2: Recruitment targets and model shares

Moment	Target	Original calibration	Current model	Current error (pp)
External migration, pooled	12.93	13.14	13.11	+0.18
Nonemployment, continuing residents	9.50	9.68	9.78	+0.28
New inward commuting, small CZs	20.16	20.16	20.12	-0.04
New inward commuting, medium CZs	19.33	19.33	19.32	-0.01
New inward commuting, large CZs	23.16	19.31	19.30	-3.86
Five-moment RMSE (pp)	—	1.72	1.73	—

Notes: Shares are percentages; errors are model minus target in percentage points. Both columns use the same 55 events, cohort hire weights, and labor coefficients. The original calibration permits county fiscal transfers. The current specification removes tax receipts from the economy and has no fiscal transfers or award financing. Its shares are recomputed without refitting the labor coefficients. Nonemployment is conditional on continuing CZ residence; the other shares use all gross recruits in the indicated sample.

The event baselines do not provide 55 independent empirical source-moment samples. The data measure hiring histories, whereas the static model measures transitions between two equilibria; these transitions do not identify cumulative five-year hires or the hires of a separate firm within the expanding tract and industry.

C.2.3 Historical migration IV diagnostic

This diagnostic predates the sector-first calibration. Its skill-specific coefficients and uncertainty are not the active common migration parameter. The simplified value proxy below does not recover

the complete current sector-first inclusive value.

From the model to the estimating equation. Equation (B.11) gives the regional share of skill-k individuals. Its log odds are

$$\log \frac{b_k^{\text{reg}}}{1 - b_k^{\text{reg}}} = \log \frac{\overline{B}_k^{\text{reg}}}{\overline{B}_k^{\text{nat}}} + \mu_k \left(\log V_k^{\text{reg}} - \log V_k^{\text{nat}} \right). \tag{C.9}$$

Treating each commuting zone c as a region and the rest of the country as the national choice, the odds are $P_{ck}/(P_k^{US} - P_{ck})$. Differencing between 2010 and 2018 removes the preference parameters, and the change in the national value, which is common to all CZs within a skill, is absorbed by the division-by-skill effects:

$$\Delta \log \frac{P_{ck}}{P_k^{US} - P_{ck}} = \delta_{d(c),k} + \mu_k \, \Delta \log V_{ck}^{\text{reg}} + u_{ck}, \tag{C.10}$$

where u_{ck} collects changes in preference parameters and amenities relative to the division. Because the outcome is in log odds, its coefficient is μ_k itself; the population elasticity in equation (B.26) multiplies it by $1 - b_k^{\text{reg}}$, which is close to one for a single CZ. The horizon is one eight-year difference, and the coefficient is not annualized.

By equation (B.3), the change in an inclusive value is, to first order, the share-weighted average of the changes in the values below it, so $\Delta \log V_{ck}^{\text{reg}}$ averages $\Delta \log \Phi_\ell$ over the CZ's choices. From equation (4), holding amenities and nonwage income fixed, the value of a working choice changes by

$$\Delta \log \Phi_\ell = \frac{w_{jk}^g}{y_\ell} \, \Delta \log w_{jk}^g - \alpha_{H,c(i)} \Delta \log q_i^R - (1 - \alpha_{H,c(i)}) s_N \, \Delta \log \mathcal{P}_\ell. \tag{C.11}$$

The historical estimating equation was

$$\Delta \log \frac{P_{ck}}{P_k^{US} - P_{ck}} = \delta_{d(c),k} + \mu_k \Delta \mathcal{V}_{ck} + u_{ck}, \qquad \mathcal{V}_{ck} = \log(e_{ck} w_{ck}) - 0.25 \log r_c. \tag{C.12}$$

Its value proxy replaces the average of these changes with the change in expected earnings, $e_{ckt} w_{ckt}$, less 0.25 times the change in rent. The coefficient on $\Delta \mathcal{V}_{ckt}$ therefore recovers μ_k when the proxy's weights match the model's: a unit weight on earnings against w_{jk}^g / y_ℓ, and 0.25 on rent against $\alpha_{H,c(i)}$. Changes in service prices, dividends and nonemployment income enter the error. The sample also covers individuals aged 25–64 rather than everyone aged 18 or older.

Instrument and inference. Local earnings respond to local population through u_{ck}, so I instrument $\Delta \mathcal{V}_{ckt}$ with $Z_{ck} = \sum_n \omega_{ckn} g_{n,-c}$, where ω_{ckn} is the 2010 share of skill-k employment in commuting zone c in industry n and $g_{n,-c}$ is national CBP employment growth in industry n outside the CZ, with industries harmonized across classification vintages. The identifying assumption is $E[Z_{ck} u_{ck} \mid d(c)] = 0$: national industry growth outside the CZ affects local population only through the value of living there, conditional on division effects. Population growth can also reflect aging,

mortality, and education transitions, which would violate this assumption if they were correlated with the industry mix. I estimate equation (C.12) by two-stage least squares with equal weights across commuting zones, using the 2010 and 2018 ACS five-year files and QWI earnings by skill. CZs with similar industry mixes are exposed to the same shocks, so Table C.3 reports standard errors clustered by three-digit industry shock as well as by commuting zone (Adão et al., 2019; Borusyak et al., 2022). The low-skill first stage is below ten, and both confidence intervals are wide. A matched 2002–2010 estimate is not available in comparable data.

Table C.3: Historical migration IV diagnostics

	High skill	Low skill
Historical estimate, μ_k	2.32^{***}	1.51^{***}
Standard error, clustered by CZ	0.75	0.51
Standard error, clustered by three-digit industry shock	1.13	1.24
First-stage Wald statistic	10.71	8.41
95% confidence interval	[1.35, 5.60]	[0.90, 4.40]
Commuting zones	558	559

Notes: Stars use the first reported standard errors, clustered by CZ, and two-sided t_{C-1} tests of a zero coefficient, where C is the skill-specific CZ count. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$. The industry-shock clustered standard errors and weak-instrument diagnostics are reported separately.

C.3 Commuting and shopping

This subsection supports the commuting and shopping parameters in Section 5.1.

C.3.1 Commuting

The commuting scale χ and the cost per minute κ are common to both skills. Within a lowest nest $\mathcal{B}$, equation (B.6) gives the share of the residence–workplace pair $\ell = (i, j)$ as Φ_ℓ^χ relative to the nest total, with no pair-specific preference parameter, and the conditional shares in equation (C.33) reproduce the baseline allocation. The baseline flows of skill k are the commuting gravity weights balanced to the observed residence and workplace margins (Appendix C.6.1),

$$F_{ijk}^0 = a_{ik}\, b_{jk} \exp(-\chi\kappa\, t_{ij}), \tag{C.13}$$

where a_{ik} and b_{jk} are the balancing factors that match the margins. Cobb–Douglas pair values factor into residence and workplace components as shown in Appendix C.6.2. These components determine the economic interpretation of the balancing factors. Changes in income and prices update the fitted shares through $\hat{\Phi}_\ell^\chi$ (Appendix B.5). Equation (C.13) is a gravity equation, and the corresponding estimating equation is

$$E[F_{ij}] = \exp\left(\alpha_i^C + \delta_j^C + \beta_c^C t_{ij}\right), \qquad \beta_c^C = -\chi\kappa, \tag{C.14}$$

where the residence and workplace effects α_i^C and δ_j^C take the place of the balancing factors.

I estimate equation (C.14) by PPML separately for each commuting zone, which accommodates the many pairs with no commuters (Silva and Tenreyro, 2006; Dingel and Tintelnot, 2026), using 2019 LODES residence–workplace flows of all workers, since the gravity weights are common to both skills, and 2020 driving times on routes of up to 180 minutes. The coefficient is identified from variation in travel time within residences and workplaces, and $\chi_c = -\beta_c^C/0.01$ under the normalization $\kappa = 0.01$ per minute. The scale χ governs where workers live and work within the CZ and is distinct from ψ_e, which governs whether residents of adjacent CZs work inside it.

C.3.2 Shopping

Using the industry aggregation in (B.48), spending from origin i at destination j in (16) is

$$\mathcal{E}_i \pi_{ij}^I = \underbrace{\mathcal{E}_i P_i^{\sigma_N - 1}}_{\text{origin}} \underbrace{\sum_{g \in \mathcal{G}_N} \xi_{jg} m_{jg} \left(p_{jg}\right)^{1-\sigma_N}}_{\text{destination}} \zeta_{ij} \exp\{-(\sigma_N - 1)\kappa t_{ij}\}. \tag{C.15}$$

Visits V_{ijc} measure trips rather than dollars. If spending per visit varies only by origin or by destination, the fixed effects absorb it. Writing the taste weight as $\zeta_{ij} = \exp\{\gamma_c \mathbf{1}[i = j] + \xi_c \mathbf{1}[\text{same county}]\}$ gives the estimating equation

$$E[V_{ijc} \mid \text{controls}] = \exp\{\alpha_{ic} + \delta_{jc} + \beta_c^S t_{ij} + \gamma_c \mathbf{1}[i = j] + \xi_c \mathbf{1}[\text{same county}]\}, \tag{C.16}$$

with $\beta_c^S = -(\sigma_{N,c} - 1)\kappa$.

I estimate equation (C.16) by PPML separately for each commuting zone, using 2019 visits and 2020 routes. The same-county term is included where supported, standard errors are clustered by origin and destination, and the routed support includes positive travel times up to 180 minutes, with five minutes assigned to trips within a tract. Visits are pooled over NAICS 44, 45, 61 excluding 6113, 722, and 81; education visits proxy for the health-care network and are counted once. The standard error of $\sigma_{N,c}$ is the coefficient's standard error divided by 0.01.

Origin and destination effects absorb how attractive each location is, but bilateral tastes and trip chaining can still covary with travel time. Within-CZ decay also does not identify how much spending leaks across commuting-zone boundaries, and baseline destination shares reflect the revenue balancing of Section 5.3 as well as the estimated decay.

From the coefficient to the elasticity. A pooled coefficient for each CZ's nontradable basket determines

$$\kappa = 0.01 \text{ per minute}, \qquad \sigma_{N,c} = 1 - \frac{\beta_c^S}{0.01}. \tag{C.17}$$

The coefficient κ is common to shopping and commuting and measures the exponential travel-cost increment per minute. Events in a CZ share $\sigma_{N,c}$, which governs substitution, markups, and free entry. The shifters T_{ij} and T_i^O incorporate shopping decay, and the destination productivities

recovered in Appendix C.6.4 align spending flows with baseline revenue. Visit data identify destinations; expenditure accounts supply spending totals, with retail expenditure measured as the merchant margin. Shopping weights are fixed across counterfactuals.

C.4 Individual budgets

This subsection supports the individual budgets in Section 5.1.

C.4.1 Income and the housing share

Workers receive wage income w_{jk}^g. Nonemployment income y_k^U is measured using public assistance, Supplemental Security Income, and Social Security payments among nonworkers. It is paid during nonemployment. Housing expenditure measures rental-equivalent services for renters and owners. Income and housing measures are per individual and use survey weights.

The population comprises individuals aged 18 or older. Skill-specific nonworker-to-worker ratios from pre-entry ACS microdata scale the observed resident worker counts. The accessible outside area pools neighboring CZs. Its population ratios for individuals aged 25–64 are adjusted to cover everyone aged 18 or older using available neighboring-state microdata; available-state age adjustments also bridge the national population stocks. These geographic and demographic approximations preserve a common finite-population decomposition.

Outside housing dividends divide observed rental-equivalent residential rental income, net of the property tax, according to outside ownership value shares and populations. Outside property tax revenue is not rebated. These dividends, outside residential rents, and the outside housing expenditure share are fixed in the counterfactuals. The local portfolio account reconciles individual income and rental income.

For each CZ, let $\widehat{E}_{H,c}$ be the weighted total pre-entry rental-equivalent housing-service expenditure and $\widehat{Y}_c^{\text{inc}}$ the total after-tax worker earnings and nonemployment income in the same sample. The residential portfolio distributes $\widehat{R}_c = \widehat{E}_{H,c}/(1+t_c)$ to the sample's adults. Sum income and the measured dividends within each observed household to obtain Y_{mc}. If E_m is its whole-household service bill and w_{mc} its survey weight times the PUMA-to-CZ fraction, the constant share and the implied service share are

$$\alpha_{H,c} = \text{wmed}_{m\in c}\left(\frac{E_m}{Y_{mc}}; w_{mc}\right), \qquad \gamma_c = s_N(1-\alpha_{H,c}). \tag{C.18}$$

Each observed household receives its survey weight once when calculating the median, and the model applies the resulting share to each individual's budget. The median targets a typical household's expenditure burden, and aggregate expenditure remains a separate diagnostic (Appendix A.7). Events sharing a CZ use the same earliest-pre-entry sample and parameter. The outside expenditure share uses the corresponding household median in its existing neighboring-CZ sample and fixed outside resource account. In the model's baseline, with total local income $Y_c^{\text{inc},0}$, producer housing

rental income satisfies

$$R_c^0 = \frac{\alpha_{H,c} Y_c^{\text{inc},0}}{1 + t_c - \alpha_{H,c}}. \tag{C.19}$$

The measured portfolio defines the empirical median's resource denominator; equation (C.19) determines rental income in the model's baseline. The portfolio divides this income according to ownership shares and current skill populations. Demanded housing at the observed residential rents anchors each tract's residential supply schedule. Constant expenditure shares approximate the housing Engel curve at the median household share, and the implied marginal service share is $s_N(1 - \alpha_{H,c})$. Appendix C.4.2 estimates that Engel curve.

C.4.2 The housing Engel curve

Housing expenditure determines how individual resources translate into demand for services and how residential rents enter location choices. The model uses a Cobb–Douglas housing share $\alpha_{H,c}$ for each CZ. The empirical calibration matches the household-weighted median housing-service expenditure share in the pre-entry sample. An auxiliary Engel-curve exercise describes the income gradient in housing shares and clarifies the distinction between a household budget share and a marginal expenditure parameter.

Estimating the Engel curve. The auxiliary exercise uses pre-entry ACS person and housing microdata for 15 CZs, selecting the earliest available pre-entry year within each CZ. Households are assigned fractionally from PUMAs to CZs; the crosswalk uses tract-population weights for 2010 PUMAs and area weights for 2000 PUMAs. All amounts are annual and expressed in 2023 dollars. Let n_m denote the number of adults in household m, ω_{mc} its household survey weight times the geographic allocation fraction, and $a_{mc} = n_m \omega_{mc}$ its adult weight. Per-adult resources in this auxiliary specification are reported household income less five percent of household wage income, divided by n_m. Positive-resource households below the within-CZ adult-weighted first percentile are trimmed. The fit uses owners and renters across the remaining income distribution, comprising 117,140 distinct households.

For renters, expenditure is annual contract rent. For owners, this particular auxiliary specification uses reported selected monthly owner costs when positive, with a rental-imputation fallback. Reported costs cover 99.97 percent of the included owner adult weight. The owner measure includes financing payments, including mortgage principal, and therefore differs from rental-equivalent service consumption. A survey-weighted renter regression of log rent on log rooms, bedrooms, structure type, and PUMA–year effects supplies the fallback and a common reference-dwelling price index q_m. Predicted rents are retransformed using a weighted smearing factor. Reference-dwelling rents are normalized to have an adult-weighted geometric mean of one.

Writing per-adult expenditure as e_m and resources as y_m, the auxiliary weighted least-squares

regression is

$$\frac{e_m}{y_m} = \alpha^{SG} + \delta_c \frac{q_m}{y_m} + u_{mc}, \qquad \delta_c = (1 - \alpha^{SG})\bar{h}_c. \tag{C.20}$$

The intercept is common across CZs and the price-to-income slopes are CZ-specific. The estimate is $\widehat{\alpha}^{SG} = 0.07$, with a conditional bootstrap standard error of 0.33 percentage points. The bootstrap resamples PUMA clusters; 195 of 199 draws yield usable estimates, holding the price and housing-imputation first stages fixed. This uncertainty measure omits first-stage and geographic-assignment uncertainty. The fitted intercept is the expenditure-share limit as q_m/y_m approaches zero, and hence an extrapolated limit rather than the observed average housing share. The corresponding expenditure relationship is

$$e_m = \alpha^{SG} y_m + (1 - \alpha^{SG}) q_m \bar{h}_c, \tag{C.21}$$

$$\left.\frac{\partial e_m}{\partial y_m}\right|_{q_m} = \alpha^{SG}, \qquad \frac{e_m}{y_m} = \alpha^{SG} + (1 - \alpha^{SG})\frac{q_m \bar{h}_c}{y_m}. \tag{C.22}$$

The derivative is structural only under the maintained demand and resource assumptions. Cross-sectional income variation can also reflect saving, measurement error, housing quality, financing arrangements, and household heterogeneity. Mean housing shares of approximately 24–31 percent in this auxiliary sample can therefore coexist with an estimated intercept near 6.70 percent.

Comparison with Tsivanidis. Tsivanidis (2026), Section 5.1, sets the long-run housing share $1-\beta = 0.24$ to match Bogotá's high-income expenditure pattern. His Appendix D.7 calibrates wages and the skill-productivity parameter to the wage premium, and then matches the GEIH average housing expenditure share of 0.31 using the housing minimum, jointly with the car-expenditure calibration. The reported post-period minimum is 1.21 in his model's units. My estimate of 0.07 is obtained from a pooled ACS regression with CZ-specific slopes. The estimators, housing measures, resource definitions, and populations differ; the estimates are not a controlled comparison of preferences between the United States and Bogotá.

Measurement and specification sensitivity are quantitatively material. A separate ACS exercise measures owner housing with rental equivalents, reconciles the residential portfolio with alternative replacements of combined investment income, and reserves households for validation. Its pooled full-sample intercepts range from 1.56 to 2.39 percent across the resource conventions, while held-out renter comparisons can favor the fixed 0.07 value. Owner service expenditures are imputed, so the combined fit does not independently validate owner consumption. These results identify sensitivity to measurement and accounting; they do not isolate a single explanation for the gap with Tsivanidis. The auxiliary Engel curve is evidence on housing demand, with its assumptions and limits reported alongside the estimate.

C.4.3 Housing-value shares by skill

Let V_a be the reported housing value of owner-occupied household a, v_a its survey weight, and f_{ak} the fraction attributed to skill k. Value is split equally between the householder and a spouse or partner when both are present; a householder without one receives the full attribution. The pre-entry ownership share is

$$\omega_{ck}^0 = \frac{\sum_{a \in c,\, \text{owner}} v_a V_a f_{ak}}{\sum_{a \in c,\, \text{owner}} v_a V_a}, \qquad \sum_k \omega_{ck}^0 = 1. \tag{C.23}$$

Pre-entry ACS microdata discipline these shares, with PUMA-to-CZ allocation providing the geographic mapping. Value shares incorporate group sizes, homeownership rates, and property values. The weights remain fixed across equilibria and allocate each skill's share of current residential rental income equally among its current residents.

C.5 Firms and space

This subsection supports Section 5.2.

C.5.1 Industry groups and baseline allocation

Tradable industry type is defined by the 2000 Census bachelor's-degree employment share at four-digit NAICS. A tradable industry is high skill when its share exceeds the unweighted median across industries, 0.21. High-skill nontradables comprise private education and health (NAICS 61 and 62); low-skill nontradables comprise private retail, accommodation and food, and other services (NAICS 44–45, 72, and 81). This industry label is distinct from individual education. The productive sectors are T_H, T_L, N_H, N_L, and both worker skills are available in each. The tradable basket comprises NAICS 31–33, 51, 52, 54, and 55; private nontradables comprise 44–45, 61, 62, 72, and 81.

LODES supplies tract employment totals over event years $-4, -3, -2$. Within each broad tradable industry, I pool Infogroup establishment employment over these years and apply its high- and low-skill industry shares to the corresponding LODES mean. All classified industries in NAICS 51, 52, 54, and 55 belong to T_H; manufacturing employment is split using its local industry mix. Missing or ambiguous manufacturing allocations use the classified manufacturing mix in the same tract, then the county, then the CZ. This rule applies to 2.89% of tradable employment across event baselines, and each allocation retains its source flag. Nontradable totals follow the fixed NAICS groups: private NAICS 61 and 62 enter N_H, and private NAICS 44–45, 72, and 81 enter N_L.

QWI private-sector employment and earnings provide the association between worker education and industry type. High-skill workers have a bachelor's degree or more; the remaining education categories form the low-skill group. QWI education categories describe workers aged 25 and above, so their industry association is used as a prior for the model's individual employment allocation. I retain industry-years with released information for both worker skills. Employment composition uses beginning-of-quarter employment; earnings use full-quarter employment to weight annualized

monthly mean earnings, expressed in 2023 dollars. The wage and employment samples retain their separate disclosure flags. County estimates pool the three pre-entry years. Where a county lacks usable information in any year for either worker skill, the corresponding CZ and event window supply the prior. This geographic fallback accounts for 0.08% of productive employment across event baselines. Coverage files retain suppressed and unclassified industries and compare released detailed employment with broad-industry controls.

Within a tract and broad sector $b \in \{T, N\}$, let L^b_{jH} and L^b_{jL} denote the two industry totals, so that broad sector b and industry type $f \in \{H, L\}$ together identify sector $g = b_f$, and let L^b_{jk} denote the existing totals by worker skill. Iterative proportional fitting allocates the industry totals across workers while preserving both margins. If p^b_{jf} is the QWI high-skill worker share for industry type f, the allocation satisfies

$$\sum_k L^b_{jfk} = L^b_{jf}, \qquad \sum_f L^b_{jfk} = L^b_{jk}, \qquad \frac{L^b_{jHH} L^b_{jLL}}{L^b_{jHL} L^b_{jLH}} = \frac{p^b_{jH}/(1-p^b_{jH})}{p^b_{jL}/(1-p^b_{jL})}. \tag{C.24}$$

The odds-ratio condition applies to positive flows; zero employment margins remain zero. Each fitted home–work flow is divided using its workplace's industry shares conditional on worker skill. The accessible outside location uses Census-2000 national industry shares conditional on education.

Let $\widetilde{w}^b_{jfk}$ be the county or CZ QWI earnings prior, and $\overline{w}^b_{jk}$ the existing broad-sector wage. Initial wages retain the QWI ratio between industry types and match the broad-sector payroll:

$$w^{b,0}_{jfk} = a^b_{jk} \widetilde{w}^b_{jfk}, \qquad a^b_{jk} = \frac{\overline{w}^b_{jk} L^b_{jk}}{\sum_{f \in \{H,L\}} L^b_{jfk} \widetilde{w}^b_{jfk}}. \tag{C.25}$$

This normalization applies to positive broad-sector employment. It preserves workplace payroll and aggregate earnings at each residence. Wages subsequently clear distinct workplace–industry–worker-skill labor markets. Skill cost shares are calibrated from each industry's payroll. Initial nontradable revenue follows its payroll and the common labor cost share. Establishment masses and baseline destination expenditure shares are divided according to these revenue shares. The two industry masses sum to the tract's initial establishment mass, and both industries enter one CES shopping basket. Each sector uses the common sector scale ν; its work and nonemployment choices use the separate common participation scale η.

C.5.2 Production cost shares

Data and industry aggregation. I use the 2012 detailed supply–use accounts of the U.S. Bureau of Economic Analysis (BEA). The source is the 2012 worksheet in `Use_SUT_Detail.xlsx`, together with the industry's classification in the workbook's `NAICS Codes` concordance.[33] All source amounts are in millions of current 2012 dollars, and industry gross output is valued at basic prices.

[33] The workbook is distributed in BEA's supply–use data package: https://apps.bea.gov/industry/release/zip/SUPPLY-USE.zip. The version retrieved on October 28, 2026, contains historical 2012 estimates organized using the workbook's 2017-NAICS concordance. The archived source and replication files preserve the data vintage.

Let $g \in \{T_H, T_L, N\}$ index the two tradable sectors and the nontradable composite, which pools N_H and N_L. The high-skill-intensive tradable group, T_H, contains NAICS 51, 52, 54, and 55. The low-skill-intensive group, T_L, contains manufacturing, NAICS 31–33. These labels describe broad production composites: both firm types employ both worker skills, and the grouping does not separately identify high-technology manufacturing. The nontradable composite contains retail trade, education, health care and social assistance, accommodation and food services, and other services (NAICS 44–45, 61, 62, 72, and 81). Real estate is excluded as an output sector but remains an input to the modeled industries. Retail output is measured by retail margins rather than the full value of merchandise sold.

Purchased intermediate inputs. For each industry i, let Y_i denote gross output, X_i total purchased intermediate inputs, and R_i purchases of real-estate services. These correspond to BEA rows `T018`, `T005`, and `531ORE`, respectively. For the set of industries $\mathcal{I}_g$ assigned to group g, define

$$Y_g = \sum_{i \in \mathcal{I}_g} Y_i, \qquad X_g = \sum_{i \in \mathcal{I}_g} X_i, \qquad R_g = \sum_{i \in \mathcal{I}_g} R_i. \tag{C.26}$$

The intermediate-input share is

$$\alpha_X^g = \frac{X_g - R_g}{Y_g}. \tag{C.27}$$

I sum monetary amounts before taking ratios, so the resulting share is weighted by industry gross output. Subtracting R_g keeps purchased real-estate services out of the outside-input composite and avoids assigning the same expenditure to both intermediate inputs and commercial space. The aggregation uses the BEA concordance, including the nonnumeric industry code `4B0000` for other retail activities.

Allocation between labor and commercial space. I allocate the remaining production-cost share according to

$$\alpha_L^g = 0.8(1 - \alpha_X^g), \qquad \alpha_C^g = 0.2(1 - \alpha_X^g). \tag{C.28}$$

Thus, $\alpha_L^g + \alpha_C^g + \alpha_X^g = 1$ and $\alpha_C^g/(\alpha_L^g + \alpha_C^g) = 0.2$ for each group. The common 80/20 split draws on the labor–floor-space calibration in Ahlfeldt et al. (2015), Hoelzlein (2023), and Miyauchi et al. (2025). Here, I apply that benchmark to the share remaining after purchased non-real-estate intermediates have been deducted. This retains sectoral differences in intermediate-input use without introducing additional group-specific labor–space parameters.

Purchased real-estate services alone do not measure the full economic cost of commercial premises, particularly when businesses own rather than rent their space. Conversely, gross operating surplus is not itself a purchase of intermediate inputs. These considerations motivate retaining the measured intermediate-input share while using a separate benchmark for the labor–space allocation.

The allocation is nevertheless a maintained approximation. The residual $Y_g - X_g + R_g$ contains

employee compensation, purchased real-estate services, gross operating surplus, and net production taxes, subject to published rounding. Operating surplus also includes depreciation and returns to capital other than commercial premises. Allocating this entire residual between labor and space therefore does not recover their separately observed national-account payments or imply that all operating surplus and taxes accrue to local workers and property owners. It maps the accounting residual into the model's restricted set of production factors.

Resulting shares. Table C.4 reports the source totals and implied allocations. I retain the unrounded ratios in the calibration.

Table C.4: 2012 BEA inputs and calibrated production shares

	T_H	T_L	N
Number of detailed industries	38	232	41
Gross output, Y_g	5,730,400	5,746,010	4,831,575
Purchased intermediates, X_g	2,376,329	3,855,036	1,943,393
Real-estate purchases, R_g	163,670	24,482	305,066
Non-real-estate inputs, $X_g - R_g$	2,212,659	3,830,554	1,638,327
Labor share, α_L^g	0.4911	0.2667	0.5287
Commercial-space share, α_C^g	0.1228	0.0667	0.1322
Intermediate-input share, α_X^g	0.3861	0.6666	0.3391
Space share within labor–space bundle	0.2000	0.2000	0.2000

Notes: Monetary amounts are millions of current 2012 dollars. T_H comprises NAICS 51, 52, 54, and 55; T_L comprises NAICS 31–33; and N comprises NAICS 44–45, 61, 62, 72, and 81. Intermediate-input shares are calculated from BEA accounts. The 80/20 allocation of the remaining share is imposed.

Interpretation in the model. The model treats the non-real-estate intermediate composite as supplied from outside the CZ at a fixed price. This is a sourcing assumption, not a geographic fact identified by the national BEA tables: observed purchases can include inputs supplied within the same CZ. Consequently, α_X^g measures purchased-input intensity in the data and expenditure leakage only under the model's sourcing closure.

For the tradable Cobb–Douglas technologies, the calibrated shares are production exponents. For the nontradable labor–space CES nest, the 80/20 allocation is a baseline cost-share target used to recover the distribution parameter at observed factor prices; expenditure shares can change when relative prices change. The allocation between the two worker skills within labor is calibrated separately.

With incumbent tradable output, commercial rents, and outside-input prices fixed, a proportional increase in both skill wages implies

$$\frac{\partial \log L_g}{\partial \log w} = -(1 - \alpha_L^g). \tag{C.29}$$

The corresponding elasticity magnitudes are approximately 0.509 for T_H and 0.733 for T_L. These are

implications of the calibrated technology and fixed-output closure, rather than elasticities estimated from the BEA accounts.

C.5.3 Establishment entry

I set $\epsilon_m = 0.40$, in line with the OLS coefficient of 0.39 from the county regression below. The regression is descriptive rather than causal, for the reasons given at the end of this subsection, but it shows that the maintained value matches how establishment counts have moved with service receipts across counties.

The free-entry condition in (20) gives the number of varieties in tract j, suppressing the sector index g,

$$m_j = \left(\frac{R_j^N}{\sigma_N \varphi_j c_j^N} \right)^{\epsilon_m}. \tag{C.30}$$

Aggregating to counties and taking logs and five-year differences,

$$\Delta \log m_{jt} = \epsilon_m \left(\Delta \log R_{jt}^N - \Delta \log c_{jt}^N \right) - \epsilon_m \Delta \log \varphi_{jt}. \tag{C.31}$$

I measure varieties by establishments, revenue by receipts, and unit-cost growth by wage growth, and I absorb common movements in input prices and entry costs with end-year effects δ_t:

$$\Delta \log m_{jt} = \epsilon_m \left(\Delta \log R_{jt}^N - \Delta \log w_{jt} \right) + \delta_t + e_{jt}. \tag{C.32}$$

The regression uses 9,798 five-year county–window observations, after aggregating the five service sectors and excluding NAICS 53, and gives a historical descriptive coefficient of 0.39 with a conventional OLS standard error of 0.01. The error contains $-\epsilon_m \Delta \log \varphi_{jt}$ and the gap between unit-cost and wage growth. Because entry also raises revenue by attracting spending, and receipts differ from the model's merchant-margin output, I read the estimate as descriptive rather than causal.

C.5.4 Accounting units and the property tax

Amounts are expressed in constant 2023 dollars. Individual income, rents, production, and tax revenue are measured over the same one-year accounting period. The property tax rate in the reported equilibria is $t_c = 0.28$. Tax revenue, equation (29), leaves the modeled economy.

I measure a sample-specific tax-to-housing-cost benchmark by adapting Couture et al. (2024, Section 4.1.3 and online Appendix C). They divide a population-weighted median of local median property-tax bills by a population-weighted median home value converted to an annual user cost. I apply this construction at the county level to the 44 distinct host counties of the 55 model events, counting each county once. The 2014 ACS five-year release covers 2010–2014; its end year equals the median event year, so this is a common-period benchmark rather than a uniformly pre-entry measure. County population, median annual real-estate taxes, and median owner-occupied home value come from ACS tables B01003, B25103, and B25077, respectively. All sample counties have

valid estimates. Following the authors' conversion for 2014, I annualize home values at 4.6%:

$$\widehat{t}^{\,\mathrm{ACS}} = \frac{\mathrm{wmed}_{c,\,\mathrm{pop}_c}(\text{median annual tax}_c)}{0.046\ \mathrm{wmed}_{c,\,\mathrm{pop}_c}(\text{median home value}_c)} = \frac{\$2{,}432}{\$7{,}700.40} = 31.6\%.$$

The underlying population-weighted median county home value is $167,400. The numerator and denominator are aggregated separately; this is not a median of household tax-to-cost ratios. The common dollar deflator cancels from the ratio. Using aggregate taxes and aggregate home values instead (ACS B25090 and B25082), with the same annualization, gives 26.1%. For comparison, taxes are 17.4% of observed annual owner costs (twelve times ACS B25089); those costs include mortgage payments, taxes, utilities, insurance, and relevant fees, so this is a different denominator. The ACS measures owner-occupied housing and taxes paid to all taxing jurisdictions, not county-government receipts alone or commercial tax rates. These measurements are empirical benchmarks; the existing equilibrium results continue to use $t_c = 0.28$ on pretax rent.

C.6 Building the pre-entry economy

This subsection supports Section 5.3.

C.6.1 Baseline allocations and preference parameters

The baseline allocation covers individuals in private tradable and service employment and in nonemployment. The observed residence and workplace margins are balanced using the commuting gravity weights. Inbound and outbound flows connect the CZ to the neighboring region. LODES does not report commuting flows by education, so the boundary flows by skill use LODES earnings bands as a proxy: the top band, with monthly earnings above $3,333, stands in for high-skill workers and the two lower bands for low-skill workers. For each tract and skill, the earnings-band flows set the share of jobs held by residents of the CZ rather than inbound commuters and the share of residents who work outside the CZ. These shares are applied to the QWI workplace totals by skill, which are measured by education. The additional outside-residence/outside-workplace worker stock is neighboring-CZ employment net of existing inbound workers. Nonworker counts and national population stocks complete the finite-population account.

Let $n^0_{\ell k}$ be the baseline mass of skill k in choice ℓ, including the national choice. For a nest v whose members $\mathcal{C}(v)$ are location pairs or broader choices, write M^0_{vk} for the baseline mass of all choices in the nest. Let $\mathcal{I}(\ell)$ collect the conditional choices specifying ℓ. The nest's baseline conditional probabilities are

$$\pi^0_{u|v,k} = \frac{M^0_{uk}}{M^0_{vk}}, \qquad M^0_{vk} = \sum_{u \in \mathcal{C}(v)} M^0_{uk}, \qquad n^0_{\ell k} = \mathcal{N}_k \prod_{(u,v) \in \mathcal{I}(\ell)} \pi^0_{u|v,k}. \tag{C.33}$$

On choices with positive baseline mass, these conditional probabilities reproduce participation, sector, area, and commuting allocations. Location-pair commuting probabilities within the CZ are

predicted by the fitted gravity system; they do not reproduce every realized bilateral count. The amenities recover these fitted probabilities in levels. At broader nests, (B.28) recovers the preference parameters given the baseline inclusive values. The shape parameters govern responses to relative utility. The scale order and the placement of the inbound commuting nest are in Appendix B.2.

Location amenities. Let a^0_{ik} and b^0_{jk} be the fitted residence and workplace factors for skill k, and ϑ^0_{jgk} the share of workplace employment in sector g. Within the CZ, fitted commuting flows are allocated across industries using these workplace shares:

$$F^0_{ijgk} = a^0_{ik} b^0_{jk} \vartheta^0_{jgk} \exp(-\chi \kappa t_{ij}), \qquad i, j \in \mathcal{J}. \tag{C.34}$$

The workplace shares sum to one across employment sectors. Because industry allocation depends on workplace and skill, the residential factor remains common across sectors. Let $\gamma = (1 - \alpha_H) s_N$ and let $y^0_{jgk} = w^{g,0}_{jk} + D^0_k$ be disposable income for a CZ resident employed at j in sector g. One normalization of the location amenities is

$$\begin{aligned} B^R_{ik} &= (a^0_{ik})^{1/\chi} (q^{R,0}_i)^{\alpha_H} (P^0_i)^{\rho\gamma}, \\ B^W_{jgk} &= (b^0_{jk} \vartheta^0_{jgk})^{1/\chi} \frac{(P^0_j)^{(1-\rho)\gamma}}{y^0_{jgk}}. \end{aligned} \tag{C.35}$$

Substitution gives $(\Phi^0_\ell)^\chi = F^0_{ijgk}$ on internal working choices. Hence normalized utility powers reproduce the fitted location-pair probabilities without a further bilateral preference parameter. The broader preference parameters are expressed in the corresponding value units using (B.28).

Boundary flows retain their separately measured totals. For an existing working choice involving the outside location, define its fixed amenity composite by

$$\mathcal{B}_\ell = (F^0_{ijgk})^{1/\chi} \frac{(q^{R,0}_i)^{\alpha_\ell} (\mathcal{P}^0_\ell)^{\gamma_\ell}}{y^0_\ell}, \qquad i = o \text{ or } j = o. \tag{C.36}$$

These are the outbound, inbound, and outside–outside composites in (B.1). They index only the local origin or local destination, or the single outside worker stock. They introduce no unrestricted residual for an internal residence–workplace pair. For nonemployment, choose B^U_{ik} so that $(\Phi^0_\ell)^\chi = U^0_{ik}$ for each nonworking choice at i; conditional residential probabilities then equal $U^0_{ik} / \sum_{a \in \mathcal{I}_r} U^0_{ak}$ in every sector. Broader nest probabilities allocate the nonworkers across sectors as in (B.31).

These inversions apply to choices with positive baseline mass.

C.6.2 Commuting gravity

Consider internal commuters in one skill–sector group and suppress its indices. The residence and workplace amenities are the fitted factors recovered in (C.35); the workplace amenity includes the sector dimension suppressed here. Let $T = D_{ck}$ be the common local dividend and set $\gamma_c =$

$(1-\alpha_{H,c})s_N$. Cobb–Douglas maximization gives

$$\Phi_{ij} = \underbrace{\frac{B_i^R}{(q_i^R)^{\alpha_{H,c}} P_i^{\rho\gamma_c}}}_{\mathcal{R}_i} \underbrace{\frac{B_j^W (w_j + T)}{P_j^{(1-\rho)\gamma_c}}}_{\mathcal{W}_j} e^{-\kappa t_{ij}}. \tag{C.37}$$

The normalized-power probabilities within the bilateral choice group are proportional to Φ_{ij}^{χ}. Thus its internal flows have the gravity representation

$$F_{ij} = a_i b_j e^{-\chi\kappa t_{ij}}, \qquad \log F_{ij} = \delta_i + \zeta_j - \chi\kappa t_{ij}. \tag{C.38}$$

Writing M for the group's commuter mass and $Z = \sum_{a,b} (\mathcal{R}_a \mathcal{W}_b)^{\chi} \exp(-\chi\kappa t_{ab})$, the factors can be chosen as $a_i = M\mathcal{R}_i^{\chi}/Z$ and $b_j = \mathcal{W}_j^{\chi}$. At baseline, positive fitted factors solve the margin equations

$$O_i^0 = a_i^0 \sum_j b_j^0 e^{-\chi\kappa t_{ij}}, \qquad D_j^0 = b_j^0 \sum_i a_i^0 e^{-\chi\kappa t_{ij}}, \tag{C.39}$$

where O_i^0 and D_j^0 are the calibrated resident-worker and workplace margins for this group. One normalization fixes their otherwise arbitrary relative scale. Given baseline budgets and prices, the fitted factors identify the corresponding residence and workplace amenity composites. The fitted gravity system reproduces these tract margins and predicts the bilateral counts. An observed zero on a feasible link is compatible with a positive fitted expected flow.

To see how the model solves a counterfactual, let π_{ij}^0 be the normalized baseline fitted flows. Within this group,

$$\pi_{ij}^1 = \frac{\pi_{ij}^0 \exp(\chi \Delta v_{ij})}{\sum_{a,b} \pi_{ab}^0 \exp(\chi \Delta v_{ab})}, \tag{C.40}$$

$$\Delta v_{ij} = \log \frac{w_j^1 + T^1}{w_j^0 + T^0} - \alpha_{H,c} \log \frac{q_i^{R,1}}{q_i^{R,0}} - \gamma_c \left[\rho \log \frac{P_i^1}{P_i^0} + (1-\rho) \log \frac{P_j^1}{P_j^0} \right]. \tag{C.41}$$

The common commute cost and fixed amenities cancel in the value change; they remain encoded in π_{ij}^0. Market clearing determines wages, rents, shopping prices, and dividends, while the higher nests determine the group's counterfactual mass. This update uses fitted probabilities, so a zero realized survey or administrative count need not prevent a positive model flow after entry.

Residence and workplace factors incorporate income, prices, amenities, and group normalization. Higher choice nests determine the group's mass; boundary-flow calibration connects it to the outside economy. Baseline internal commuting probabilities are fitted to travel times and tract margins. Bilateral counts are predictions, with positive expected flows on feasible links between positive margins. This covariate-based approach follows the calibration principle examined by Dingel and Tintelnot (2026), who distinguish estimated probabilities from noisy realized counts in granular

commuting data. Exact-hat updates can use these fitted probabilities.

The functional-form restriction can be seen by substituting the auxiliary Stone–Geary demand into the same joint-choice structure. Resources in (C.37) would become $w_j + T - q_i^R \bar{h}_c$, and log flows would contain the additional term $\chi \log(w_j + T - q_i^R \bar{h}_c)$. This origin–destination interaction generally cannot be absorbed by separate fixed effects. It arises from purchasing power: the same minimum housing bill absorbs a larger fraction of resources at a lower-paying workplace. The Cobb–Douglas specification makes that purchasing-power relationship multiplicatively separable while retaining individual and ownership budget consistency.

C.6.3 Production costs and payroll

Appendix B.7 derives unit cost and factor demands. The following calibration applies to each service sector $g \in \mathcal{G}_N$.

The baseline shares $(\alpha_L^N, \alpha_C^N, \alpha_X^N) = (0.53, 0.13, 0.34)$ are the nontradable-composite shares in Table C.4, common to N_H and N_L. They are the shares α_L^g, α_C^g, and α_X^g of both service sectors $g \in \mathcal{G}_N$ in (17). Because production is Cobb–Douglas, these shares hold in every tract and at any factor prices, so no tract-level inversion is needed. The outside category is purchased intermediate inputs other than real-estate services. Property taxes enter through the tax-inclusive commercial rent.

The model's labor bill equals payroll:

$$W_{jg} = \sum_k w_{jk}^g L_{jk}^g = \alpha_L^g C_{jg} = \alpha_L^g R_{jg}. \tag{C.42}$$

The same wage determines firms' labor costs and individuals' employment income. The labor share above is applied directly to payroll.

Observed revenue, unit cost, and establishment mass determine the baseline fixed-input shifter through the zero-profit condition:

$$\varphi_{jg} = \frac{R_{jg}^0}{\sigma_N c_{jg}^0 (m_{jg}^0)^{1/\epsilon_m}}. \tag{C.43}$$

The inversion uses the specified shopping elasticity, production technology, and property tax. Reported entrant wages provide a validation comparison for the common workplace–skill tradable wages used in (24).

C.6.4 Service revenue and productivity inversion

The inversion recovers the service productivity and operating-cost shifters that make household spending, the shopping network, and service revenue consistent in the baseline equilibrium. It adapts the approach of Oh and Seo (2023) and proceeds in four steps. Revenue targets are inferred from payroll. The service expenditure share is set so that household spending covers these targets in

aggregate. Destination productivities allocate that spending across tracts. Operating-cost shifters then reconcile revenue per establishment with free entry. The inversion is performed separately for each event's commuting zone, and I suppress the CZ index below. Superscript 0 denotes the baseline, i indexes shopping origins, j indexes destinations, and $g \in \mathcal{G}_N = \{N_H, N_L\}$ indexes the two service sectors.

Revenue targets. Service sales are not observed by tract, so I infer them from payroll. Let W_{jg}^0 denote annual service payroll and m_{jg}^0 the number of establishments in tract j and sector g. Variable production and establishment operation use the same technology, so under free entry labor payments equal the labor share α_L^N of revenue (equation (C.42)). The revenue targets are therefore

$$R_{jg}^0 = \frac{W_{jg}^0}{\alpha_L^N}, \qquad R_j^0 = \sum_{g \in \mathcal{G}_N} R_{jg}^0, \qquad R^0 = \sum_j R_j^0. \tag{C.44}$$

These targets follow from payroll and the production specification; they are not independently observed sales.

Aggregate expenditure. Let Q_i^0 denote the income remaining after housing, including dividends, that is assigned to shopping origin i, before the service expenditure share is applied. An employed individual assigns the fraction ρ of this income to her residence and $1-\rho$ to her workplace, and a nonworker assigns all of it to her residence. Inward commuters therefore contribute to the budgets of their workplace origins, while income assigned to origins outside the CZ is excluded from Q_i^0. Service expenditure originating at i is $\mathcal{E}_i^0 = s_N Q_i^0$.

Let $\bar{\pi}^O$ be the measured share of these budgets spent outside the CZ, weighted by spending. The service expenditure share at which household spending covers aggregate revenue is

$$s_N^{\text{req}} = \frac{R^0}{(1-\bar{\pi}^O)\sum_i Q_i^0}. \tag{C.45}$$

Because individuals cannot spend more than their income after housing on services, I cap the share at one and define the pooled external budget $\mathcal{E}^{\text{ext}}$ as the remaining shortfall,

$$s_N = \min\{s_N^{\text{req}}, 1\}, \qquad \mathcal{E}^{\text{ext}} = R^0 - (1-\bar{\pi}^O) s_N \sum_i Q_i^0. \tag{C.46}$$

Thus $\mathcal{E}^{\text{ext}} = 0$ whenever household spending can cover the revenue targets. Otherwise, $\mathcal{E}^{\text{ext}}$ is the smallest additional budget that reconciles the aggregate accounts. It represents purchases financed outside the modeled household budgets.

Spatial allocation and productivity. Given total spending, the inversion recovers where it goes. Let Γ_{ij} and Γ_i^O denote the initial shopping weights for destination j and for the outside alternative. These weights carry the shopping network's travel-cost structure and its baseline

normalization of destinations. For a vector of destination multipliers $\delta = (\delta_j)_j$, the shares of origin i's spending that go to j and outside the CZ are

$$\pi_{ij}(\delta) = \frac{\Gamma_{ij}\delta_j}{\Gamma_i^O + \sum_{j'} \Gamma_{ij'}\delta_{j'}}, \qquad \pi_i^O(\delta) = \frac{\Gamma_i^O}{\Gamma_i^O + \sum_{j'} \Gamma_{ij'}\delta_{j'}}. \tag{C.47}$$

The outside weight is fixed, which pins down the level of the destination multipliers.

The external budget follows the distribution of household purchases, so households finance the fraction $1 - \mathcal{E}^{\text{ext}}/R^0$ of each tract's revenue. I recover δ by solving

$$\sum_i \mathcal{E}_i^0 \pi_{ij}(\delta) = \left(1 - \frac{\mathcal{E}^{\text{ext}}}{R^0}\right) R_j^0 \qquad \text{for each active destination } j. \tag{C.48}$$

Summing these equations across destinations shows that

$$\frac{\sum_i \mathcal{E}_i^0 \pi_i^O(\delta)}{\sum_i \mathcal{E}_i^0} = \bar{\pi}^O, \tag{C.49}$$

so the inversion preserves aggregate shopping leakage while allowing each origin's outside share and its bilateral destination shares to change.

I solve the system by iterative proportional fitting. Each iteration scales every destination multiplier by the ratio of its target to its predicted household revenue, with a common normalization that keeps the outside weight fixed. The calculation keeps the shopping network's existing support, so every destination with a positive revenue target must be reachable from origins with positive budgets.

The multipliers map into productivity through CES demand. With elasticity $\sigma_N > 1$, a destination's attractiveness is proportional to its price raised to the power $1 - \sigma_N$, and at fixed input prices and establishment counts, higher productivity lowers price proportionally. Relative to the initial normalization, the recovered multiplier therefore corresponds to

$$\frac{A_{jg}^0}{A_{jg}^{\text{ref}}} = \delta_j^{1/(\sigma_N - 1)}, \qquad \frac{p_{jg}^0}{p_{jg}^{\text{ref}}} = \delta_j^{-1/(\sigma_N - 1)}. \tag{C.50}$$

A destination that receives more revenue than the shopping network alone would send it is thus inferred to have more productive service firms, whose lower prices draw more spending. The productivity adjustment is common to the two service industries within a tract, and their baseline shares of the tract's spending are set separately as

$$\vartheta_{jg}^0 = \frac{R_{jg}^0}{R_j^0}. \tag{C.51}$$

Together, the tract multipliers and industry spending weights reproduce each tract–industry revenue target. The recovered productivities are conditional on the maintained shopping preferences and cost specification, and they do not separate physical productivity from unobserved attractive-

ness of a destination.

Operating costs and baseline consistency. With CES markup pricing, variable profits equal revenue divided by σ_N, so free entry determines baseline operating expenditure per establishment,

$$f_{jg}^0 = \frac{R_{jg}^0}{\sigma_N m_{jg}^0}. \tag{C.52}$$

I recover the operating-cost shifter that delivers this expenditure at baseline input prices and productivity (equation (C.43)), which preserves the observed establishment counts. Baseline amenity normalizations absorb the resulting change in service price levels, so the inversion also preserves baseline worker allocations. The baseline equilibrium therefore reproduces the employment, payroll, revenue, and establishment-count targets.

Counterfactual expenditure and revenue. The recovered productivity and operating-cost shifters, the service expenditure share, and the total external budget are held fixed in counterfactuals, while establishment counts and prices are endogenous. Let hats denote ratios to baseline values, and let π_{ij}^0 and $\pi_i^{O,0}$ be the fitted baseline shopping shares. The counterfactual share of origin i's spending received by tract j and sector g is

$$\pi'_{ijg} = \frac{\pi_{ij}^0 \vartheta_{jg}^0 \widehat{m}_{jg} \widehat{p}_{jg}^{1-\sigma_N}}{\pi_i^{O,0} + \sum_{j',g'} \pi_{ij'}^0 \vartheta_{j'g'}^0 \widehat{m}_{j'g'} \widehat{p}_{j'g'}^{1-\sigma_N}}. \tag{C.53}$$

Outside opportunities stay fixed, but the share spent outside the CZ changes as inside prices and varieties change.

Given counterfactual origin budgets $\mathcal{E}'_i$, household purchases and total revenue are

$$\begin{aligned} R_{jg}^{HH\prime} &= \sum_i \mathcal{E}'_i \pi'_{ijg}, \\ R'_{jg} &= R_{jg}^{HH\prime} + \mathcal{E}^{\text{ext}} \frac{R_{jg}^{HH\prime}}{\sum_{j',g'} R_{j'g'}^{HH\prime}}. \end{aligned} \tag{C.54}$$

The external budget is fixed in nominal terms, but its allocation follows current household purchases. The revenue targets are imposed only in the baseline. In counterfactuals, revenues are determined by equilibrium spending.

C.7 Equilibrium computation and verification

The solver uses residence, sector, participation, and workplace nests in that order, with common labor coefficients and tract-specific residential supply.

Individual choices are evaluated using fixed baseline predicted probabilities and changes in utility, as in (B.29). Fixed amenities and commuting costs cancel from utility ratios and remain

encoded in those probabilities. Equation (B.42) normalizes the available choices in each equilibrium.

Each equilibrium jointly solves for wages by sector and skill, residential and commercial rents, service establishment mass, and portfolio dividends, all represented in logs. A matrix-free Newton method uses GMRES steps and a line search. Continuation in the size of the entry is available when a direct full-entry solve fails. Reported equilibria satisfy the full counterfactual specification.

Saved reference states are re-verified before reuse, and newly solved states satisfy equilibrium and accounting residual tolerances of 10^{-8}. Each saved state allocates positive disposable income to every available individual choice.

Dollar equivalent variation is evaluated after the equilibrium solve. A bracketed scalar root-finder finds the transfer at reference prices that matches the entry opportunity index; this calculation does not solve the equilibrium again. Safeguarded interpolation and bisection give the same equivalent-variation solution. Initializing nearby counterfactual exercises from verified states reduces the equilibrium iterations without relaxing the residual checks.

No exercise solves how the award is financed, so the award does not enter any equilibrium. Shopping rules and the combined home-shopping/inbound counterfactual use their own no-entry equilibria.

Of 69 requested events, 55 have eligible prepared inputs. Ten lack prepared baselines, one violates the nesting condition, one lacks required inputs, and two awards are excluded from the model analysis.

Numerical convergence establishes that the reported states satisfy the specified equations; it does not establish uniqueness or empirical validity.

C.8 Exposure for the validation scorecard

For Table 6, each worker's exposure is assigned entirely to the home tract. The full model's post-entry residence shares of tradable workers at the entrant's site, within skill group, allocate entrant jobs across residences. These shares are held fixed across shopping scenarios. Outside-CZ residents contribute no within-CZ exposure, and the workplace tract is excluded without reallocation. This saved exposure does not reconstruct the pre-entry addresses required by the empirical benchmark. Slopes use equal event–tract weights after within-event demeaning, so event weights are proportional to centered exposure sums of squares. Repeated tracts in a shared CZ remain separate event observations.

D Employment, Welfare, and Incidence: Supporting Results

This appendix supports Section 6 in the order of its subsections. The sample contains 55 events in 38 CZs. Observed-event summaries weight events equally. All money is annual 2023 dollars; "resident" refers to the model's adult population, including nonworkers. Appendix E reports the award-accounting and location comparisons for Section 7.

D.1 Employment and the payroll accounting chain

This subsection supports the employment and payroll results in Section 6.1. Let J_e be the promised job increase in the entrant's tract and industry, $\Delta T_e^{\text{other}}$ the change in other tradable jobs, and ΔN_e the change in service jobs. The observed full model satisfies

$$\Delta L_e^{\text{net}} = J_e + \Delta T_e^{\text{other}} + \Delta N_e, \qquad M_e^N = 100\frac{\Delta N_e}{J_e}, \qquad M_e^{\text{net}} = 100\frac{\Delta L_e^{\text{net}}}{J_e}. \tag{D.1}$$

The service multiplier per net tradable job is $100\Delta N_e/(J_e + \Delta T_e^{\text{other}})$ when its denominator is positive. The reported median is taken over the event-level ratio, rather than a ratio of separately reported medians.

Let ΔP_e be the additional payroll at the entrant's site. Each ledger entry is $100\Delta X_e/\Delta P_e$. Resident earnings equal workplace payroll less inbound workers' earnings plus residents' earnings outside the CZ. Disposable income adds nonemployment payments and private rental dividends. Resident service budgets are $s_{N,c}(1-\alpha_i)$ times income at each residence. Home- and workplace-origin budgets can differ from resident budgets because workers cross the CZ border. Allocating these origin budgets over shopping destinations gives local spending and spending outside modeled destinations. Identities are checked in levels within events; medians across events are not an additive accounting chain.

Table D.1: Payroll, resident income, and service spending per $100 of additional payroll at the entrant's site

Account	Median	P10	P90
CZ workplace payroll	61.98	49.40	73.39
Resident labor earnings	30.17	18.82	40.59
Nonemployment payments	0.16	-0.25	0.54
Rental dividends	5.80	3.72	8.07
Resident disposable income	37.31	22.44	48.75
Resident housing expenditure	7.43	4.77	10.34
Resident service budget	20.82	12.65	28.56
Net workplace-origin budget imports	5.51	3.60	8.33
Local-origin service budget	25.96	19.66	36.55
Spending outside modeled destinations	9.99	7.44	15.82
Spending at local service firms	15.73	7.39	24.07

D.2 Welfare and its components

This subsection defines the welfare measure used throughout Section 6 and the decomposition reported in Section 6.1. For skill k and place of residence a, a county or the CZ, let V_{ak}^t be the model's inclusive value over feasible residential and employment opportunities. The cash equivalent

EV_{ak} solves

$$V_{ak}^0(y^0 + \mathrm{EV}_{ak}, p^0) = V_{ak}^1(y^1, p^1), \qquad \mathrm{EV}_a = \sum_k N_{ak}^0 \mathrm{EV}_{ak}, \qquad \overline{\mathrm{EV}}_a = \frac{\mathrm{EV}_a}{\sum_k N_{ak}^0}. \tag{D.2}$$

The transfer is uniform across the baseline alternatives of a given skill. Prices and primitives on the left are held at their baseline values. Reported welfare per entrant job is EV_c/J_e; its wage equivalent divides that ratio by the employment-weighted post-entry wage in the entrant's tract and industry. Both ratios are formed within events before reporting their medians. The measure values a place's opportunities and does not track the welfare of a fixed cohort. Regional indices are nonlinear, so host-county and rest-of-CZ welfare are evaluated directly, not obtained by subtracting or adding tract welfare totals.

The four accounting blocks are wages, rental dividends, housing prices, and service prices and variety. With $K = 4$ and $v_e(B)$ the cash equivalent when blocks in B take their entry values, the Shapley contribution is

$$\phi_{eb} = \sum_{B \subseteq \mathcal{K} \setminus \{b\}} \frac{|B|!(K - |B| - 1)!}{K!} \left[v_e(B \cup \{b\}) - v_e(B) \right], \qquad \sum_b \phi_{eb} = \mathrm{EV}_{ce}. \tag{D.3}$$

Hybrid evaluations change the stated prices or resources without solving another equilibrium. Mean percentage contributions average $100\phi_{eb}/\mathrm{EV}_{ce}$ across events. Mean dollar contributions are calculated separately and need not have those percentage shares.

Table D.2: Components of residential-opportunity welfare

Component	Share (%)	All residents	High skill	Low skill
Wages	131.4	7.99	16.92	5.55
Rental dividends	115.9	6.98	11.89	5.60
Housing costs	-77.2	-4.66	-8.34	-3.63
Service prices and variety	-70.2	-3.80	-6.79	-3.02
Total	100.0	6.50	13.68	4.50

Notes: The share column averages event-level percentage contributions. The remaining columns are mean annual dollars per baseline resident, using the indicated skill group where relevant. Components sum to total welfare within each event and skill group. All-resident dollar amounts weight skills by that event's baseline population shares.

D.3 Commuting and shopping restrictions

This subsection supports the comparisons in Table 7. Write Y_e^{ab} for the entry-minus-no-entry response, where $a = 1$ imposes home-only shopping and $b = 1$ holds inbound headcounts fixed by skill. Every specification receives the same calibrated full-model productivity shock. The original promised-job denominator is held fixed, and actual employment at the entrant's site can differ. The joint interaction is

$$I_e = Y_e^{11} - Y_e^{10} - Y_e^{01} + Y_e^{00}. \tag{D.4}$$

All contrasts are computed within the same event before taking medians or percentiles. Fixed inbound headcounts do not freeze workplace allocations. Home-only shopping has its own solved no-entry comparator; the joint specification uses that comparator's inbound totals. Restricted welfare uses the opportunity values of the corresponding specification.

Table D.3: Paired mechanism effects relative to the full model

Outcome	Comparison	Median	P10	P90
Service jobs	Home-only shopping	-2.49	-4.42	-1.00
Service jobs	Fixed inbound	-4.27	-6.91	-2.93
Service jobs	Both restrictions	-5.91	-8.57	-3.88
Service jobs	Interaction	1.28	0.76	2.09
Net jobs	Home-only shopping	-0.34	-1.94	1.19
Net jobs	Fixed inbound	-15.83	-24.30	-10.29
Net jobs	Both restrictions	-15.63	-24.37	-10.30
Net jobs	Interaction	0.69	-0.11	1.37
Welfare	Home-only shopping	-0.04	-0.24	0.00
Welfare	Fixed inbound	0.40	0.15	1.10
Welfare	Both restrictions	0.22	0.08	0.88
Welfare	Interaction	-0.03	-0.19	0.00

Notes: Employment is per 100 original promised jobs; welfare is annual dollars per baseline resident. Restriction rows subtract the full-model response. Interaction rows use equation (D.4). Percentiles describe the 55 paired event effects.

D.4 Spatial incidence, exposures, and regression checks

This subsection supports Section 6.2. It describes the illustrative map, the within-county variance share, the two exposures and the regressions, and the accounting behind the host county's service-job change in the illustrative market.

The illustrative event is Event 025. To select it without using outcomes, I first keep the earliest event year in each CZ, breaking ties by event identifier. Among these 38 reference events I select the one closest to median log baseline employed residents, again breaking ties by identifier. The maps use a common CZ geography. Each outcome's five magnitude classes use the 20th, 40th, 60th, 80th, and 100th percentiles of absolute tract changes, mirrored around zero. The units and numerical scales remain separate across outcomes. Undefined population or employment denominators are not imputed. The income-concentration statistic ranks all tracts by changes in total disposable income and selects the largest $\lceil 0.1 n_e \rceil$ tracts. It reports medians of signed event-level shares; these can be negative or exceed 100 percent.

Let $y_{ie} = 100\Delta N_{ie}/J_e$, w_{ie} be baseline service employment, and $\bar{y}_{ge}$ and $\bar{y}_e$ be weighted county and CZ means. The within-county share is

$$W_e = \frac{\sum_g \sum_{i \in g} w_{ie}(y_{ie} - \bar{y}_{ge})^2}{\sum_i w_{ie}(y_{ie} - \bar{y}_e)^2}. \tag{D.5}$$

The complementary between-county sum of squares uses $\sum_g (\sum_{i \in g} w_{ie})(\bar{y}_{ge} - \bar{y}_e)^2$. Zero total

variation makes the share undefined. Main-text results report the median event share. Equal-tract weights and exclusion of the expansion tract are reported below.

For an entrant in industry f at tract s, let $\omega^0_{r|sf}$ be the share of the industry's baseline payroll at that tract earned by workers living at r, including outside residents; S^0_{ri} is the baseline shopping share from origin r to destination i. Define $b_r = s_{N,c}(1-\alpha_r)$ and the home-shopping weight ρ. The two exposures in dollars per \$100 of payroll are

$$H^0_{ies} = 100\rho \sum_{r\in c} \omega^0_{r|sf} b_r S^0_{ri}, \tag{D.6}$$

$$W^0_{ies} = 100(1-\rho) S^0_{si} \sum_{r\in c\cup\{\text{outside}\}} \omega^0_{r|sf} b_r, \qquad P^0_{i|sf} = (H^0_{ies} + W^0_{ies})/100. \tag{D.7}$$

Payroll shares weight workers by their skill-specific baseline cash wage; headcount shares alone would not measure a dollar of payroll. Outside workers contribute to workplace purchases, using their outside housing share. All ingredients precede the expansion.

The fitted specification is

$$y_{ie} = \alpha_{e,g(i)} + \beta_H H^0_{ies} + \beta_W W^0_{ies} + X^{0\prime}_{ie}\gamma + u_{ie}, \tag{D.8}$$

with the blocks included as indicated in the table. Controls are log one plus service jobs, log average service wage, and log residential rent. Every column uses the same complete-case sample and equal observation weights. Standard errors use CZ-clustered CR1 covariance with absorbed fixed effects counted in the finite-sample correction. Within R^2 removes event-by-county means. These standard errors characterize a descriptive projection across event economies, not model-parameter or numerical-solver uncertainty.

Section 6.2 explains the host county's service-job change in the illustrative market with an exact accounting decomposition, which separates spending propagation from conversion into jobs. For service industry g, write revenue as $R_{jg} = \sum_i \pi^I_{ijg}\mathcal{E}_i$ and jobs as $N_{jg} = q_{jg}R_{jg}$, where q_{jg} is jobs per dollar of revenue. With bars denoting arithmetic averages of the two equilibria,

$$\Delta N_{jg} = \bar{q}_{jg} \sum_i \bar{\pi}^I_{ijg} \Delta\mathcal{E}_i + \bar{q}_{jg} \sum_i \bar{\mathcal{E}}_i \Delta\pi^I_{ijg} + \bar{R}_{jg}\Delta q_{jg}. \tag{D.9}$$

These terms add exactly but are not independent causal mechanism effects. Aggregating the selected event gives the following county decomposition.

D.5 Standardized entry and cross-CZ comparisons

This subsection supports Section 6.3 and the industry comparison in Section 7.3. The absolute design targets 500 additional jobs in the observed tract and industry; the proportional design targets 0.005 times baseline employed CZ residents, including outbound workers. Each is solved in TH and TL, giving 110 event–industry observations per design. The additional equal-payroll

Table D.4: Commuting and shopping exposure and tract service-job changes, including the entrant's tract

	(1)	(2)	(3)	(4)
Home-route exposure		0.362*	4.151***	4.263***
		(0.189)	(0.518)	(0.533)
Work-route exposure			-5.562***	-5.592***
			(0.716)	(0.717)
Log(1 + baseline service jobs)	0.015***			-0.028***
	(0.005)			(0.008)
Log baseline service wage	-0.006			-0.115***
	(0.027)			(0.028)
Log baseline residential rent	0.014			0.007
	(0.020)			(0.009)
Event × county effects	Yes	Yes	Yes	Yes
Within-county R^2 (%)	0.1	0.5	36.0	36.2
Tract–event observations	30,842	30,842	30,842	30,842
CZ clusters	38	38	38	38

Notes: The same specifications as Table 8, estimated on all 30,842 tract–event observations, including each entrant's own tract. All columns include event-by-county effects and give observations equal weight. Parentheses report CZ-clustered standard errors. Stars indicate $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$ from two-sided t_{37} tests using the reported clustered standard errors.

Table D.5: Within-county variation and fit

Sample	Variance weight	Within (%)	Fit (%)
All tracts	Service jobs	97.9	36.2
All tracts	Equal	99.8	36.2
Excluding expansion tract	Service jobs	79.5	75.1
Excluding expansion tract	Equal	91.7	75.1

Notes: Within is the median within-county variance share. Variance weights do not change the regression weights, which remain equal across included tract–event observations. Fit uses both exposures and all controls.

Table D.6: Service-job accounting for the illustrative event

Area	Budgets	Destinations	Jobs/revenue	Total jobs
Host county	404.98	-414.35	-66.18	-75.55
Rest of CZ	258.06	265.51	-157.53	366.05

Notes: Entries are jobs, summed over tracts and both service industries. Budgets, destinations, and jobs/revenue refer to the three terms in the preceding identity.

experiment targets $50 million of incremental annual payroll at the entrant's site, including wage changes for existing workers there; its employment normalization uses its realized job increase. No standardized design averages across alternative sites.

The variance comparison retains each event's observed industry and site, yielding 55 matched

observations. For design d, the reported percentage is

$$100\left[1-\frac{\mathrm{Var}_e(M^N_{e,d})}{\mathrm{Var}_e(M^N_{e,\text{observed}})}\right]. \tag{D.10}$$

A negative value denotes increased dispersion. This comparison changes shock size in general equilibrium; it is not an additive causal variance decomposition into independent market and shock components.

The cross-CZ regressions include industry effects and cluster by CZ. The main sample weights all event–industry observations equally. Predictors are standardized after removing industry effects, and reported fit is the fraction of within-industry variation explained. Shopping leakage weights each local origin's outside share by its home plus workplace service budget. Residential supply elasticity is weighted by baseline residents. The augmented specification is

$$M^N_{egd} = a_{gd} + \widetilde{X}'_{ce}\beta_d + \gamma^S_d\widetilde{\sigma}_{N,c} + \gamma^C_d\widetilde{\chi}_{ce} + u_{egd}, \tag{D.11}$$

where e denotes the event calibration in CZ c, g the entrant's industry, and d the standardized shock design. The vector X_{ce} contains log baseline employed residents, the inbound employment share, shopping leakage, residential supply elasticity, and the service expenditure share. Tildes denote centering within industry and division by the resulting standard deviation. Column (5) omits the two elasticity terms; column (6) adds both on the same sample.

The shopping elasticity $\sigma_{N,c}$ comes from the frozen model inputs used for the saved equilibria and is constant across events in a CZ. The commuting elasticity χ_{ce} comes from the saved baseline; it varies slightly across event years in four CZs, so I retain each simulation's actual parameter rather than imposing a common value afterward. Both elasticities enter in levels before standardization. These coefficients are conditional associations across calibrated economies, not effects of changing one parameter while holding the economy fixed. In particular, the calibrated inbound scale also depends on χ_{ce} through $\psi_e = q_{\text{size}(e)}\chi_{c(e)}$, where size$(e)$ is the CZ-size group. Table D.10 uses the earliest event per CZ, with identifier tie-breaking, so repeated calibrated years are not treated as additional markets in that sensitivity check.

Table D.7: Standardized shocks by entrant industry

Panel A. Jobs, welfare, and payroll

Shock	Industry	Service jobs	Net jobs	Welfare	Payroll/job
500 jobs	TH	12.35	47.54	1.75	102,252
500 jobs	TL	6.59	40.83	1.06	84,885
0.5% employment	TH	14.68	55.36	15.25	109,105
0.5% employment	TL	9.00	48.63	10.53	92,632
$50m payroll	TH	12.29	47.45	1.84	102,244
$50m payroll	TL	6.72	40.81	1.31	85,217

Panel B. Award per net job and welfare per public dollar

Normalization	Industry	Target jobs	Award/job ($)	EV/award	Events
500 target jobs	High	500.0	72,337	0.89	55
500 target jobs	Low	500.0	89,786	0.70	55
0.5% of employed residents	High	4,081.1	16,711	5.30	55
0.5% of employed residents	Low	4,081.1	17,683	3.40	55
$50m additional payroll	High	489.0	77,215	1.00	55
$50m additional payroll	Low	586.7	79,213	0.78	55

Notes: Medians across 55 event economies in each row. Service and net jobs are per 100 entrant jobs; welfare is annual dollars per baseline resident. Payroll/job is additional payroll per additional job at the entrant's site, in annual dollars. TH and TL denote high- and low-skill-intensive tradable industries. The equal-payroll rows use actual equilibrium job gains in their job denominators. Panel B uses each event's observed site and recorded award. Award/job divides the full ten-year award by actual net CZ jobs, and EV/award divides annual CZ welfare by the annual installment.

Table D.8: Service multipliers and market characteristics: 500-job shock

	(1)	(2)	(3)	(4)	(5)	(6)
Log baseline employed residents	-2.01**	-3.27***	-4.18***	-1.59*	-3.61***	-1.09
	(0.84)	(0.75)	(0.92)	(0.90)	(1.05)	(1.18)
Inbound employment share		-2.44***			-1.37	-1.39
		(0.54)			(1.03)	(0.90)
Shopping leakage			-3.24***		-1.94*	-1.20
			(0.67)		(1.13)	(0.92)
Residential supply elasticity				0.62	0.50	0.44
				(0.88)	(0.98)	(0.83)
Service expenditure share					0.43	0.55
					(0.56)	(0.54)
Shopping elasticity, $\sigma_{N,c}$						-2.18
						(1.34)
Commuting elasticity, χ_{ce}						-0.36
						(0.70)
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Within-industry R^2 (%)	9.3	19.3	22.6	9.8	24.0	27.2
Event–industry observations	110	110	110	110	110	110
CZ clusters	38	38	38	38	38	38

Notes: Definitions, standardization, and CZ-clustered standard errors match Table 9; only the shock normalization changes. Stars indicate $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$ from two-sided t_{37} tests using the reported clustered standard errors.

Table D.9: Market characteristics, net employment, and welfare

	Net jobs 500 jobs	Net jobs 0.5%	Welfare 500 jobs	Welfare 0.5%
Log baseline employed residents	-5.90***	-2.02	52.15	91.76**
	(1.73)	(2.40)	(35.78)	(43.39)
Inbound employment share	4.21***	1.31	-31.17	-87.38***
	(1.32)	(1.79)	(21.59)	(31.52)
Shopping leakage	-0.87	1.93	-0.58	41.32
	(1.86)	(2.35)	(24.93)	(31.96)
Residential supply elasticity	-1.68	-0.56	-19.29	-21.59
	(1.06)	(1.38)	(22.84)	(31.27)
Service expenditure share	0.11	0.73	-5.95	12.99
	(0.89)	(1.23)	(11.99)	(15.28)
Shopping elasticity, $\sigma_{N,c}$	-0.13	1.55	-9.48	20.38
	(1.42)	(1.76)	(30.75)	(39.12)
Commuting elasticity, χ_{ce}	-1.07	-0.48	-61.57***	-54.16**
	(0.84)	(1.22)	(13.67)	(24.91)
Within-industry R^2 (%)	38.7	12.0	28.0	45.4
Size-only fit (%)	33.0	6.0	7.3	30.2
Observations	110	110	110	110

Notes: All columns include industry effects and all seven predictors, including shopping and commuting elasticities as in column (6) of Table 9. Net employment is jobs per 100 entrant jobs. Welfare is thousands of annual dollars per 100 entrant jobs. Coefficients are per standard deviation of the predictor; parentheses report CZ-clustered standard errors. Fit is measured after removing industry effects. Stars indicate $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$ from two-sided t_{37} tests using the reported clustered standard errors.

Table D.10: Sensitivity to one reference event per commuting zone

Shock	Outcome	Observations	Size fit (%)	Full fit (%)
500 jobs	Service jobs	76	1.9	27.5
500 jobs	Net jobs	76	28.3	36.5
500 jobs	Welfare	76	20.8	41.2
0.5% employment	Service jobs	76	4.2	38.5
0.5% employment	Net jobs	76	11.6	17.4
0.5% employment	Welfare	76	34.7	60.7

Notes: Each design contains 38 reference event economies and two industries. Fit is within-industry R^2, expressed in percent. Full fit includes all seven characteristics, including the saved shopping and commuting elasticities. Reference selection uses only event year and identifier, not counterfactual outcomes.

E Counterfactual Costs, Locations, and Entrant Types

This appendix supports Section 7 in the order of its subsections. Appendix E.1 defines the cost and welfare ratios behind Section 7.1 and reports their sensitivity to additionality and to fixed floor space. Appendix E.2 describes the alternative sites behind Section 7.2. Appendix E.3 repeats the commuting and shopping restrictions across those locations, and Appendix E.4 compares sites at equal payroll. Appendix E.5 describes the industry swap behind Section 7.3. All dollar values are in 2023 prices. Welfare is annual residential-opportunity equivalent variation, and "resident" denotes a baseline adult, including nonworkers. Summaries weight events equally. TH and TL denote the

high- and low-skill-intensive tradable industries.

E.1 Cost and welfare ratios

This subsection supports Section 7.1. The award enters no equilibrium, so each event's costs are computed from its recorded award and the assumed county share of one half. Incremental host-county tax receipts are taken from the host county's outcome, not from the CZ total.

Definitions. For event e, let A_e be the recorded award, $a_e = A_e/10$ its annual installment, and ΔL_e^{net} the change in CZ workplace employment in equation (D.1). Let W_e and W_e^H be the annual CZ and host-county welfare gains and $\Delta\mathcal{T}_e^H$ the additional host-county property tax. The measures in Section 7.1 are

$$C_e = \frac{A_e}{\Delta L_e^{\text{net}}}, \qquad R_e^{\text{public}} = \frac{W_e}{a_e}, \tag{E.1}$$

$$F_e = \tfrac{1}{2}a_e - \Delta\mathcal{T}_e^H, \qquad R_e^{\text{CZ}} = \frac{W_e}{F_e}, \qquad R_e^H = \frac{W_e^H}{F_e} \quad (F_e > 0), \tag{E.2}$$

$$a_e^* = 2\big(W_e + \Delta\mathcal{T}_e^H\big). \tag{E.3}$$

Location comparisons replace ΔL_e^{net} and W_e with their values at each evaluated site, holding A_e and either the promised jobs or the observed incremental payroll fixed. Under the payroll normalization, the realized job increase at the site replaces J_e in the net-job identity. Industry comparisons do the same for each tradable industry at the observed site.

Cost per job-year. The main cost is $A_e/\Delta L_e^{\text{net}}$. Dividing the annual installment by the same employment change gives $a_e/\Delta L_e^{\text{net}}$, with a median of \$4,356. If the new employment level lasts for all ten payment years, total undiscounted job-years are $10\Delta L_e^{\text{net}}$, and this annual-payment ratio is also the award per job-year. The model provides a comparative-static employment change rather than an estimated persistence path. I do not assume a transition or discount rate when reporting the award per job.

Additionality. For additionality probability p, the award is paid regardless of whether it changes the firm's decision. Only the induced fraction contributes incremental jobs, welfare, and taxes. Hence the expected county outlay is $F_e(p) = a_e/2 - p\Delta\mathcal{T}_e^H$, and the annual CZ fiscal break-even award is

$$a_e^*(p) = 2p\big(W_e + \Delta\mathcal{T}_e^H\big). \tag{E.4}$$

Ratios with $F_e(p)$ in the denominator are reported only when it is positive. This avoids interpreting a negative denominator as a negative return. It also means that the county-ratio medians at different p use different subsamples. Gross welfare per annual award uses the same 55 events at every p.

Table E.1: Sensitivity to the probability that the award induces entry

Measure	$p = 1$	$p = 0.25$	$p = 0.02$
Award per induced net job ($)	43,555	174,220	2,177,753
CZ welfare / annual award	1.85	0.46	0.04
CZ welfare / net county outlay	1.07	0.50	0.05
Host welfare / net county outlay	0.31	0.16	0.01
Events with positive county outlay	10	28	49
Events with nonpositive county outlay	45	27	6
Awards above CZ fiscal break-even	5	19	47

Notes: Medians except the final three count rows. All 55 observed events have positive net employment and CZ welfare gains. County-denominator ratios use the positive-outlay counts shown below them. Host-county welfare is negative in 9 observed events and is retained with its sign. The $p = 0.25$ and $p = 0.02$ columns are sensitivity assumptions, not event-specific estimates of additionality. A negative net county outlay indicates that credited tax receipts exceed the assumed county contribution; the solved model still discards all receipts.

Fixed floor space. Section 7.1 notes that the cost per net job depends on crowd-out. Table E.2 uses the observed-site restrictions on residential and commercial floor-space supply. Each keeps the productivity shock calibrated in the full model and the same award. The restricted model's employment at the entrant's site can therefore differ from the promised jobs. The net employment multiplier divides by the original promised target for comparability, while cost per net job divides by the actual net CZ employment change in that restricted equilibrium.

Table E.2: Space-supply restrictions and the cost of net employment

Model	Net jobs / 100	Award/job ($)	Change (%)	EV/award
Full model	50.54	43,555	0.0	1.85
Fixed residential space	48.72	43,999	5.0	1.16
Fixed commercial space	37.42	64,151	32.3	1.14
Both space supplies fixed	35.88	66,524	38.9	0.61

Notes: Medians over the same 55 observed events. "Net jobs / 100" uses 100 original promised jobs. "Change" is the median within-event percentage change in cost per net job relative to the full model. EV/award is annual CZ welfare divided by the annual award. These are restrictions on physical space supply, not a fixed-capital experiment specific to tradable producers. They hold the productivity shock fixed, rather than recalibrate it to the original target jobs.

E.2 Alternative sites

This subsection supports the equal-jobs comparisons in Section 7.2. The original alternative sites for a CZ are up to 20 tracts drawn in a fixed pseudorandom order from the geography common to its repeated events, excluding every observed event tract. Each event's own observed tract is then added. The draw uses no counterfactual outcomes and gives repeated events in a CZ the same alternatives. A sampled tract with no baseline employment in the entrant's industry cannot host the entry and is not replaced within that original CZ draw.

For the host-county comparison, eligibility requires positive baseline employment in the event's target industry. I keep every eligible original county alternative and add eligible tracts in the same

fixed pseudorandom order until there are at least ten alternatives or the county's eligible pool is exhausted. The observed tract is excluded from the alternative count; another event's observed tract can be an alternative for this event. Neither selection nor eligibility uses solved employment, welfare, costs, or required productivity. The expanded comparison adds 189 equilibria and contains 570 host-county alternatives across all 55 events. 51 events have at least ten alternatives. Two events have only five eligible alternatives each, and two have nine each; all available alternatives are evaluated. These are limits of the available county geography, not failed solves.

Together the original and additional sites give 1,335 solved event–site equilibria. Another 4 event–site pairs have no baseline employment in the entrant's industry in the original CZ draw. No solved site is dropped because of its outcome. Both geographic comparisons cover all 55 events; the median host-county site count, including the observed site, is 11. The whole-CZ statistics retain the original draw so that supplementing the county does not change that comparison's sampling weights.

Table E.3: Coverage and scale of the evaluated location experiments

Measure	Minimum	Median	Maximum
Evaluated sites per event: Host county	6	11	21
Evaluated sites per event: CZ	18	21	21
Promised jobs / baseline industry jobs at site	0.021	5.783	4,555.352
Required productivity factor	1.002	1.196	2.661

Notes: Site counts summarize all 55 events. Shock-size and productivity statistics summarize all 1,335 supported event–site cases. Baseline industry employment sums both worker skills in the target tract. No target-to-baseline size screen is imposed. These diagnostics describe model support, not the physical availability of parcels or the feasibility of expansion.

In the fixed-job design, the productivity increase is calibrated to the same event-specific job increase at the entrant's site. Baseline employment in the entrant's industry varies sharply, and the median ratio of promised jobs to that employment is 5.78, with a maximum of 4,555.4. Thus some experiments substantially expand an industry that is initially small at the site. Their outcomes are retained and the ratios are recorded for every site in the underlying data. The largest equilibrium residual is 9.95×10^{-09}, the largest accounting residual is 1.50×10^{-09}, and the largest absolute gap from the job target is 5.22×10^{-05} jobs. Every reported equilibrium has equilibrium and accounting residuals no greater than 10^{-8} and meets the job target within $\max(10^{-4}, 10^{-7} J_e)$.

Table E.4 restricts evaluated host-county alternatives to promised jobs at most ten times baseline target-industry employment. The observed site remains the reference even if its own ratio exceeds ten. This sensitivity uses the evaluated sample and does not guarantee ten alternatives. Paired summaries include only events with a remaining alternative.

For each event, within-event welfare percentiles use linear interpolation. The observed-site rank is $100(r_e - 1)/(n_e - 1)$, where r_e is its ascending rank with average ranks for ties and n_e is the number of evaluated sites. The employment–welfare association is Spearman's rank correlation. The P90/P10 welfare ratio is reported only if the within-event P10 is positive. Negative-welfare sites are retained, and there are 4 among the 1,335 evaluated cases. Sites with nonpositive net

Table E.4: Host-county comparisons with a baseline employment size cap

Measure	Baseline support	$J_e/L^0 \leq 10$
Events with an alternative	55	54
Sites including observed	11	8
Observed welfare percentile	30.0	50.0
Welfare gain at median site (%)	17.6	0.0
Welfare gain at P75 site (%)	56.9	12.1
Cost saving at best evaluated site (%)	26.1	14.8

Notes: Medians of event-level comparisons. Both specifications select alternatives using baseline data only. The size-cap specification restricts the evaluated alternatives to $J_e/L^0_{sf_e} \leq 10$ and retains the observed site as the reference. Site counts include that reference. Welfare is for CZ residents, and costs use the original award.

job creation would likewise remain in coverage and welfare rankings, with award-per-net-job costs undefined. The current sample has 0 such sites. The lowest-cost site is the evaluated site with the most net CZ jobs at the fixed award.

Site characteristics. Table 12 uses two characteristics of the entrant's industry at each evaluated site, both measured in the no-entry baseline. The inbound share is the fraction of the industry's workers at the site who live outside the CZ. Work-route retention is the share of those workers' workplace-origin service spending that goes to destinations inside the CZ. Within each event, I split the evaluated sites, including the observed site, at the median of each characteristic and report the median percentage difference of each site's CZ welfare and net jobs from the event's median evaluated site. The groups use the whole-CZ candidate set to preserve the original geographic draw. Within events, the median rank correlation with residents' gain is -0.43 for the inbound share and 0.03 for retention, and the inbound share's median rank correlation with net jobs is -0.12.

E.3 Commuting and shopping restrictions across locations

This subsection extends the restrictions in Appendix D.3 to the location comparisons in Section 7.2. It uses all 1,280 supported alternative event–sites, including the county additions, and adds 3,840 equilibria. The full model and each of the three restrictions are compared at identical sites. The host-county set contains 625 event–sites and the original CZ set 1,146, including the 55 observed sites in each. Both comparisons cover all 55 events. The original four unsupported event–site pairs remain outside the comparison. Neither eligibility nor retention depends on restricted employment or welfare outcomes.

Shocks, comparators, and employment. Let a^*_{es} denote the productivity shock calibrated to the original promised jobs in the full model at event e and site s. Each restriction receives this same a^*_{es}, without recalibration. Home-only shopping uses its solved no-expansion comparator with the same primitives, without reinverting the model. Fixed inbound commuting holds total CZ inbound headcounts fixed by skill at that specification's no-expansion level; the joint restriction uses the home-only comparator's headcounts. It does not freeze the allocation of workers across workplaces.

The associated access wedges are nonpecuniary, and fiscal transfers remain zero. All new equilibria must pass the original 10^{-8} market and accounting certificates; scaled inbound-headcount residuals must also be at most 10^{-8}. The stored productivity shock must match its full-model reference exactly. The largest absolute inbound-headcount discrepancy among the new equilibria is 0.0016 workers.

Employment in the entrant's tract and sector can therefore differ across specifications. I report its change from each specification's own no-expansion comparator alongside the location comparisons. Employment outcomes are per 100 original promised jobs throughout, so their denominator does not change when a restriction changes target employment. Welfare is annual CZ residential-opportunity EV relative to the corresponding comparator, divided by the event's unchanged annual award. These are equilibrium restrictions, not a decomposition that holds the realized expansion fixed or assigns additive causal shares to the two channels.

Dispersion and rankings. For outcome Y, let S_e be the event's candidate set, including its observed site, and m index the specification. I calculate

$$D_e^m(Y) = 100\,\frac{\mathrm{sd}_{s\in S_e}(Y_{es}^m)}{\mathrm{sd}_{s\in S_e}(Y_{es}^{\mathrm{full}})}, \tag{E.5}$$

using the population standard deviation. The P90–P10 index replaces both standard deviations by their corresponding within-event quantile gaps. An index of 100 means unchanged dispersion, below 100 means less, and above 100 means more. Quantile gaps remain meaningful when some outcomes are negative. Each reported index is the median of event-level indices, giving equal weight to events rather than to candidate sites.

Rank correlations are within-event Spearman correlations with midranks. Observed-site percentile ranks run from zero to 100 as in Appendix E.2; the tables report the median absolute change from the full-model rank. A best-site change means that the two sets of maximizing sites do not overlap. For this comparison, outcomes within $\max\{10^{-8}, 10^{-6} \times \text{within-event range}\}$ of the maximum are treated as tied. This convention avoids using exact floating-point equality to break ties; it is not a tighter-tolerance numerical sensitivity exercise. The underlying data retain raw ranges, signed rank changes, undefined-comparison counts, and the full-model outcome loss from choosing a restricted model's maximizing site. The four-way interaction is also retained separately, rather than assuming that suppression effects add.

Table E.5: Mechanisms and location effects within host counties

Specification	SD index	P90–P10 index	Rank ρ	Best changes	$\lvert\Delta$ obs. rank$\rvert$
Net CZ jobs per 100 promised jobs					
Home-only shopping	95.3	94.6	0.98	1	0.0
Fixed inbound commuting	77.2	77.4	0.92	24	10.0
Both restrictions	75.1	75.4	0.92	24	10.0
Service jobs per 100 promised jobs					
Home-only shopping	93.3	93.4	0.98	6	0.0
Fixed inbound commuting	94.8	94.5	0.98	6	0.0
Both restrictions	90.1	89.3	0.97	10	0.0
CZ welfare / annual award					
Home-only shopping	100.8	101.2	1.00	1	0.0
Fixed inbound commuting	102.5	101.7	0.99	1	0.0
Both restrictions	102.9	102.8	0.99	3	0.0

Notes: All 55 events and 625 event–sites, including observed sites. SD and P90–P10 indices equal 100 in the full model. Rank correlation is with the full-model outcome at identical sites. "Best changes" counts events; other entries are medians of event-level statistics. Observed rank changes are in percentile points. Jobs refer to the entire CZ, and welfare is annual CZ residential-opportunity EV divided by the recorded annual award. No sites are excluded because of their counterfactual outcomes.

Table E.6: Mechanisms and location effects in the original CZ sample

Specification	SD index	P90–P10 index	Rank ρ	Best changes	$\lvert\Delta$ obs. rank$\rvert$
Net CZ jobs per 100 promised jobs					
Home-only shopping	94.7	94.9	0.98	8	0.0
Fixed inbound commuting	77.9	79.1	0.91	37	5.0
Both restrictions	75.9	75.4	0.90	39	5.0
Service jobs per 100 promised jobs					
Home-only shopping	94.4	94.3	0.98	1	5.0
Fixed inbound commuting	94.5	94.8	0.99	4	5.0
Both restrictions	90.6	90.4	0.97	5	5.0
CZ welfare / annual award					
Home-only shopping	100.5	100.9	1.00	0	0.0
Fixed inbound commuting	102.6	101.6	1.00	2	0.0
Both restrictions	102.6	102.2	0.99	2	0.0

Notes: The original 1,146 supported event–sites for all 55 events, including observed sites. County top-ups do not enter this comparison. Definitions and units are the same as in Table E.5.

Table E.7: Employment and welfare under fixed productivity shocks

Specification	Target jobs	Δ target	Net jobs	Services	EV / award
Host county					
Full model	100.00	0.00	53.46	16.33	1.91
Home-only shopping	100.21	0.21	53.21	13.81	1.84
Fixed inbound commuting	99.94	-0.06	34.82	11.82	2.12
Both restrictions	100.10	0.10	34.57	10.05	2.03
Original CZ					
Full model	100.00	0.00	54.07	15.90	1.87
Home-only shopping	100.23	0.23	53.25	12.50	1.79
Fixed inbound commuting	99.95	-0.05	35.68	10.54	2.04
Both restrictions	100.14	0.14	35.55	8.67	1.99

Notes: Medians of within-event means across the same candidate sites, giving equal weight to all 55 events. Target, net CZ, and CZ service-job gains are per 100 original promised jobs. The target-change column subtracts the corresponding full-model gain. Welfare is annual CZ EV relative to the recorded annual award. Every gain is relative to the specification's own no-expansion baseline. Baseline employment levels, entry levels, unnormalized gains, and target-employment quantiles are retained in the underlying data. The full-model target gain equals promised jobs up to numerical solver tolerance; no employment target is imposed under the restrictions.

Table E.8: Mechanism comparisons with the 10-times-baseline size cap

Specification	SD index	P90–P10 index	Rank ρ	Best changes	$\lvert\Delta$ obs. rank$\rvert$
Net CZ jobs per 100 promised jobs					
Home-only shopping	95.8	97.1	0.99	6	0.0
Fixed inbound commuting	79.9	80.3	0.90	19	10.6
Both restrictions	77.4	78.7	0.90	19	0.0
Service jobs per 100 promised jobs					
Home-only shopping	94.1	93.5	0.98	4	0.0
Fixed inbound commuting	95.9	95.8	0.98	5	0.0
Both restrictions	92.1	92.7	0.96	7	0.0
CZ welfare / annual award					
Home-only shopping	102.2	101.5	1.00	0	0.0
Fixed inbound commuting	101.4	101.2	1.00	1	0.0
Both restrictions	103.2	103.2	1.00	1	0.0

Notes: The host-county comparison retains observed sites and alternatives whose promised jobs do not exceed ten times original baseline employment in the target industry. The same cap and candidate set apply to all four models; the criterion does not use restricted outcomes. This leaves 407 event–sites across 54 events with at least one alternative. Definitions are otherwise the same as in Table E.5.

Table E.9 relates outcomes across each event's host-county sites to baseline characteristics of the site, its pre-entry commuting catchment, and the shopping of those workers. It supports the discussion of why sites differ at the end of Section 7.2.

Table E.9: Site characteristics and outcomes within host counties

	CZ	Host county share of	
Site characteristic	welfare	service jobs	welfare
Service employment in the site tract	-0.83	-0.70	-0.49
Jobs in the site tract	-0.79	-0.39	-0.21
Jobs per resident in the site tract	-0.74	-0.32	-0.18
Existing employment in the entrant's industry	-0.67	-0.14	0.07
Payroll per entrant job	0.62	0.19	-0.24
Workers living outside the CZ	-0.48	-0.03	-0.25
Workers living in the host county	0.30	0.19	-0.21
Home-origin shopping in the host county	-0.05	0.10	-0.43
Workplace shopping in the host county	-0.24	-0.01	-0.40
Concentration of workers' residences	0.08	0.10	-0.09
Concentration of home-origin shopping	-0.05	-0.03	-0.18
Concentration of workplace shopping	-0.28	-0.23	-0.10
Events	55	40	55

Notes: Each entry is the median, across events, of the within-event Spearman rank correlation between the site characteristic and the outcome, over each event's evaluated host-county sites including the observed site. Events with fewer than five sites with the outcome are omitted. Welfare is CZ residential-opportunity EV. Host-county shares divide the host county's service-job change or welfare index by the CZ's; the welfare indices are evaluated separately and are not additive. Commuting and shopping characteristics use pre-entry flows in the entrant's industry, and concentration is the Herfindahl index across residence or destination tracts. These are associations across sites, not separate effects.

E.4 Alternative sites at equal payroll

This subsection supports the equal-payroll comparison in Section 7.2. The equal-payroll comparison uses the 381 supported host-county alternatives for 51 events from the original CZ draw, retaining the observed site once in each candidate set. Both normalizations use that same subset; the subsequent county additions are excluded. The three originally unsupported host-county alternatives remain unpriced. The four events without a supported alternative in the original draw are excluded from paired summaries; their observed references are retained in the underlying data. No additional sites are sampled for this comparison.

Payroll target and outcomes. Let o_e be event e's observed site, k_e its tradable industry, and s a worker skill. For any candidate site ℓ, incremental annual payroll is

$$\Delta P_{e\ell} = \sum_s \left(w^{\text{after}}_{\ell k_e s} L^{\text{after}}_{\ell k_e s} - w^{0}_{\ell k_e s} L^{0}_{\ell k_e s} \right), \qquad P^*_e = \Delta P^{\text{observed}}_{e o_e}. \tag{E.6}$$

Each alternative's productivity increase is calibrated so that $\Delta P_{e\ell} = P_e^*$, subtracting that alternative's own baseline payroll. This includes wage changes for workers already in the entrant's tract and industry; it is neither total post-entry payroll nor a separately observed entrant's wage bill. The original industry, award, baseline, model parameters, and shared commercial-space market remain fixed. Employment and its skill composition adjust. Realized job gains at the site range from 0.247 to 1.620 times the original promised jobs across alternatives. All 381 alternatives pass equilibrium and accounting checks at 10^{-8}; the largest relative discrepancy from the payroll target is 2.71×10^{-08}. The 55 observed-site equilibria already meet their own payroll targets and are reused.

Let $Y_{e\ell}^J$ and $Y_{e\ell}^P$ denote an outcome under fixed jobs and fixed payroll, and Y_e^o its observed-site value. The common candidate set $\mathcal{S}_e$ includes o_e, where both designs use the same saved equilibrium. For each outcome I calculate

$$D_e^d(Y) = \max_{\ell \in \mathcal{S}_e} Y_{e\ell}^d - \min_{\ell \in \mathcal{S}_e} Y_{e\ell}^d, \qquad I_e(Y) = 100 \frac{D_e^P(Y)}{D_e^J(Y)}, \qquad d \in \{J, P\}. \tag{E.7}$$

The main table reports the median of $I_e(Y)$, not a ratio of separately computed median ranges. Both designs use outcome levels; dividing both ranges by the same P_e^* gives the identical index. Dividing by each site's realized job gain would change the comparison and is not used. The outcome is net CZ jobs, service jobs, or annual CZ welfare. Holding the award fixed also makes each event's welfare-range index identical to the index for welfare per annual award.

For cost comparisons I minimize $A_e / \Delta L_{e\ell}^{\text{net},d}$ over the same sites and compare the result with the observed cost. Every equal-payroll alternative adds net CZ jobs. The 13 alternatives with negative CZ service-job changes and the 1 with negative CZ welfare are retained. The highest-welfare-site improvement is computed separately and need not use the lowest-cost site.

Scope of the mechanism comparison. At any alternative site, the saved outcomes obey

$$Y_{e\ell}^J - Y_e^o = \left(Y_{e\ell}^J - Y_{e\ell}^P\right) + \left(Y_{e\ell}^P - Y_e^o\right). \tag{E.8}$$

The first contrast changes payroll and employment at a fixed alternative site; the second changes location at the observed expansion's payroll. This identity is path-dependent accounting, not an additive causal decomposition into scale and transmission mechanisms. In particular, a 29 percent median reduction in the welfare range is not an equally sized causal contribution of wages. The equal-payroll location contrast still allows commuting, shopping, wages, labor supply, incumbent competition, housing, and the distribution of income to respond.

Table E.10: Dispersion and site rankings after equalizing payroll

Welfare / jobs in	Outcome	Range index	Smaller	Rank corr.	Best changes
Entire CZ	Service jobs	66.2	46	0.95	15
Entire CZ	Welfare	70.6	42	0.89	22
Entire CZ	Net jobs	100.1	25	0.62	42
Host county	Service jobs	81.2	42	0.99	8
Host county	Welfare	84.1	40	0.93	25
Host county	Net jobs	105.3	25	0.74	38

Notes: All rows use the same 51 events and host-county candidate sites. The first column indicates where the outcome is measured, not a different site sample. Range index is the median of $I_e(Y)$ in equation (E.7). "Smaller" counts events whose range falls at equal payroll. Rank correlation is the median within-event Spearman correlation between the two designs; "Best changes" counts events whose maximizing site changes. Counts are out of 51. CZ and host-county welfare are separate opportunity indices and are not additive.

E.5 Entrant industry

This subsection supports Section 7.3. The industry swap keeps each event's observed site, promised jobs, and recorded award and solves the equilibrium once with each tradable industry, leaving the skill mix of each industry's workers endogenous. All 55 pairs are solved. Panel B of Appendix Table D.7 reports the same two industries at common job and payroll targets, including the award per net job and welfare per public dollar.